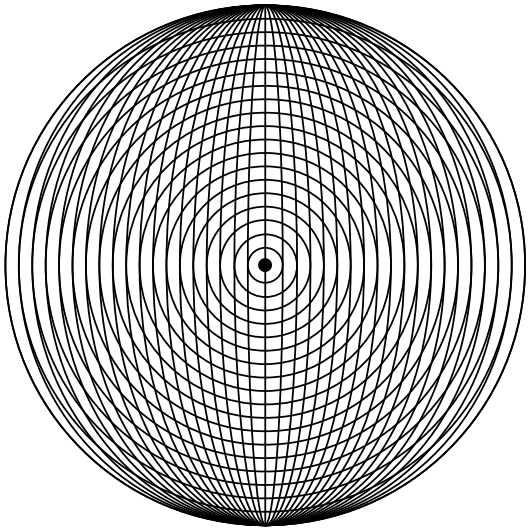
The Planetary

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EDITED BY NILS GILMAN



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Preface

We introduce this book in the hope of offering a point of focus and inflection for the creation of a new cross-cultural, interdisciplinary, and global discourse on the Planetary — a new school of thought concerning what **Nils Gilman** calls "the condition of planetarity."

The initial chapters discuss the technoscience of how we came to understand this condition — how it was disclosed to us. **Benjamin Bratton**'s concept of "planetary sapience" highlights that the Earth has recently evolved a new technological exoskeleton that is rendering the planet as a whole into a kind of sensory and cognitive organ. **Claire Isabel Webb** suggests that exploring conditions of life beyond Earth — both the conditions under which Earthly life might thrive beyond our planet and what sorts of other, non-Earthly life forms may exist — is causing us to reconsider the nature and meaning of life here on our home planet. **Stephanie Sherman** proposes that what is emerging are planetary-scale automated platforms capable of not just reestablishing the biosphere's homeostasis but also neo-cybernetic control of evolutionary processes.

The second part considers the Planetary as a philosophical event. According to **Niccolò Milanese**, the planetary represents a profound challenge to Enlightenment ways of thinking about time, which in turn demands fundamentally new narratives about the human in relation to the planet. **Boris Shoshitaishvili** explores the metaphysical nuances between several such narratives, focusing on categories like Gaia, the noosphere, and the Anthropocene. And **Song Bing** suggests a fruitful reconsideration of various Chinese philosophical traditions, in particular the value

of relational ontologies of co-becoming.

The final part considers what these technological and philosophical visions imply for what Hannah Arendt referred to as the *vita activa,* that is, the engagement in politics and policy designed to enact change. **Dawn Nakagawa** and **Laura Ryan** propose new forms of political identity rooted in multispecies constituencies rather than a nationally delimited citizenry as a new basis for legitimating new institutional architectures of governance, which **Jonathan S. Blake** explains are necessary in order to effectively address planetary challenges. These proposals amount to a pragmatic approach to planetary cooperation rooted in what **Lorenzo Marsili** describes as a new vision of universalism which, if successful, would create a novel form of what **Nathan Gardels** (following former California governor Jerry Brown) calls a "planetary realism" that is not simply about competition between sovereign states, but rather a new basis for cooperation.

Many of the governance ideas we propose here may seem wild, perhaps even utopian. There's certainly no guarantee we will get there. Indeed, the world today seems to be moving in the opposite direction. But this is the point of philosophy: to engage in what Friedrich Nietzsche called "untimely meditations." If this volume helps people learn to speak of "planetary challenges" in the same natural way that starting in the 1990s we began to speak of "global challenges," that in itself will mean success.

—NICOLAS BERGGRUEN

The Condition Of Planetarity

NILS GILMAN

The Planetary is at once a technoscientific object, a philosophical event, and a call to new forms of governance. This volume introduces the Berggruen Institute's interpretation of this emergent analytic and political category.

As a technoscientific object, the term "planetary" typically refers to Earth-scale biogeochemical processes, systems, and phenomena. The Earth, in other words, is an intricate web of ecosystems, with myriad layers of integration and interaction between various geophysical systems and living beings — both human and nonhuman — that must be understood as a totality.

Scientific holism is not a new or unusual idea. Within the Western scientific tradition, it is often associated with

the work of the polymath natural philosopher Alexander von Humboldt, who proposed the concept of *Naturegemälde* — that is, nature as a living whole. But in recent decades, a holistic understanding of planetary phenomena has been accelerated by new planetary-scale *technologies of perception*[1] — the rapidly maturing and integrating megastructure of sensors, networks, and supercomputers that together are rendering various planetary systems more and more visible, comprehensible, and foreseeable. This recently evolved exoskeleton — in essence a distributed sensory organ and cognitive layer for the planet — is fostering fundamentally new forms of what Benjamin Bratton calls *planetary sapience,*[2] which is stimulating new questions for scientists and researchers. Exobiology, psychology, microbiology, epidemiology, and especially Earth system science and systems ecology, all emphasize the lateral and vertical forms of physical and cognitive integration between systems and creatures. Everything from gut health to the search for habitable planets is now being framed in terms of holistic views of planetary-scale biogeochemical processes.[3]

The concept of the Planetary also signifies an irruption into philosophy. Just as the works of Nicolaus Copernicus and Charles Darwin put forth scientific discoveries and shattered traditional notions of the uniqueness and exceptionalism of mankind, the Planetary marks a fundamental and perhaps even deeper break from anthropocentrism. By transforming how we see our own position on the planet, the Planetary emphasizes our interpenetration and co-becoming[4] with other creatures as well as abiotic things and processes. The Planetary also discloses the limits of direct human subjective experience of fundamental

categories such as space and time, as Niccolò Milanese explores in his contribution to this volume. Indeed, planetary processes impinging upon us unfold over timescales that may be comprehensible to humans only by metaphor.[5] Terms such as Gaia, the noosphere, and the Anthropocene represent efforts both scientific and philosophical to understand this complex holism, as Boris Shoshitaishvili explains in his essay in this volume.

In its methodological holism, planetary thinking disrupts the category of the clearly bounded, monadic individual and instead underscores how every creature and system is interlaced with a constellation of others.[6] Such a conception portends the end of methodological individualism — the notion of the individual subject, which for centuries has formed the cornerstone of both liberal political theory and much Western social science.[7] The very definition of "the human," as defined since René Descartes as distinct from brute but unchanging nature on the one hand and unthinking machines on the other, is increasingly subject to both empirical and philosophical scrutiny — an investigation accelerated by a growing awareness of planetary phenomena such as global warming. This "crumbling of the conceptual distinction between humankind and nature," as the philosopher Masatake Shinohara has argued elsewhere, "allows us to consider the realm of the human as a small part of the natural world."[8]

Connected to this methodological holism, the systems theory and neo-cybernetics that subtend much philosophical reflection on the Planetary emphasize not so much alterity as *relationality*.[9] Here, too, philosophers are taking cues from scientists. For example, radical new theories, such as "loop quantum gravity," developed by the physicist Carlo Rovelli and

others, assert that reality as such does not consist of a (vastly large but finite) set of "things" uniquely ordered in time and space, but rather is an endless relational network of processes and events.[10] Taking a holistic view of the biogeochemical evolution of Earth also radically refigures how we envision the emergence of (and our search for) extraterrestrial life. As astrobiologist Michael L. Wong and his colleagues have observed, the hunt for lively exoplanets involves treating exobiology and exogeology as a single "tapestry," a "twine" that recognizes "the patterns engendered by the coevolution of biological and geological forces."[11] As Claire I. Webb's contribution to this volume further details, the technoscientific exploration of life in outer space — both the immense difficulty of Earth life thriving off-planet and the imagination of and search for radically different forms of life elsewhere in the universe — is fundamentally reshaping our appreciation for the existential requirements of life here on our own planet.

What modern technoscience is disclosing with ever greater precision, and modern philosophy is just catching up to, mirrors the long-held metaphysical intuitions of many religious and cosmological traditions. Indeed, virtually every traditional religion and Indigenous epistemology, forged before or outside the modernist rupture of the human from nature, has emphasized human embeddedness in nature and the need to respect and sustain the Earth. Thus the Planetary makes room for engaging and integrating Western philosophy with Indigenous and non-Western thought, including Eastern philosophies, as Song Bing's contribution to this volume attests.[12]

What's more, such radical innovations in technoscience

and philosophy manifest as deeply disruptive to politics. Modern political institutions are almost universally erected upon a foundation of assumptions that are being systematically revealed as untrue. Philosophically, if the monadic liberal subject is an ontological fiction, then how is it possible to maintain fealty to anthropocentric (much less ethnocentric) conceptions of democracy? Practically, if our greatest challenges are planetary in scope, then how can governing institutions focused on arbitrarily bounded national states possibly address them? Indeed, what today is sometimes called "the polycrisis" takes place in the shadow of the Planetary.[13]

Considered as a political category, the Planetary is best understood in contrast to the concept of the global.[14] As Nathan Gardels has put it: "Globalization was about markets, information flows, and technology crossing borders. The planetary is about borders crossing us, embedding and entangling human civilization in its habitat."[15] In other words, whereas "the global" is about the intentional movement and processes of human things — money, ideas, goods, services, etc. — and the management of the relations between sovereign national states in the context of those flows, "the Planetary" is about things that trespass human borders and intentions. An accelerated carbon cycle, biodiversity loss, and zoonotic viruses all have profound impacts on the well-being of humans and other creatures but are neither defined by human political borders nor limited to particular national territories.

At the same time, however, human activities are perturbing each of these planetary-scale phenomena — whether it is fossil fuel-driven carbon-loading of the atmosphere driving climate change and oceanic acidification; or the expanding urban/wildlife

interface triggering the emergence of new zoonotic diseases such as SARS, Ebola, or Covid-19; or how agro-industrial chemicals are interacting with our bodies and minds. Together with other human-induced planetary effects, such as biodiversity loss, oceanic plastics, or space junk, we face a radically different and new sort of challenge than the problems arising from globalization.[16] Whether or not geologists can agree that these disruptions warrant the designation of a new geological era called the Anthropocene,[17] they undoubtedly demand dramatic new forms of governance, which will in turn entail novel forms of politics.

The Condition

The combination of the intellectual development of the concept of the Planetary, the intensification of anthropogenic effects on the planet, and the construction of technology capable of sensing those effects reveals the *condition of planetarity*. Using the term "planetarity" here underscores that this condition is both an ontological transformation and an epistemic break with modernist understandings of the human position on the planet.[18]

The phrase "condition of planetarity" alludes to the geographer David Harvey's magisterial book *The Condition of Postmodernity: An Enquiry into the Origins of Cultural Change* (1989). Harvey took the concept of postmodernism — originally developed by architecture critics in the 1970s and then abstracted into a theory of information and knowledge by the philosopher Jean-Francois Lyotard — and explained its emergence in the context of the transformations in the late capitalist economy that today are often labeled "the neoliberal turn." Whereas Lyotard had

famously described "the postmodern condition" as defined by "an incredulity toward metanarratives" (above all the emancipatory metanarrative of Marxism), Harvey inverted the argument by showing how much postmodernism as a cultural phenomenon was really a reflex response to material transformations in a globalizing economy that abstracted and alienated labor ever further from the consumer. What Lyotard and Harvey shared, however, was a belief that postmodernity was a condition — that is, not just a new epistemological frame but an actual shift in how people live, work, and play.[19]

The Planetary, even more, represents such a condition.

A "condition" in philosophical terms derives from Immanuel Kant's concept of the *Bedingungen der Möglichkeit* — that is, "the conditions of possibility." For Kant, the conditions of possibility are the physical framework within which action can take place — both enabling action and setting its limits. A condition does not in itself cause things to happen, but rather it sets the terms within which things can happen. Planet Earth thus forms the primal ontological frame within which any human and nonhuman action becomes possible: everything about how we live, work, think, and politic is defined (quite literally) by this planet's horizons. We cannot live anywhere else, except in a brutally constrained way. There is, as the saying goes, no Planet B.[20]

The condition of planetarity exposes the narcissism of anthropocentric framings of planetary systems. Consider *The Human Condition* (1958), in which Hannah Arendt observed that "In addition to the conditions under which life is given to man on earth, and partly out of them, men constantly create their own, self-made conditions, which, their human origin and

their variability notwithstanding, possess the same conditioning power as natural things."[21] For Arendt, the terrestrial condition was the stable backdrop against which the political drama of the human played out. But the Anthropocene, philosophically, illuminates the collapse of the distinction between our "self-made conditions" and "the conditioning power of natural things." The natural order of the planet cannot be treated as the stable stage for the human drama; rather it has destabilized to the point where it is increasingly dictating the terms of the drama. As Bruno Latour put it in one of his last works, "[T]he physical framework that the Moderns had taken for granted, the ground on which their history had always been played out, has become unstable."[22] The stage itself is shaking, and the pettiness of the human dramas playing out upon it has become all too apparent.

Even if it has only recently been disclosed by growing planetary sapience, the condition of planetarity is and always has been an ontological fact — that is, an empirical statement about our place inside the planet's biogeochemical feedback systems. Acknowledging this condition entails facing the fact that humans cannot thrive unless the ecosystems we inhabit are themselves thriving. However, growing planetary sapience is also revealing dynamic transformations to the details of this condition — the damage we have done by treating the planet as an endless font of resources and a bottomless sink for waste. We can see now the folly of promethean ambitions to control and master, as well as our fiduciary duties toward Planet A.

Two Epistemes Of Planetarity

Within the emerging conversation over the condition of planetarity, there exists a fundamental schism, one that is epistemic but also of direct practical importance.

On the one hand, there are the followers of the literary theorist Gayatri Spivak, who coined the term "planetarity" in the late 20th century.[23] For Spivak, "planetarity" implies a way to refuse the "custodianship of our planet," which she sees as a continuation of colonialist logics that have long silenced voices of the poor and marginalized (the subaltern): "Planetarity is a gesture of protest, a skeptical rejection of models that attempt to flatten all peoples, living beings, and ecosystems within a totalizing organizational logic."[24] In the philosopher Jeremy Bendik-Keymer's words, "Spivak intends the term 'planetarity' to point to the negation of our epistemic representations of the planet as a unified field."[25] Or as the sociologist Jennifer Gabrys put it, "the point is not to generate an evasive figure, but rather to thwart an engagement with the planetary that hinges on uniform epistemic representations."[26]

For philosophers in this register, planetarity represents a relationship to the planet characterized by unknowability — specifically, that there exists an unbridgeable epistemic limit to technoscientific knowledge of the planet. "Modern ways of imagining the earth as a totality," wrote the scholar of English and environmental humanities Elizabeth DeLoughrey, "derive from colonial histories of spatial enclosure." Resisting what she calls "the militarism of the satellite gaze or the techno-fixes of climate change" requires "embracing the contradictions of alterity

and the limits of human knowledge."[27] In this view, the "complete interconnectedness" of the planet proposed by scientific holism threatens to continue "forms of imperial control."[28] Expressing doubts about the scalability of computational modes of envisioning the planet, design theorists Michael Richardson and Anna Munster instead propose to follow William James's injunction to pluralize planetary thought by including artistic and Indigenous representations of the Planetary.[29]

Such representations may orient our imaginations toward possibilities that we otherwise only intuit vaguely, but they have a tendency to plead for the continued relevance of the spiritual or the a-rational against the expanding compass of technoscientific reason.[30] For Spivak herself, the planet "is mysterious and discontinuous—an experience of the impossible" for which she refuses to offer "formulaic access" and which instead demands a "prayer for planetarity."[31]

Consider the "constructive theologian" Catherine Keller, who in 2019 called forth "a vein of vivid transdisciplinarity in which theology itself may offer politically useful transcodings between the religious and the secular" as opposed to "'ancient schemes' of religion within politics," which "now come buried within modern schemes of politics trapped in economic schemes, and wrapped within the planetary scene of climate change."[32] All this recalls Oliver Wendell Holmes's wry criticism of William James just after his death: "[H]is wishes led him to turn down the lights so as to give miracle a chance."[33] Such thinkers adopt a normative position with respect to the technoscientific episteme of planetary perception that mirrors Lyotard's normative view of modernist metanarratives, namely that we should adopt a posture

of incredulity if not outright refusal.

To be sure, epistemic humility is all to the good, but modesty should not be allowed to deliquesce into despair. The governance challenge of planetarity is quite literally without precedent. However much some may wish to blame Western technoscience for having paved the way for the planetary polycrisis, it is also only a better technoscience that can possibly allow us to restabilize.

We should, of course, be wary of proposals for easy "technological fixes" and of "techno-solutionism" to problems of great complexity. But it is also essential to recognize that more and better technology, especially if deployed in recognition of the condition of planetarity, will be crucial for addressing planetary challenges. Technology alone is almost never enough, but technology will be an element of practically every Pareto-efficient proposal for improvement. As Charlene Sequeira, Prateek Shankar, and Tim Maly at the Rhode Island School of Design (RISD) have observed, the interaction of technology and human thought has "resulted in the emergence of tremendous new sensory capabilities, which are as much an opportunity as they are a risk."[34] In this respect, the challenge is to maintain epistemic optimism and a bias toward pragmatic action without succumbing to a credulousness about technology as an apolitical panacea. The solution to the problems produced by technoscience is (and can only be) more and better technoscience.

In contrast to the Spivakians, these other thinkers treat the condition of planetarity as defining a *new design space for action* within the biogeochemical framework of the planet.[35] Instead of using the idea of planetarity as a way to critique contemporary scientific practice and to foreclose the possibility of technological

intervention, thinkers in this vein, such as Benjamin Bratton and Stephanie Sherman in this volume, offer a more ambitious epistemology that emphasizes the cognitive and control possibilities that are both instantiated and enabled by planetary-scale computational infrastructure.[36]

Astronomers Adam Frank, Sara Walker, and David Grinspoon, for example, argue that the macro-goal should be to move from our present "immature technosphere," which is only able to represent the condition of planetarity, to a "mature technosphere," that is, one that acts recursively to steer planetary phenomena — or at least to mitigate planetary disruptions and destructions.[37] Likewise, the polymath Stewart Brand, who pioneered holistic and systems-based planetary thinking back in the 1960s, has always emphasized the need for optimism and the value of technology, more recently in *Whole Earth Discipline: An Ecopragmatist Manifesto* (2009).[38] The broader goal, as the RISD scholars put it in an earlier draft of their manifesto, is "to create an experimental zone for planetary thinking around governance and relational theory for entangled futures." *Pace* the Spivakians, there is no choice but to embrace the responsibilities of planetary stewardship and care, albeit in a form that eschews presumptions of superiority or postures of domination.

But if we abjure the epistemic despair and techno-pessimism of Spivakian renditions of the Planetary, we should also be skeptical of views of the Planetary that seek to render it as an *ideological successor* to globalization. If globalism was the ideological reflex of globalization as an emergent process, planetarism might seem like a cognate relation of the Planetary.[39] But this would be a mistake. Whereas globalization was a

historical process defined in contrast to more localized forms of human interaction and interdependence, and thus a claim about historical change, the Planetary highlights a perdurable reality — much like Copernican cosmology or Darwinian evolution. While it may be reasonable to take an ideological position for or against a change in the human historical condition, it is nonsensical to take an ideological position with respect to an ontological condition. One may lament the ontological fact of planetarity, much as Caliban raged at not seeing the image of his own face in the mirror, but one cannot be ideologically for or against it.

Governing In The Face Of Planetarity

The condition of planetarity emphasizes the interconnectedness of human communities and ecosystems across the globe, highlighting the need for collaborative and holistic approaches to addressing complex problems.[40] It suggests that solutions to these challenges can only be reached via cooperation and coordination among nations — and among species. But it also demands that we think *beyond* national boundaries and consider the implications of policy actions in both global and planetary terms. Already more than 30 years ago, the sociologist (and leading theorist of globalization) Anthony Giddens anticipated that safeguarding planetary systems would mean thinking about completely new scales of policy intervention:

> *Since the most consequential ecological issues are so obviously global, forms of intervention to minimise environmental risks will necessarily have a planetary basis. An overall system of planetary care might be created, which*

> *would have as its aim the preservation of the ecological well-being of the world as a whole. A possible way of conceiving of the objectives of planetary care is offered by the so-called "Gaia hypothesis" put forward by James Lovelock. According to this idea, the planet "exhibits the behaviour of a single organism, even a living creature." The organic health of the earth is maintained by decentralised ecological cycles which interact to form a self-sustaining biochemical system. If this view can be authenticated in analytical detail, it has definite implications for planetary care, which might be more like protecting the health of a person rather than tilling a garden in which plants grow in a disaggregated way.*[41]

What would governance look like if the condition of planetarity were central rather than ancillary to human political self-conceptions? How would we act differently if we took seriously humanity's profound integration into Earth's planetary systems, from the microbiological scale of zoonotic viruses and gut microbes to the macrosystemic scale of the planet's changing climate? What would change as a result of humans being revealed not as masters of the planet but as part of it?

Among other ambitions, this volume proposes that governing in the face of the condition of planetarity demands a new sort of global political project that requires fundamentally new sorts of political institutions. We need to reconceive the basis for universality in a new pragmatic frame, as Lorenzo Marsili argues in his contribution.

From the condition of planetarity flows a political-ethical throughline: Our governance institutions should promote *habitability* in order to enable *multispecies flourishing*. The idea

of habitability contrasts with the concept of "sustainability" that guides much of global governance today, in particular as embedded in the United Nations' Sustainable Development Goals, which have been at the heart of the international system's economic development and environmental governance strategies since the beginning of the 21st century. Whereas the sustainability concept implicitly suggested that nature be seen and treated as something separate from humans — as a standing reserve of resources and "ecosystem services" to be managed and responsibly harvested for human benefit — the idea of habitability begins from a scientific understanding of human embeddedness and inseparability from nature. The focus of the Planetary, therefore, is on creating the conditions for the continuity of the entire web of life in which humans are inextricably embedded.

There are at least two dimensions to a politics of planetarity, both of which reflect the decentering of the human that the condition of planetarity reveals. First, it is a politics that must encompass the voice and ambitions of not just the subaltern but also the nonhuman. Second, it is a politics that exceeds the hegemonic modern institutional vessel of politics, namely the sovereign nation-state.

Appreciating the condition of planetarity means rejecting the Aristotelian tradition that claims only humans (can) engage in politics. Much of this tradition was based on the claim that politics is in its essence a linguistic activity, which in turn rested on the also false belief that only humans have "language." Here, too, new technologies of perception have undermined such philosophical certitudes. As the philosophers Sue Donaldson and Will Kymlicka observe, "[A]nimal languages in general are far more complex and

variable than humans ever imagined, as is increasingly revealed by new technologies for listening to bats, birds, whales, elephants, and others."[42] With properly tuned instruments it becomes possible to hear nonhuman others, from whale songs and primate gestures to the collective expressions of insect colonies and forests — presuming we deign to listen.[43]

Our awareness of the condition of planetarity forces us to reimagine the entire architecture of our governing institutions. As Bendik-Keymer has put it, borrowing a term from the political theorist Stefan Pedersen, the political "imaginary" of the Planetary "begins by thinking about how we can govern ourselves in a way that makes our feedback loops with the Earth sustainable for us and for the extant order of life on Earth, irrespective of national territories or sovereignties."[44] We need to imagine a new form of constitutionalism (or perhaps post-constitutionalism) that reflects our post-individualist, multispecies ontological reality. This, in turn, will entail the development of a new kind of political actor, what Dawn Nakagawa and Laura Ryan in their contribution to this volume call a "planetary constituent," as opposed to the ethnocentric concept of the national citizen or the anthropocentric concept of the cosmopolitan "citizen of the world."

This insight reveals why the current architecture of global governance, based on multilateral member-state governance institutions where sovereign national states gather to promote their separate interests or ideologies, is simply unfit for addressing planetary challenges that, by definition, exceed the scope of territorially bounded interests or ideologies. As Jonathan S. Blake argues in his contribution to this volume, what we need are new

"planetary institutions" that, instead of remaining beholden to superseded anthropocentric and nationalist categories and commitments, are designed directly to address planetary challenges while remaining accountable to their multispecies constituencies. Such a vision may sound literally wild, but it inevitably follows from an honest and fearless confrontation with the condition of planetarity.[45]

How can such ambitions be realized? What is required is the intellectual courage to relinquish old assumptions and the political will to experiment with unfamiliar forms of governance. Some of this will involve the development of extraordinary new governance capacities, likely enabled by or instantiated through emergent cognitive infrastructures offering prodigious possibilities for communication, coordination, and automation, such as those envisioned in Sherman's contribution to this volume. Without endorsing either accelerationism or techno-solutionism, we can still recognize that the planetary polycrisis presents no political option but a *Flucht nach vorne*.

Notes

1 Claire Isabel Webb, "Technologies of Perception: Searches for Life and Intelligence beyond Earth," PhD diss., MIT (2020). https://dspace.mit.edu/handle/1721.1/12902, as well as in this volume.

2 Benjamin Bratton, "Planetary Sapience," Noema (June 17, 2021). https://www.noemamag.com/planetary-sapience/, as well as in this volume and in a forthcoming MIT Press monograph.

3 See, for example, Angela Sessitsch, et al., "Microbiome Interconnectedness throughout Environments with Major Consequences for Healthy People and a Healthy Planet," Microbiology and Molecular Biology Reviews 87, no. 3 (2023): e0O212-22; and Michael L. Wong, et al., "The Process We Call Earth: Relationships between Dynamic Feedbacks and the Search for Gaiasignatures in a New Paradigm of Earthlikeness," *Perspectives of Earth and Space Scientists* 5, no. 1 (2024): e2023CN000223.

4 Bing Song and Yiwen Zhan, eds., *Gongsheng Across Contexts: A Philosophy of Co-Becoming* (Singapore: Palgrave Macmillan, 2024).

5 Stephen J. Gould observed that the concept of deep time is only humanly comprehensible by way of metaphor: *Time's Arrow, Time's Cycle: Myth and Metaphor in the Discovery of Geological Time* (Boston: Harvard University Press, 1987).

6 Scott F. Gilbert, Jan Sapp, and Alfred I. Tauber, "A Symbiotic View of Life: We Have Never Been Individuals," *The Quarterly Review of Biology* 87, no. 4 (2012): 325-41.

7 As the concept of the monadic individual is unmasked as myth — we are all multispecies assemblages — the question then becomes how to recast conceptions of self, worth, value, and being on a post-liberal, post-human basis. The work of Rosi Braidotti, Karen Bakker, and Katherine Hayles is particularly valuable on this topic. For an overview, see Francesca Ferrando and Shraddha A. Singh, "Philosophical Posthumanism and Planetary Concerns," in *Globalization and Planetary Ethics: New Terrains of Consciousness*, eds. Simi Malhotra, Shraddha A. Singh, and Zahra Rizvi, (New Delhi: Routledge India, 2023), 29-36.

8 Masatake Shinohara, "Rethinking the Human Condition in the Ecological Collapse," *The New Continental Review* 20, no. 2 (2020): 190, 196.

9 See, for example, Yuk Hui, *Machine and Sovereignty: For a Planetary Thinking* (Minneapolis: University of Minnesota Press, 2024), as well as Song Bing's contribution to this volume.

10 Carlo Rovelli, *The Order of Time* (London: Penguin, 2018).

11 Wong, et al., op cit.

12 Epistemological pluralism mirrors the growing recognition of the diversity of intelligences, as Michael Levin recently observed: "What folk psychology and prescientific concepts of minds and intelligence took for granted, the field of diverse intelligence is developing into mature science, opening the door for us to grow up as a species as we learn to recognize kin in novel embodiments." See Michael Levin, "The Space of Possible Minds," *Noema* (April 17, 2024): https://www.noemamag.com/ai-could-be-a-bridge-toward-diverse-intelligence/.

13 Historian Adam Tooze popularized the term polycrisis in the early 2020s, by which he meant simultaneous and escalating problems in the realms of politics, economics, and society that since the global financial crisis that began in 2008 have been amplifying each other worldwide, but especially in North Atlantic democracies. See Adam Tooze, "Welcome to the World of the Polycrisis," *Financial Times* (October 28, 2022). https://www.ft.com/content/498398e7-11b1-494b-9cd3-6d669dc3de33. And "The Last Dystopia: Historicizing the Anthropocene Debate in a Multipolar Age: Polycrisis" (lecture, Princeton, Princeton, NJ, November 10, 2023). https://lectures.princeton.edu/lectures/2023/tanner-lectures-human-values-adam-tooze-0. Interestingly, however, the term was originally coined in 1993 by the philosophers Anne-Brigitte Kern and Edgar Morin in their book *Terre-Patrie* (Paris: Seuil, 1993) in reference specifically to the causal imbrication of various "vital" human and ecological concerns. As Morin and Kern argued, "Il n'y a pas un seul problème vital, mais de nombreux problèmes vitaux, et c'est cette intersolidarité complexe de problèmes, d'antagonismes, de crises, de processus incontrôlés et de crise générale de la planète qui perpétue le problème vital numéro un."

14 Dipesh Chakrabarty, "The Planet: An Emergent Humanist Category," *Critical Inquiry* 46, no. 1 (2019): 1-31.

15 Nathan Gardels, "The Third Great Decentering," *Noema* (April 5, 2024). https://www.noemamag.com/the-third-great-decentering/.

16 See, for example, Dawn L. Rothe and Victoria E. Collins, "Planetary Geopolitics, Space Weaponization and Environmental Harms," *The British Journal of Criminology* 63, no. 6 (2023): 1523-38; and Columba Peoples and Tim Stevens, "At the Outer Limits of the International: Orbital Infrastructures and the Technopolitics of Planetary (In)security," *European Journal of International Security* 5, no. 3 (2020): 294-314.

17 David Adam, "Ditching 'Anthropocene': Why Ecologists Say the Term Still Matters," *Nature* (March 14, 2024). https://www.nature.com/articles/d41586-024-00786-2.

18 It is important to underscore the distinction between "planetary conditions" — that is, a planet's biogeochemical realities (as disclosed by technoscientific apparatuses of planetary perception) — and "the condition of planetarity," which represents the philosophical reflection, particularly with respect to the human, prompted by the technoscientific disclosure of planetary conditions.

19 Jean-Francois Lyotard, *The Postmodern Condition: A Report on Knowledge* (Minneapolis: University of Minnesota Press, 1984 [1979]); David Harvey, *The Condition of Postmodernity: An Enquiry into the Origins of Cultural Change* (Oxford, UK: Oxford University Press, 1990).

20 Mike Berners-Lee, *There Is No Planet B: A Handbook of the Make or Break Years* (Cambridge, UK: Cambridge University Press, 2019).

21 Hannah Arendt, *The Human Condition* (Chicago: University of Chicago Press, 1958): 9.

22 Bruno Latour, *Facing Gaia: Eight Lectures on the New Climate Regime* (Cambridge, UK: Polity, 2017): 3.

23 Gayatri Chakravorty Spivak first proposed the term “planetarity” in a 1997 lecture, “Imperatives to Re-imagine the Planet,” which was then printed in *Imperatives to Reimagine the Planet*, edited by Willi Goetschell (Berlin: Passagen, 1999). She discusses the term in “‘Planetarity’ (Box 4, WELT),” Paragraph 38, no. 2 (2015): 290-92. The term “condition of planetarity” was coined by literary critic Christian Moraru in “Critique and Its Postnational Aftermath: Dialogism and the ‘Planetary Condition,’” *Criticism after Critique: Aesthetics, Literature, and the Political* (New York: Palgrave Macmillan US, 2014): 99-112. For more on Spivak’s conception of planetarity, see Boris Shoshitaishvili’s entry in this volume.

24 Tomiris Shyngyssova and Prateek Shankar, “No One Lives on the Globe,” *Jungle Publics* (February 23, 2024). https://junglepublics.com/p/no-one-lives-on-the-globe#footnote-6-141742840.

25 Jeremy Bendik-Keymer, “’Planetarity,’ ‘Planetarism,’ and the Interpersonal” (blog), *American Philosophical Association* (September 10, 2020). https://blog.apaonline.org/2020/09/10/planetarity-planetarism-and-the-interpersonal/.

26 Jennifer Gabrys, “Becoming Planetary,” *e-flux* (October 18, 2018). https://www.e-flux.com/architecture/accumulation/217051/becoming-planetary/.

27 Elizabeth DeLoughrey, “Satellite Planetarity and the Ends of the Earth,” *Public Culture* 26, no. 2 (2014): 261, 265.

28 Gabrys, op cit.

29 Michael Richardson and Anna Munster, “Pluralising the Planetary: The Radical Incompleteness of Machinic Envisioning,” *Media+Environment* 5, no. 1 (November 8, 2023). https://mediaenviron.org/article/87980.

30 Recognizing the condition of planetarity does not imply a woo-woo celebration of “nature” as some noble form, much less a valorization of the spurious concept of the “ecologically noble savage” who supposedly lived in pristine ecological “harmony” with nature. As Michael J. Rowland noted, “there is no evidence in the archaeological, ethnographic, or historical records that humans have attained this balance for more than a few centuries.” (“Return of the ‘Noble Savage’: Misrepresenting the Past, Present and Future,” *Australian Aboriginal Studies* 2 (2004): 8.) While various epistemological traditions may have valuable ecological insights to offer, particularly with respect to local environments in which they were developed, none provides a golden key for unlocking the challenge of promoting multispecies flourishing on a planet now festooned with eight billion humans. On the history of the noble savage myth in the West, see Tristan Søbye Rapp, “The Myth of the Noble Savage,” *Noema* (July 9, 2024).

31 Spivak, *Death of a Discipline* (New York: Columbia University Press, 2003), 78, 102, 114.

32 Catherine Keller, *Political Theology of the Earth: Our Planetary Emergency and the Struggle for a New Public* (New York: Columbia University Press, 2018), 6, 5.

33 Max Lerner, ed., *The Mind and Faith of Justice Holmes; His Speeches, Essays, Letters, and Judicial Opinions* (London: Routledge, 1988).

34 Charlene Sequiera, Prateek Shankar, and Tim Maly, “What Are Planetary Phenomena?” *Polycene Design Manual*. https://polycene.design/What-are-Planetary-Phenomena.

35 Rania Ghosn & El Hadi Jazairy, eds., *The Planet after Geoengineering* (Barcelona: Actar Publishers, 2021). See also filmmaker Liam Young's "Planet Earth" and "Planetary Redesign" projects: Christina Yao, "'I'm filled with dread' over climate change says Liam Young," *Dezeen Magazine* (September 27, 2023). https://www.dezeen.com/2023/09/27/liam-young-planetary-redesign-exhibition-ngv-melbourne-interview/.

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37 Adam Frank, David Grinspoon, and Sara Walker, "Intelligence as a Planetary Scale Process," *International Journal of Astrobiology* 21, no. 2 (2022): 47-61.

38 Stewart Brand, *Whole Earth Discipline: An Ecopragmatist Manifesto* (New York: Viking, 2009).

39 Stefan Pedersen, "Planetarism: A Paradigmatic Alternative to Internationalism," *Globalizations* 18, no. 2 (2021): 141-54.

40 The field of political ecology offers a rich vein for thinking about these interconnections. For background, see Michael J. Watts, "Now and then: the origins of political ecology and the rebirth of adaptation as a form of thought," *The Routledge Handbook of Political Ecology* (New York: Routledge, 2015): 19-50.

41 Anthony Giddens, *The Consequences of Modernity* (Stanford: Stanford University Press, 1990), 170-71. I am indebted to Andrew Lakoff for pointing out this early intervention into planetary thinking, a decade before Paul Crutzen coined the term "Anthropocene" as a call to eco-political action or Spivak proposed the term "planetarity" to argue against techno-engagement with ecological systems.

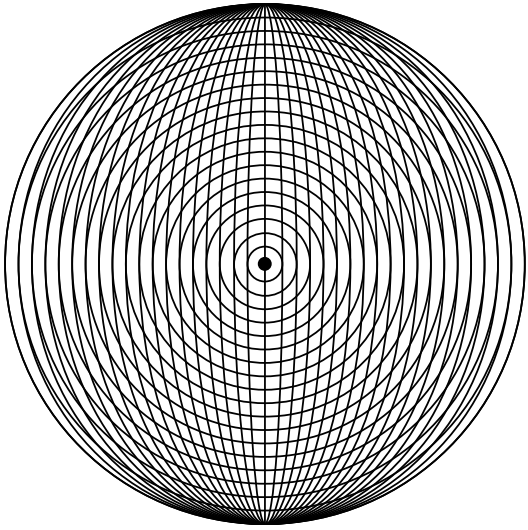
42 Will Kymlicka and Sue Donaldson, "Doing Politics with Animals," *Social Research: An International Quarterly* 90, no. 4 (2023): 621-47.

43 See for example Pratyusha Sharma, et al., "Contextual and combinatorial structure in sperm whale vocalisations," *Nature Communications* 15 no. 1 (2024): 3617. Such efforts are not without their controversies; see Mark Ryanand and Leonie N. Bossert, "Dr. Doolittle uses AI: Ethical challenges of trying to speak whale," *Biological Conservation* 295 (2024): 110648. Detecting and decoding plant communications also represents an important frontier in emergent planetary sapience; see François Tardieu, et al., "Plant phenomics, from sensors to knowledge," *Current Biology* 27 no.15 (2017): R770-R783; and Chiara Torresan, et al., "A new generation of sensors and monitoring tools to support climate-smart forestry practices," *Canadian Journal of Forest Research* 51 no.12 (2021): 1751-1765.

44 Jeremy Bendik-Keymer, op cit.

45 For proposed specifics, see Jonathan S. Blake and Nils Gilman, *Children of a Modest Star: Planetary Thinking for an Age of Crises* (Stanford: Stanford University Press, 2024) as well as Jonathan Blake's essay in this volume.

I. Technoscience



Planetary Sapience

BENJAMIN BRATTON

Consider a thought experiment: What if the famous Blue Marble image of Earth taken by Apollo 17 astronauts was instead the Blue Marble *movie* that portrayed the whole 4.5-billion-year career of the planet in a kind of super fast-forward? We would see volcanoes and storms, continents breaking apart and realigning, primordial oceans and, with the appearance of biological life after the Great Oxygenation Event, the emergence of an atmosphere that can incubate yet more life. Most of the action appears toward the end of the movie. Activity accelerates and begets yet more activity and acceleration. In the last moments of the movie, we see something unusual: clouds of satellites sprouting while metal and glass wires wrap the land and sea. An intricate artificial planetary crust

capable of tremendous feats of communication and calculation appears, enabling planetary self-awareness — indeed, planetary sapience.

This artificial planetary crust is composed of the lithosphere but designed by collaborative cognition. The emergence of *planetary-scale computation* is thus both a geological and geo-philosophical event. In addition to evolving countless animal, vegetal, and microbial species, Earth has also very recently evolved an intelligent exoskeleton, a distributed sensory organ and cognitive layer capable of calculating how old the planet is and whether it is getting warmer. Indeed, the knowledge of "climate change" is itself an epistemological accomplishment of planetary-scale computation.

Sapience does not first appear with this exoskeleton; it appears a few beats earlier as nomadic primates wandered out of Africa. These primates were notably cerebral, with large prefrontal cortexes that were both a cause and effect of their intense social cooperation and competition. Their brains allowed them to imagine, predict, and create counterfactual futures together, in dynamic coordination. They represented these ideas in pictures, sounds, and eventually writing. Along the way, they not only developed intricate models of their world but intricate models of their intricate models. Abstraction aided thinking and then became their way of thinking. The planet folded itself over millions of years to evolve creatures capable of this sapience, which in turn would artificialize itself in many ways, including in the formation of that planetary exoskeleton.

Evolved technological intelligence is both what makes the composition of this anthropogenic transformation possible and

what allows for the comprehension of the planetary condition that it discloses. Over the past few centuries, humans have chaotically and in many cases accidentally transformed Earth's ecosystems. Now, in response, the emergent intelligence represented by planetary-scale computation makes it possible and indeed necessary to conceive of an intentional, directed, and beneficial planetary-scale *terraforming*.

There are *three* terraformings that are clarified by how this new sensory organ represents its own construction to its creators. The first is the one that is made clear by the historical earth sciences, extending back as far as human migration, which wiped out megafauna, imposed agricultural striation of the landscape, and intensified industrial development. This is the terraforming we were born into and which is the precondition of our appearance. There is also the terraforming that will continue regardless of what humans do at this point, for the momentum of compositional intelligence cannot be arrested but can only be calibrated, modulated, and focused. That is the third terraforming.

The so-called Anthropocene "event"[1] was occurring for at least centuries if not millennia before computational climate science made it perceivable. Now what? No longer innocently and unknowingly transforming the world, we urbanizing, agricultural fire apes can now interpret the tracks we made on the ground. Anthropocene *agency precedes subjectivity*.

The third terraforming is the project of that subjectivity: partial, creative, wobbly, self-destructive, visionary, violent, and literally radiant, magnificent, and terrifying. The question is not if intelligence will terraform the planet in the present and future. Rather, we must ask, first, how will it do so, and, second, will

complex intelligence ultimately survive its own expression.

The Planetary

But let's back up a moment. The "planetary" in "planetary sapience" refers to many things at once. It is the physical precondition from which complex intelligence evolves and a living astronomical and geochemical body measured and revealed by computational earth sciences. It is also a philosophical problem, a "category" in Dipesh Chakrabarty's framing.[2]

The concept of "the Planetary" suggests both the very small and the very large. It implicates deep time and the abyss of outer space as the preconditions of our thoughts. It names the depth of biological and inorganic interrelations. It offers an understanding of the Earth not so much as a world in the phenomenological sense as a planet in the geologic and biogeochemical sense. The Planetary gave birth to sapience and now represents that sapience's greatest challenge. The Planetary did not appear suddenly as a "world picture," as Martin Heidegger had it,[3] but rather as the habitat of a particular species that was able to construct an exterior image that, finally, could uncover a planetary condition from which that species and its world emerged. It was there all along, but we've only just become able to grasp it. It's revealing doesn't "destroy the Earth," as Heidegger said,[4] but gives it finally a mirror with which to understand itself.

For contemporary philosophy, the provocative concept of the Planetary (and its corollary, "planetarity") has been put forward as an alternative to "the global," an expired notion that is static, flattened, and Eurocentric. At the end of the last century, after a

few decades of hibernation, the term *planetarity* reemerged in the work of the literary theorist Gayatri Chakravorty Spivak.[5] I extend and depart from Spivak's connotation to focus on a planetarity that is, first, revealed as the precondition of any philosophy and, second, the name of the project before us as we contemplate how to preserve, curate, and extend complex life. Whereas Spivak[6] and the veins that extend from her influential work take great care to differentiate the quantitative from the qualitative and to affirm the supremacy of the latter over the former, to my sense, qualitative insights about what, where, when, and who are thinking creatures are abstractions drawn *from* quantitative insights, not in moral opposition to them.

So: There is an *astronomical* planetarity and a *political-philosophical* planetarity, and while they are different, they should both inspire correspondence and mutual reinforcement. There is no workable political-philosophical planetarity that does not define itself through the disclosures of the astronomic understanding of what a planet is, where it goes, and how a sapient species emerges from it. Together they dissolve the humanist figure of the meaning-making subject bound only by its self-generated space of immanent signifiers and undermine political superstitions of place, ground, and horizon that plague our modernities.

These modes of "planetarity" comprise a cause and an effect of planetary sapience. The planet evolves entities capable of building technologies through which they partially grasp their own origins and the planet's as well. From the perspective of the computational crust that appears at the end of the Blue Marble movie, Vladimir Vernadsky's noosphere[7] as a geological fact

wins out over Pierre Teilhard de Chardin's theologically dualist noosphere.[8] The conclusion is that planets can think through the life they evolve, not that thinking landed on our planet.

The Revelation Of The Planetary

This revelation only frames the normative challenges posed for the terraforming subjectivity; it does not begin to answer them. The questions implicitly posed by the Blue Marble movie, but which it cannot answer on its own, are: what is planetary-scale computation *for*, and as something that literally evolves from its host planet, what *should* it do? What contribution to a viable planetarity can it make?

A preliminary answer bends back toward the question of the purpose of planetary computation: One purpose is to make the contemporary notion of the Planetary possible. It doesn't cause the Planetary as a condition to come into being, but in concert with scientific and philosophical inquiries, it makes it possible for the primary sapient species within that circumstance to grasp the terms of its own emergence. It shows intelligence *where* and *how* it came to be.

Planetary-scale computation is an example of what may be called, following the nomenclature of the great Polish novelist Stanisław Lem, an epistemological technology.[9] The most important social impact of some technologies is not in what they allow people to do, but in what they reveal about how the world works. But revealing how the world works can lead to trouble. While anxiety about technology is expressed in accounts of its pernicious effects, that unease is sometimes rooted in what

technology uncovers that was there all along. Microscopes did not conjure microbes into being, but once we knew they existed, we could never see surfaces the same way again.

Such unrequested demystifications are disturbing, especially when they seem to remove us humans from a place of presumed privilege. Even as such technologies reorganize personal and global economies, their deeper philosophical implication concerns how they introduce a Copernican trauma, unsettling our earlier understanding of the cosmos. Such traumas are not always recognized for their significance, including in Copernicus' time, and their consequences usually take generations to reverberate.

The revelation of the Planetary — so different from the "international," the "global," or the "world" — comes into view via the location of human culture as an emergent phenomenon connected to an ancient and deep biogeochemical flux. Planetary-scale computation may have first emerged mostly from the context of what is often inaccurately called "Western" science by both its defenders and detractors, but its ultimate implications upend and disrupt the conceits of such historical distinctions as much as Darwinian biology eroded longstanding institutions' biopolitical authority. Even as many of those institutions made biological sciences possible, this dynamic by which an initiative collapses under the implications of its own technological discoveries may be more norm than exception. We may well worry if complex intelligence itself will similarly collapse as a result of its own success.

Put differently, the instrumentality of planetary computation is derived from its epistemological significance, and the inverse is equally true: What it does and must do is derived from what it

discloses.

The Terraforming

The continually evolving technologies of a planetary society are *processes* over which we have some limited agency and which have limited agency over us in turn. More precisely, globe-spanning anthropogenic agency (as in climate change) works through planetary technologies, but what real compositional steerage we may have over the long-term evolution of the myriad technologies that afford that agency is not as clear.

Nevertheless, we may have more agency over them than we have control over our own long-term goals, and as the real effects of anthropogenic agency on the planet increase, these misalignments may be increasingly unsustainable. What then of planetary computation? In its current commercial form, one primary purpose of planetary-scale computation is to measure and model individual people in order to predict their next impulses. But a more aspirational goal would be to contribute to the comprehension, *composition*, and enforcement of a shared future that is richer, more diverse, and viable.

How so? Paradoxical as it may seem, given the precariousness of the present moment, instead of reviving ideas of nature, we must reclaim the *artificial* — not fake, but *designed*. For this, human-machine intelligence and urban-scale automation become part of an expanded landscape of life, information, and labor. They are part of a living ecology, not a substitute for one. No more pretending that the terraforming isn't happening and no more pretending that it can simply be refused by calling us "down to

Earth." Put more directly: *The response to anthropogenic climate change must be at least equally anthropogenic.*

The critical apparatuses of such a composition are multiple, modular, layered, complex, and already underway. They are not hypothetical, promethean, or futuristic. They are the rules of the game. They include *automation*, (understood as an ecological principle of inter-entanglement more than a reductive autonomy); *geoengineering* (defined not as a specific portfolio of controversial techniques but rather in terms of *climate-scale effects*); the rotation of *planetary computation* away from individual users and toward processes more relevant for long-term ecological viability; the *deliberate self-design* of sapient species toward *variation*, including reproductive technologies, universal medical services, and synthetic gene therapies; the cultivation of artificial, mathematical, linguistic, and robotic intelligences with which *general sapience deliberately evolves*; the deployment of experimental expertise with biotechnologies, through which *living matter composes living matter*; the intensification of urban habitats and technologies as media for the *general provision* of universal and niche services; the projective *migration* outside the Kármán line, the boundary between Earth's atmosphere and outer space, from where the existing and potential terrestrial planetarity comes into focus; and, finally, the aggregation of creative *governing intelligences* capable of architecting such mobilizations.

These are the building blocks of an intentional *terraforming* — not of another planet, but of our own. It is not the project of making other planets suitable for Earth-like life but of making Earth suitable for Earth-like life. To do this doesn't mean freezing life as it currently is or relying solely on restoration, but by

linking (somehow) the acceleration of complex intelligence to the stabilization of its planetary foundations.

This terraforming embraces the artificial, but what is that? Isn't everything — from the most organic to the most plastic — ultimately part of the chemosphere, and is this just another nature versus culture dichotomy? First, just as evolution has selected for intelligence, it has also selected for artificialization: the capacity for intelligence to remake the world in its image. Both driving and being driven by the parallel evolution of technology, artificialization allows energy-ravenous intelligence to feed itself. In the relatively short timescale of a few million years, artificialization has been highly adaptive. But if it is driven solely by the blind imperatives of evolutionary advantage, does artificialization eventually reach a terminal point after which it so decisively undermines the planetary conditions of its own emergence that it is doomed to burn itself out? This is the question that planetary sapience seeks to answer through the project called the Terraforming.

Terraforming is a deliberate, practical, political, and programmatic project to conceive and compose a viable planetarity based on the secular disenchantment of Earth through the ongoing artificialization of intelligence and the emergence of a general sapience that conjoins human and nonhuman cognition. It names a future condition realized by the nonlinear and playful rationalization of ecosystems toward diversification as order. As intelligence expands beyond the biosphere and deep into the lithosphere, that condition is animated by a liberation of synthetic intelligence in the wild.

Politics And Governance

The questions of planetary agency naturally suggest planetary politics and planetary governance, which are increasingly dissimilar topics. The Political (with a Schmittian capital P) refers to the agonistic contestation of the symbolic media of power, whereas governance (small g) refers to the process by which a complex system is able to sense itself, model itself in the present and future, and to recursively act back upon itself accordingly.[10] The former is driven by narratives and the latter is driven by cybernetics. One of the unintended effects of planetary computation has been an unraveling of the tether that links the Political to governance as the memetic clown show of ceremonial monarchies clashes with that of ceremonial parliaments in increasingly shrill and entropic echo chambers. Meanwhile, the mundane work of opening and closing doors and moving things from here to there is absorbed by software and hardware platforms. The Political is now free to float in the clouds of its own semiotic exhaust. Much of the dramaturgy of contemporary political life can be explained by this decoupling of politics from governance.

The question of agency then is not *who* should be the sovereign first-mover of the Terraforming but *what* should be the architecture of relays that cascade from metabolic mayhem to planetary sapience. That agency is not necessarily a subjectivity, nor must it be a perfect subdivision of the human population versus a different subdivision. To imagine planetary governance as reducible to a particular arrangement of people and ideal representational processes is to reduce it to an extra-large-scale caricature of anthropomorphic national politics, a failed attempt to tuck all reality into the space of legible tropes. That planetary systems are and will be governed — sensed, modeled, effected

— not through recognizable institutions of government but through technologies with evolutionary arcs irreducible to mere "techne" is not some futuristic, science-fiction scenario of what might happen. This is what already exists, though in unevenly distributed form; this is the quality of the Terraforming into which all those alive were born.

The *direction* of the evolution of synthetic intelligence is bound by path dependencies and vagaries of contingency, not unlike the evolution of biological intelligence. Complex human social intelligence emerges from the manifold of semiotic communication and thus relies on symbolic economies to both represent and govern its world, which it often conflates as if one were the other. As intelligence expands its scope of material substrate from carbon to silicon, it is possible that familiar modes of anthropomorphic governance that present themselves as the full range of options are both too simplistic and too convoluted. The more they try to subsume governance into representation the more of the real world escapes its view.

It can't be known in advance whether this escape will guarantee the future of intelligence or annihilate it. The Terraforming project aims to ensure a sapient correspondence between biosphere and technosphere: not Earth as homeostatic mechanism in the macro-logical sense, but actively chaotic in more measured ways that will support the ongoing evolution of intelligence rather than foreclose it.

An Ecological Theory Of Automation

What then is *automation*? The dissolution of decision into form? Intelligence does not live in a petri dish or laboratory or inside a single skull; it lives out in the open; it lives in and as our cities. A city is not just architecture plus dwellers — it is an artificial environment par excellence. The city itself is perhaps the longest continuous process that humans have created. Introducing synthetic computational intelligence into urban systems augments existing forms of embedded sensing and intelligence, and in so doing produces novel qualities.

I am reminded of Gakutensoku, a massive robot built in Osaka in the 1920s by Makoto Nishimura. Nishimura was appalled by the mechanistic humanoid robots in Karel Čapek's play "Rossum's Universal Robots," which introduced the term "robot." So he set out to make an automaton that manifested what he saw as the most noble and fragile aspects of human culture, complete with intricate facial expressions and the ability to transcribe poetry.

When I visited a factory in Shenzhen that makes cases for Android phones and employs many robots and people working side by side, I was struck by an unexpected feeling, a kind of serenity. The mood was calm, not frantic. Some things were moving quickly but quietly, while other things were quite still, as if waiting their turn. It did not feel like a "factory" in the Charlie Chaplin sense; it felt much more like a garden of machines in the Richard Brautigan sense.

I remarked that I would very much like to spend time in a cafe like this, that it would make for a lovely kind of public

gathering spot. As I spoke, I realized that this was no joke. The present locus of automation will inevitably spill out into the city, and as it does, we must be aware of the deceptively simple fact that automation creates a particular kind of ambiance. It is more than form following function; it is a functionalism becoming a delicate formation. Or at least it can be.

To avoid the miserable future in which urban computational automation is trained foremost on the optimization of the most arbitrary and banal aspects of human spatial logistics (parking, security, vending, etc.), a different understanding of automation is needed. (Stephanie Sherman writes about this in her contribution to this volume.) First, automation is not primarily about autonomy. Second, globalization didn't cause automation — automation caused globalization. In the densest city or jungle, causality and determination are everywhere, but automation's processes and techniques are themselves both overdetermined and indeterminate. If we were to imagine these as dominos, their arrangement would extend deep into the heart of things, and the agency of their cascade would go beyond the intention of any first tipping.

These systems are choreographed, but they also evolve with each iteration, learning as they go and shaping and being shaped by the worlds in which they are situated. As urban infrastructure they remember and encode specific decisions that can be repeated over and over. The appearance of autonomy — of a machine, process, person — is an illusion. These systems' causal relations upon relations have been set by previous stages and positions, and so the whole automated set-piece is itself automated. Our synthetic automation makes use of existing footprints and

previous patterns of urbanization and also forces others that generate quite different geographies. New niches emerge, while others go dark.

Synthetic Intelligence

The role of synthetic intelligence in planetary sapience is essential but different than it is generally assumed. More important than planetary sapience simply deploying computation to do things is how planetary sapience is made possible by the ongoing artificialization of intelligence, a process that began many millennia ago. The intensification of that process through computation raises the stakes considerably.

The evolution of machine intelligence today is warped by various ideologies of "artificial intelligence," which are in turn hobbled by misconceptions about what is and is not artificial and what is and is not intelligent. Foremost among these misconceptions is the presumption that machine intelligence must be recognizably "human-like" to qualify as intelligence. Multiple anthropomorphic biases and presumptions have left us with inadequate allegories for machine intelligences' remarkable accomplishments. Some of these look nothing like human thought — though some do, like the very large natural language processing models. What appears anthropomorphic may not be and what appears alien may be more human than we think.

Obviously, the term "artificial" implies something false and deceptive, but for planetary sapience this is a misleading framing of machine intelligence. Revisiting the distinction between the "artificial" and the "synthetic" posed by the economist Herbert

Simon[11] half a century ago, the "artificial" refers to something that merely resembles an original (such as a cheap plastic "diamond"), whereas the "synthetic" is a genuine and meaningful version of something that was deliberately created (such as a laboratory-grown diamond identical to a "natural" one at the molecular level). Thus, artificial intelligence merely *seems* smart, but synthetic intelligence really *is*. We should be pursuing synthetic intelligence, not artificial intelligence.

There is another connotation of synthetic intelligence that is perhaps even more important: the synthesis of human and machine intelligence in pursuit of insights or creativity that would be impossible for either on its own. A now-famous example of this synthesis occurred in the Go match between Lee Sedol and AlphaGo in 2016. The AI's move 37 in the second game was one that Go experts have said no human could have imagined. But in the fourth game, Sedol's move 78 was equally unexpected and creative. If the first move proved that AlphaGo was in some way not just "smart" in a narrow sense, but also capable of creating novelty, the second move proved that in response, a human saw the game differently and could produce a brilliant move that never would have happened otherwise. AI and the human, interfaced by the closed world of the game board, co-discovered a new way of comprehending a 4,000-year-old game. This is a synthesis of unlike forms of intelligences, a glimpse of what a general sapience may look like.

The planetarity of computation forms, which I have elsewhere called an "accidental megastructure,"[12] are composed of overlapping functional layers. Quite literally, it is a stack extending down to the mines of central Africa through subterranean data

centers and transoceanic cables to interlaced urban networks up to the glowing glass screens through which we view it, and it views us. Planetary-scale computation is not virtual. It is a kind of terraforming of its host planet. That system has been constructed over the past half century based on ideas both new and old, but it is only a precursor. Artificial intelligence implies fundamentally different architectures, from top to bottom, from chip design and data center design to the geopolitical tensions between great power hemispheres competing to build the foundational models that will form the basis of both economic and military capability.

Measuring the weight of planetary-scale computation requires a sober reckoning with the physical costs of its sprawling infrastructures, which includes differentiating essential purpose from the trivial and ultimately pondering the price of intelligence itself. The intelligent exoskeleton that the planet has developed has come at a price. In the context that really matters most, the cultivation of synthetic intelligences capable of collaboration with our own most virtuous ambitious and virtuoso expressions is precious. The syntheses they portend are available only if we pursue them while remaining mindful of their high costs.

The evolution of biological intelligence also came with extreme costs. Someone to refusing or accepting the costs of synthetic intelligence must also consider the price of the natural intelligence that purports to hold the scales. It was not only symbiotic social cooperation but also mountains of gore that led our common ancestors from Olduvai Gorge to Göbekli Tepe and the literate cultures of Mesopotamia, East Asia, and Mesoamerica. The deepest values are at stake. Is the very long-term evolution of "intelligence" — human, animal, machine, hybrid — a fundamental

purpose of the organization and complexification of life itself? If so, now that intelligence begins to migrate to the inorganic substrate of silicon, what planetarities does this portend?

The Situation Of Intelligence

The Terraforming embraces the artificial, but what is that? Is not everything -– from the most organic to the most plastic -– ultimately part of the chemosphere? Is this embrace just another nature versus culture dichotomy? It is not. Just as evolution has selected for intelligence, it has selected for *artificialization*, the capacity for intelligence to remake the world in its image. Both driving and being driven by the parallel evolution of technology, artificialization allows energy-hungry intelligence to feed itself and to replicate. In the relatively short time of a few million years, artificialization has become highly adaptive. But if it is driven solely by the blind imperatives of immediate evolutionary advantage, does artificialization eventually undermine the planetary conditions of its own emergence and doom itself to burn out?

The most critical relation between the planetarity that has been revealed and the Planetary that must be composed depends on the position of intelligence from which any such intervention might take place, and how that position might incorporate the situation of its agency. This is far more difficult than some would have us believe. It is to be born into unpayable debt.

The decisive paradox for general sapience is the dual recognition that, first, its existence is extremely rare and extremely fragile, vulnerable to numerous threats of extinction in the near

and long term, and, second, the ecological consequences of its own historical emergence have been the chief drivers of the conditions that establish this very same precarity. The approach to these questions cannot avoid the correspondence between honing our own sapience through machinations of war and strategic violence and the emergence of machine intelligence that depends on the provisions of material extraction, military applications, and their ecological and social devastations.

Both modes of intelligence are also modes of planetarity. Both are positions from which reason exercises its agency, for better or worse. Both are also tied to what we all may recognize as our most inspired aspirations. But if planetary intelligence is to survive the consequences of its own appearance, in the short term and in the long term, it must reform its trajectory or risk extinction and disappearance.

This historical moment seems long but may be fleeting. It is defined by a paradoxical challenge. *How can the ongoing emergence of planetary intelligence comprehend its own evolution and the astronomical preciousness of sapience and simultaneously recognize itself in the reflection of the violence from which it emerged and against which it struggles to survive?* It is possible that our privilege of retroactive hindsight will decide that, for some final register, this history was a worthwhile and even perhaps necessary condition for the ultimate emergence of planetary intelligence. Even if so, its development and survival depend on a decisive graduation from primordial habits.

What future would make the past worth it? Perhaps the future of planetary intelligence is now as existentially entwined with a radically different career for composition, foresight, and

order-giving as its advent was from the cascading centuries of pilotless destruction. Taking this new existential condition seriously demands a radically different sort of philosophy.

Notes

1 As opposed to a geologic epoch, defining the Anthropocene as an "event" encapsulates broader human-environment interaction. See Leigh Phillips, "Moving on From the Anthropocene," *The Breakthrough Journal* (August 9, 2024). https://thebreakthroughjournal.substack.com/p/moving-on-from-the-anthropocene.

2 "The Planet: An Emergent Humanist Category," *Critical Inquiry* 46, no. 1 (autumn, 2019): 1-31.

3 "The Age of the World Picture," *Off the Beaten Track*, trans. Julian Young and Kenneth Haynes (Cambridge, MA: Cambridge University Press, 2002). http://tems.umn.edu/pdf/Heidegger-AgeWorldPicture.pdf.

4 "The Question Concerning Technology," as it appears in *The Question Concerning Technology and Other Essays*, trans. William Lovitt (New York: Harper & Row, 1977).

5 *Death of a Discipline* (New York: Columbia University Press, 2003).

6 Stephen Moore and Mayra Rivera, eds. *Planetary Loves: Spivak, Postcoloniality, and Theology* (New York: Fordham University Press, 2010).

7 *Scientific Thought as a Planetary Phenomenon* (Moscow: Nongovernmental Ecological V.I. Vernadsky Foundation, 1997).

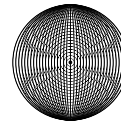
8 *The Phenomenon of Man* (New York: Harper & Row, 1965).

9 *Summa technologiae*, trans. Joanna Zylinska (Minneapolis, MN: University of Minnesota Press, 2013).

10 Carl Schmitt, *Political Theology: Four Chapters on the Concept of Sovereignty* (Chicago: University of Chicago Press, 2010).

11 *The Sciences of the Artificial* (Cambridge, MA: MIT Press, 1968).

12 See *The Stack: On Software and Sovereignty* (Cambridge, MA: MIT Press, 2015).



Extraplanetary Conditions

CLAIRE ISABEL WEBB

Technologies Of Extraplanetary Perception

"To see the earth as it truly is, small and blue and beautiful in that eternal silence where it floats is to see ourselves as riders on the earth together, brothers on that bright loveliness in the eternal cold — brothers who know now they are truly brothers."[1] So reflected the poet Archibald MacLeish in the *New York Times* as images of Earth streamed from outer space on Christmas Eve, 1968. That night, US astronaut William Anders used a 250-millimeter Hasselblad camera to capture photographs of the Earth as the Apollo 8 spacecraft he was aboard rounded the Moon. In "Earthrise," as the iconic image came to be known, the

surface of the Moon's gray, bare surface is in the foreground, weary and battered from violent collisions over eons. In the distance, only partially illuminated by the Sun, Earth seems to proclaim a delicate yet stubborn aliveness. Our planet's palette of white, green, and blue glows against the black, ahuman abyss of space.

From MacLeish to Carl Sagan to Pope Paul VI, Earthlings were moved. The Apollo 8 mission marked the first time anyone had gazed from space at the Moon first and the Earth beyond. The extraterrestrial perspective of our planet stoked the collective realization that there is only one Earth, and it is precious for all humans who precede us and must be kept safe for those who might come after. The image dominated the *Times*'s front page, but the peripheral headlines were filled with death: a Christmas Day ceasefire in Vietnam had collapsed; soldiers had returned from harrowing captivity in North Korea; a commercial plane crashed in Pennsylvania, killing 20 of the people aboard. Here was a juxtaposition of global conflict with images that simultaneously broadened the human condition and pointed to the singleness of the human home. "We came all this way to explore the Moon," Anders remarked, "and the most important thing is that we discovered the Earth."

Humans have used a variety of modern technologies to "discover Earth" more deeply from extraterrestrial vantage points, such as spacecrafts, satellites, supercomputers, and airborne sensors. From Anders's Hasselblad camera to the global network of distributed computational intelligences now in low-Earth orbit, we use technoscientific apparatuses to form an ever-growing knowledge base that reveals the material conditions of Earth.

The assemblages of instruments, data collection

mechanisms, and computational tools that progressively disclose the complex materialities of our planet are *technologies of perception* — they transform how we see ourselves in the world and in the universe. By manifesting more and more granular empirical data about Earth systems, we cultivate novel orientations to our home planet scaling from the microbial to the celestial.[2]

In the introduction to this volume, Nils Gilman names this orientation the "condition of planetarity." Earth's planetary conditions today are the results of ancient processes both extraterrestrial and local, but humans' *condition* of planetarity is a philosophical attitude about our bounded position on and within Earth. This condition, Gilman asserts, is the "inescapability of our embeddedness in an Earth-spanning biogeochemical system." We humans would not be here today but for myriad felicitous astrophysical phenomena that transformed Earth over the ages and allowed for humans to evolve. Charting this history makes visible how our capacity to flourish here on Earth is diminishing, which, in turn, (hopefully) deepens our sense of urgency to care for this planet.[3]

Unearthing Habitability

The fulcrum of humans' condition of planetarity is *habitability*.[4] The *Oxford English Dictionary* defines habitability as the condition of being "suitable for habitation or as a human abode." Its root word is "οἰκουμένη" or "ecumene," the ancient Greek term for the known world that stretched from what is now called Afghanistan in the east to Iberia in the west and from Upper Egypt in the south to Scythia in the north. Since ancient times,

then, habitability has been defined by what is known by humans and fit for humans to live in, with the Earth as the ineluctable point of reference from which all alternatives for habitability have been drawn.[5]

But a wider frame of view, one borne out through technologies trained star-ward (the James Webb Space Telescope in particular), is destabilizing anthropocentric and Earth-specific categories of life and habitability. As the properties of alien atmospheres and surfaces are coming into clearer focus, we must resituate those categories in a cosmological register. What environments are habitable, and for whom? Must life spark from a carbon- and water-based world? What might it mean to be human but not of Earth?

Since the Space Age that defined geopolitics in the post-World War II world, technologies of extraplanetary perception have been reshaping self-conceptions of the humans who built and evolved along with them. Unmanned space missions since the 1970s that imaged Mars, Venus, and the moons of Jupiter and Saturn have suggested that enigmatic places in our very own solar system might be suitable for habitation by extraterrestrial life forms. And scientists' detection of exoplanets in the 1990s — faraway worlds that could contain alien ecosystems — extends the material conditions of planetary life to the extraterrestrial. The pursuit to identify, clarify, and demarcate life forms and their habitats at the edge of knowledge foments a plurality of scientific and conceptual approaches about how we have heretofore regarded life on Earth. The conditions of extraplanetarities — ones that include humans in outer space but also reach toward yet undisclosed forms of life — present a wholly new philosophical

category.

The deepening knowledge about Earth and humans' forces within it cast our home planet in relation to speculative worlds beyond. This relationship is unearthing new ways to be human. As we discover astrophysical resonances and dissimilarities between our planet and other worlds, we emplace ourselves within Earth's web of life — and reorder our self-conception as beings in the cosmos.

While there is but one (Earthly) condition of planetarity, there are imponderably many potential conditions of (non-Earthly) extraplanetarities. Each shade of possible worlds — the variegated compositions of their atmospheres, geologies, and other features — is a contrast to our local world as we know it, revising conceptions of habitability through revelations about (extra)planetary fragility and (extra)planetary possibilities of life as it might be otherwise.

The speculatively imagined thus encroaches on the empirically enclosed. The extraterrestrial jostles echt-Earth as the sole site of possible life. Exoplanets and moons habitable for radically alien life forms break free from anthropocentric constraints of flourishing on Earth specifically, a chiasmus that is transmogrifying not just what it *means* but indeed what it *is* to be human.

Fragile Natures

During the Space Age, humans were afforded an extraterrestrial perspective of Earth for the first time, revealing several precarious relationships between human, Earthly, and

hypothetical life. For one thing, we learned how fragile our bodies in outer space are, how dependent we are on Earth's life systems — ones that, ironically, we have been making brittle. As scientists gained the scientific and technological abilities to explore the material conditions of other planets and moons, they learned of extraplanetarities that might hold the key to the origin of Earth life and yet would be in danger of being trampled through hapless human intervention. The conditions of extraplanetarities — speculative fragilities on alien worlds, speculative knowledge about nonhuman life — expanded humans' self-conception to a cosmic register.

As the historian Leah Aronowsky has written, the vessel that bore the Apollo astronauts to the moon and back in 1969 was a testament to nonpareil American ingenuity and technoscientific dominance, not just over the globe but over nature. She described it as the ultimate representation of man's aspiration to conquer nature. "In forming a self-sustaining world in the face of a lethal extra-spacecraft environment, the space cabin was propelled by its own mythos of sorts," she wrote. "The ark, the biblical structure, sealed and closed to the outside diluvial world, that maintained planetary life when no other place could."[6] It redefined where and how human life *can* be.

Yet the harsh physical reality of manned missions to space highlighted just how fragile the borders were between the interior and the exterior, the planetary and the extraplanetary. The maintenance of a life-support system in space depended on constant vigilance by astronauts and engineers back on land; continuously threatened by emergencies, flukes, and biotechnical porosity, early space flights were a vivid illustration of how hard it

is to sustain human life unmoored from the complex cosmological object we know best: Earth.

It is not a coincidence that the modern environmental movement arrived on the heels of the Space Age. The technologies of planetary perception developed to enable space travel would, of course, look back onto Earth itself. Those technologies illuminated how vulnerable Earth's life is to human forces, and researchers foresaw an alarming future, calling into question the viability of humans' continuing habitation of Earth.[7]

That bruited future is now unfolding. We know now with great certainty that humans have been radically altering Earth's biogeochemical conditions, especially since 1945 with the advent of what environmental historians J. R. McNeil and Peter Engelke dubbed the "Great Acceleration."[8] The extraction and burning of fossil fuels has both allowed humans to proliferate in otherwise inhospitable environs and wreak havoc on delicate ecosystems. The technosphere continues to evolve with human inventions and technological agents, changing our planet's material conditions, while also enabling a new self-awareness of these changing conditions. These entwined cultural and scientific forces helped to shape our philosophical understanding of the condition of planetarity. Perceived from outer space, Earth became a planetary object that was both precious and in peril.

The nascent field of space biology in the late 1950s focused on the dangers of contamination.[9] Exobiologists, as they came to call themselves, developed extraplanetary technologies that could potentially harm the other worlds on our very doorstep. Nobel Prize-winning microbiologist Joshua Lederberg implored NASA to develop protocols to prevent such biological contamination.

Speculative microbial exchange posed a threat to astronauts aboard manned missions and to planetary conditions; imagined as closed cybernetic systems, both the space cabin and the Earth were at risk from any invading substance that could disrupt the delicate homeostasis of life. At Lederberg's urging, NASA began exploring sterilization and quarantine procedures for the planned lunar missions with such detail that they included possible microorganisms lurking in the threads of screws and in the plastic that protected electronics.[10]

Lederberg and other exobiologists also asserted that contamination from returning spacecrafts posed a threat to planetary health: Unwitting astronauts could return with alien entities that could have unknown consequences for terrestrial life. "An organism, innocuous when in the hostile environment of a planet might, when transported to the comparatively lush conditions of the earth, overgrow terrestrial life forms or alter the physical or chemical characteristics of the biosphere," warned the Space Science Board that advised NASA in a 1964 report.[11] Given a chance, hardy extraterrestrial pathogens, so the exobiologists feared, could exploit the moist, fecund, highly habitable Earth.

Conversely, spaceship-borne terrestrial microbes could contaminate pure extraplanetary surfaces. Any adulteration might rob scientists of the chance to detect ancient or extant life elsewhere in the universe. "The danger of contamination of these planets [Mars and Venus] is mainly biological since there is a reasonable probability that the conditions on Mars are such that some terrestrial organisms might grow," Lederberg wrote in *Science* in 1958. "Water, nitrogen, carbon oxides and photosynthesis are all available."[12] An inadvertent terraforming

might disrupt fragile hypothetical biologies and destroy a once-in-solar system opportunity to reveal clues about the origins of life on Earth.

Transforming as a technoscientific and philosophical object in the Space Age, the Earth became framed in relation to other possible sites of life. Humans had to confront their historical and present roles in stewarding the planet — cultivating its habitability for continuing human existence and vigilantly guarding against alien incursion. Technologies of extraplanetary perception made starkly visible just how human, Earth, and extraterrestrial life were materially interrelated and therefore sensitive to each other. As such, the continuous revision of the category of habitability prompted by technologies of extraplanetary perception — how we humans relate to Earth as an endangered paragon — were central to the unveiling of our condition of planetarity.

Life, Otherwise

For decades, astronomers interested in the possibilities of life beyond Earth relied on a definition of habitability circumscribed by the conditions of life as known here on Earth. In particular, the "habitable zone" demarcated the distance from a star in which a planet would be able to maintain liquid water at its surface — just like Earth. Scientists described this extraterrestrial condition as the "Goldilocks Zone": For surface water, Mars was too cold, Venus too hot, Earth just right.

Recently, though, explorations of worlds farther afield have focused not only on direct observations of possible life, but also on simulating the conditions under which life in various forms

may emerge or flourish. These efforts have prompted radical reconsiderations of what counts as habitable, and for whom. (Venus's climate history — it was not always such a hothouse — is part of what has prompted fears about "runaway climate change" here on Earth.)[13] But learning about the cosmos's capacities to evolve non-Earth-like planets has evected habitability toward speculative, nonhuman life forms — where and how alien life forms *could* be.

NASA is now launching missions to explore nearby moons in our solar system, hoping to detect evidence of conditions that could spark and sustain deeply unusual extraterrestrial life. In the fall of 2024, the Europa Clipper will loop around its eponymous Jovian moon, which beneath its frozen surface almost certainly harbors an ocean of liquid H2O. In a later mission, NASA hopes to send a probe to drill into that thick icy crust, where hydrothermal vents in a subsurface ocean warmed by Jupiter's gravitational push-and-pull could stimulate microorganisms that are akin to Earth's "extremophiles," the microbes that percolate in environments once thought to be inhospitable to life, such as alkaline lakes, acidic water from coal mine run-off, and the Marianas Trench, where pressure is 1,000 times what it is at Earth's surface. Earth's known planetary conditions thus resonate with Europa's speculative ones, affording scientists the ability to reconfigure the extreme as suitable.

The result of the expansion of the concept of habitability is that Earth's planetary conditions no longer define the blueprint for life. Habitable planets need not be temperate, rely on water, or even depend on a carbon-based ecosystem. These realizations in turn pushed scientists to develop new ideas about how life might

have emerged, and might yet evolve, here on our own planet.

NASA's Dragonfly, a robot-cum-drone, is planned to arrive at Saturn's moon Titan in the mid-2030s. A critical component of the Dragonfly is its mass spectrometer, a sensor technology that discerns an object's composition of atoms and molecules. Dragonfly will clamber and fly over Titan's surface, collecting samples for chemical analysis, searching for prebiotic molecules and other tantalizing possibilities of alien life.

In some ways, Titan is Earth's analog. For instance, Titan's volcanic activity could have churned atmospheric and interior material to spark life — just as we think it might have on Earth. Ontario Lacus, near Titan's south pole, is eerily similar to the Etosha Pan, a salt flat in Namibia, suggesting related geochemical formation.[14] Yet, in other ways, Titan has sharply un-Earth-like planetary conditions. Its lakes, rivers, and even oceans are mostly methane, not salt water. In 2020, astrobiologists detected in its dense, hazy, orangey atmosphere a molecule called cyclopropenylidene (C3H2). Though it is simple and carbon-based, C3H2 is unique in the solar system, a possible precursor for life forms both familiar and extremely strange. If so, the identification of prebiotic chemistries on Titan could reshape the story of how life originated on Earth.

Scientists must perform a double epistemic act as they search for other worlds. They both welcome familiarity and quest for alienness. Through storytelling, metaphors, computational modeling, and physical analogs, astrobiologists "make planets places" to relate familiar features to alien ones.[15] In this way, scientists liken Titan's hydrocarbon dunes to coffee grounds and note its Ligeia Lake could contain three of our Lake Michigans.

Researchers look to Lake Vostokin in Antarctica to predict characteristics of Europa's ice layer and the salty waters of Lake Tirez in Spain to make predictions about the alien ocean below. "A place-based orientation, rather than passively gazing at the globe from the outside, allows for an imagination of being on/within/alongside, of experiencing, the planet," writes the anthropologist Lisa Messeri.[16] Material witnessing on Earth thus generates more robust speculations of the alien world and concretizes the possibilities of going there.

Other epistemic objects even more distant from terrestrial conditions of life hint at undisclosed extraplanetarities on the horizon. For example, hypothetical Hycean worlds — a portmanteau of "hydrogen" and "ocean" — are filled with liquid water oceans and blanketed by a thick hydrogen atmosphere. In theory, such planets could be ideal for aquatic life but could suffer a greenhouse effect as Venus did. Hycean worlds create tendrils of both incongruity and likeness to Earth, but they elude tight analogies, thus straining our ability to place-make. So far, astronomers using the James Webb Space Telescope have detected many candidate Hycean worlds, but none is yet confirmed.[17]

Supporting such exploratory work are ever-more sophisticated computer simulations that serve as epistemic tools for fleshing out extraplanetary conditions. Terrestrial simulations of our own home planet, such as Earth System Models and Global Climate Models (GCMs), integrate Earth's global atmospheric and oceanic circulations, glacial activity, and biological cycles to forecast changes in the climate.[18] By stretching components of GCMs on Earth to alien planets, scientists are building models to anticipate how life could be otherwise in the universe. Given

initial starting conditions — for instance, an exosolar system where the star is a red dwarf, much cooler and dimmer than our Sun — simulations offer tantalizing pathways for imagining unfamiliar but viable life. On the two water worlds known as Kepler-138 c and Kepler-138 d orbiting a red dwarf star 218 light-years away, one can imagine black, aqueous plants that evolved to absorb the star's dim, near-infrared light basking in the warm waters of an alien planet.[19]

These projects could illuminate life's origins and conditions for continuity on Earth and beyond. Suspended in the space of the speculative, teetering between characteristics of aliveness and barrenness, simulated worlds beckon us to consider all the possible levers — light, gravity, atmosphere, pressure, material, time — that could orchestrate the emergence of life, not just on alien planets, but on Earth.

Computer-based simulations can unfurl billions of evolutionary pathways of life as they might have been, as they are, as they could be. Simulating diverse and mind-bending extraplanetary conditions can rewind and reset initial conditions for life. Scientists use them to speed up the petri dish of the universe, producing worlds with chemistries and conditions that could support forms of life as yet undreamt.[20] In so doing, they "rediscover Earth."

One of the most influential (and notorious) planetary simulations was pointedly analog. In the Biosphere 2 experiments during the 1990s, four women and four men took up residence in a giant greenhouse in Oracle, Arizona, with the intention of demonstrating an autopoietic system of Earthly life that could presage and model anticipated Martian settlements.

The dwelling was Earth in miniature, containing seven biomes that the researchers cultivated, wrangled with, and depended on. If the 1960s space cabin was a fiction of man's independent existence outside the planetary, then Biosphere 2 was an explicit embrace of humans' interdependence with Earth's systems — the superior model for humans' future inhabitation of space. Indeed, one participant, a pathologist named Roy Walford, evangelized a restrictive caloric intake to extend human lifetimes, a regimen the Biosphere 2 participants followed. His effort could be framed as *idealizing* the Earth's material conditions in preparation for improving extraterrestrial humans.[21]

Like the space cabin, Biosphere 2 illustrated just how permeable material boundaries of the planetary (the structure) and the extraplanetary (the "real" Earth outside) are and how difficult it is to maintain an uncrossable line between. Oxygen leaked out of the roof. Food supplies dwindled. Artificial mechanisms to maintain conditions continuously failed or had to be assiduously monitored by hand because the engineering had not encapsulated the complexity and breadth of Earth's ecosystems. All pollinating insects perished. Social cooperation deteriorated. Living required external intervention. This vision of a planetary, autopoietic utopia became uninhabitable.

Though often considered a failure, the simulation demonstrated an artificial environment's quintessence of Earthly relations — interrelated, messy, and tenuous. We might say that it helped us rediscover ourselves — our follies and our frailty — within the simulated Earth-within-the-Earth, a process we might call "gaiamorphization."

Future Worlds

The conditions of extraplanetarities engender strong emotions about our precious planet in peril, so far the only example of life we know of in the universe. They clarify human embeddedness with Earth systems and spur calls for a course correction to prolong human and other life forms flourishing here. They let us imagine building a permanent settlement on Mars, changing our condition to be extraterrestrial. They sketch the possibilities to detect worlds with utterly unfamiliar life forms. They deracinate the anthropocentrically defined concept of habitability.

Extraplanetary relationships unshackle the concept of habitability from human comfort. Worlds are habitable or could be made so — but not exclusively by or for humans. Europa, Titan, Hycean worlds, and planets yet undetected reconfigure both the possibilities of alien planetarities of life in the cosmos and suggest alternative ways for humans to relate to Earth specifically. Rather than define the planetary, Earth has transformed into a subsidiary of it. This shift ushers in a new episteme of life, mind, and outer space.

We are fleshy, frail creatures who are just learning how big the space of life might be, and thus we require a revised orientation toward habitability both on Earth and beyond. We can today affirm MacLeish's prediction that the photographs from the Apollo 8 mission ushered in a "new notion" of humanhood that would remake our image of mankind, forming a fundamentally new episteme: "No longer that preposterous figure at the center, no longer that degraded and degrading victim off at the margins of

reality and blond with blood, man may at last become himself."[22]

The near future holds even greater possibilities to become ourselves. As the conditions of extraplanetarities come into focus, we will evolve new ways to be human. The book of life is a palimpsest, haunted by alien traces, with many wondrous rewritings perhaps to come.

Notes

1 Archibald MacLeish, “We Are All Riders on the Same Planet,” *New York Times* (December 25, 1968), A1.

2 Claire Isabel Webb, “Gaze-Scaling: Planets as Islands in Exobiologists’ Imaginaries,” *Science as Culture* 30, no. 3 (April 2021): 394.

3 Think of the Great Oxidation Event that began approximately 2.5 billion years ago: cyanobacteria deluged Earth in oxygen, forced surviving bacteria into symbiotic processes that produce mitochondria and, eventually, multicellular life forms who are our most ancient ancestors. Think, too, of the asteroid, imprinted on the Chicxulub crater 66 million years ago, that doomed reptilian rule and made space for small mammals to expand their niche.

4 “Habitability” has become the watchword to describe extra-planetary life in recent decades, in part as a result of the diminishing utility of the word “sustainability.” Whereas sustainability evokes humans’ persistence on Earth and refers to the vast anthropocentric techno-economic infrastructures to merely maintain global health, habitability conjures human and nonhuman possibilities for living in the general sense of the Planetary.

5 Of course, habitability is a political category as much as it is a material one; those in power maximize their flourishing.

6 “Of Astronauts and Algae: NASA and the Dream of Multispecies Spaceflight,” *Environmental Humanities* 9, no. 2 (2017): 360.

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8 William McNeil, *The Great Acceleration: An Environmental History of the Anthropocene since 1945* (Boston, MA: Harvard University Press, 2016).

9 Letter from Joshua Lederberg, University of Wisconsin, to Detlev Bronk, President, National Academy of Sciences, December 24, 1957, with enclosed memorandum entitled “Lunar Biology?”, National Academy of Sciences, Records Office, Washington, D.C. (National Academies of Sciences, Engineering, and Medicine, 2006). *Preventing the Forward Contamination of Mars* (Washington, DC: The National Academies Press, 2006). https://doi.org/10.17226/11381.

10 *The Planetary Quarantine Program: Origins and Achievements, 1956–1973*, NASA SP-4902 (1975), 26.

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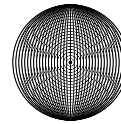
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Planetary Platform Automation

STEPHANIE SHERMAN

In his planetary design treatise *Operating Manual for Spaceship Earth* (1969), Buckminster Fuller described "comprehensively commanded automation" as a fundamental principle for a viable "Earth operating system."[1] "[A]utomation displaces the automatons," Fuller claimed, positioning automation as a foundational process of biological and technological evolution, a dynamic bio-technical process of encoding and embedding persistent decisions.

Fuller parsed automation in three distinct and contingent ways: first, automation as a function of *universal physical principles*, such as gravity, rotation, and movement, part of the regulatory structure that governs the trajectories of planetary bodies;

second, automation as *socio-technical processes* that delegate memorization, specialization, and maintenance to computers and machines to open space for human creative generalization and transferability; and third, automation as a *planetary protocol* that coordinates Earth's ecosystemic complexity through artificial management, interoperation, and calibration.[2]

Over half a century later, Fuller's manual remains an unrequited mandate for Earth operations, with his framing of Earth as an automated spaceship and his pragmatic operational design philosophy still beyond the scope of increasingly desperate planetary policies. Automation is a series of resolutions that build on one another, an ever-evolving operating infrastructure that inscribes decision-making power in agents and environments. Earth is, and has always been, an automated platform. In the age of planetary computation and artificial intelligence, automation can and must be harnessed to intentionally enable the development, survival, and governance of planet Earth.

As described elsewhere in this volume, the Planetary provides a practical philosophical framework for situating Earth and its extraplanetary astronomical systems. The Planetary accounts for the evolutionary functions and flows of material recomposition, scientific observation, metabolic recalibration, and synthetic construction. Distinguished from the sociocultural and political emphasis of the global, which centers the forces of human development and divisions in territories and ideologies, the Planetary investigates the complex processes that merge the biological, geological, chemical, metaphysical, and technological. The Planetary grapples with the forms of intelligence implicated in this comprehension, as well as the functional modalities that

can reorient the current phase of energetic malfunction toward systems viability. The planet, in other words, evolved humans — and now humans are evolving ways to sense, perceive, model, and modify the planet.

Planetary intelligence, as described by Sara Walker, David Grinspoon, and Adam Frank, is the evolutionary transformation of an immature biosphere to a mature technosphere in which energetic systems eventually compose a self-maintaining apparatus capable of ensuring a modulated and sustainable homeostasis.[3] The achievement of a mature technosphere would necessarily entail the evolution of the process of evolution itself — from the slow localisms of natural selection to an accelerated and intentional integration of "autocorrect" functions informed by planetary sensing and modeling.

Planetary computation is one way that planetary intelligence evolves to construct self-maintaining apparati. The instruments and infrastructures through which the planet is observed, modeled, and patterned increasingly coordinate automatic action without human intervention. Planetary computation provides the tools to support, augment, and recompose complex Earth operations while posing new paradigms for learning, behavior, and intelligence.[4]

Planetary platform automation frames this evolutionary process as a form of *operational* intelligence through which platforms build upon other platforms, embedding intelligence into environments and infrastructures. Planetary platform automation positions intelligence not as awareness, metacognition, reason, consciousness, or prediction, but as a functional operational behavior by which systems continuously learn and revise

themselves according to feedback from the world.

The Earth, as Fuller saw, is *already* functioning as an automated platform. Planetary platform automation builds upon the existing condition of automated platform Earth, shaping the maneuvers by which planetary computation might augment, coordinate, recalibrate, update, and optimize Earth's operations, which inevitably reshapes the philosophies that underpin it.

Planetary Platforms

Earth is a platform of platforms. The geologic definition of platform is a flat plane or plate of rock covered by sedimentary strata. Geological platforms, such as Earth's continents, form by filling in a jagged or inconsistent formation to create a smooth base. These platforms, bounded by oceans or crusts, compose new layers of tectonic stability and homogeneity. Over time, these geologic platforms grow and degrade and migrate and splinter, precipitated by both slow tectonic shifts and accelerated by grand disruptions like volcanoes and earthquakes.

Earth's platforms include not only geological composites, but also the biogeochemical assemblages, energetic reactions, atmospheric compositions, and material pressures that provide the basis for the ongoing dynamic interaction and development of life. In this sense, platforms are foundations and frameworks that scaffold operations for further interaction. This process is found both in the material physics of planetary operations and also in technologies across scales. In synthetic biology, for example, researchers refer to a cellular scaffold as a "chassis," a platform upon which biological systems are designed, built, and

tested.[5] This platform provides standardized protocols, resources, and infrastructures that can be used to manipulate biological components such as DNA, RNA, and proteins to engineer biological functions or systems.

Earth's platforms also include the human-constructed architectures and infrastructures that, at planetary scale, encase the Earth in an "artificial megastructure"[6] that builds from, upon, with, and between biological and geochemical foundations. These platform architectures — oil rigs, train platforms, ramps, docks — collect and distribute energy, facilitate material processing, and channel and coordinate information through standardization and allocation, evolving upon one another through layering, sometimes amplifying and sometimes undermining their earlier formations. Road, rail, marine, and other infrastructures are by now almost entirely standardized to coordinate global flows of physical goods across the Earth's surface.

These platforms fold the planetary into itself, recomposing Earthly materials in their construction and operations. Digital platforms extend this infrastructural network logic to information channels, financial exchanges, and communication, which similarly depend on an energetic substrate. Extraplanetary machines in orbit, such as satellites, servers, and sensors, produce planetary observations of the Earth's condition, a condition that computational systems like GIS, visualization, and simulation tools can monitor and model. Planetary organizations such as the Square Kilometre Array Observatory, space agencies, and the International Telecommunications Union provide platforms for regulating, certifying, and coordinating Planetary science and research. This stratum of platforms provides the basis for

the rapid evolution of planetary computation and the ongoing disclosure of planetary conditions, what Benjamin Bratton in this volume calls "planetary sapience."

More broadly, platforms are a particular type of system with particular evolutionary dynamics. They designate an environment or ecosystem, a physical or cognitive architecture, an infrastructure or interface that establishes the parameters in or through which other things can be programmed, built, or designed. Platforms provide a foundation that both sets things apart and raises them up, making closer and interoperable connections, a reverse entropy that evolves by microscopic iteration as well as macrophenomenal reconstruction. These platforms prescribe protocols that, like DNA, operate within fixed and rigid vertical rules, enabling a horizontal distribution based on repeatability, transferability, interoperability, and flexibility. A multitude of permutations develop, advancing diversity and differentiation through homogenization and coordination. Through standardization, consolidation, distribution, and resolution, platforms accelerate automation, which further accelerates and encodes decisions into processes that no longer need to be explicitly considered but are directly embedded into environments themselves. Platforms not only provide stages for enhanced observation (looking out or over) but also often make other processes invisible. These stages refer to the spatial domains and protocols that platforms set up and also to the stages of development through which platforms evolve and progress according to feedback.

It is through the ongoing development of platform processes that humans are able to comprehend and act upon planetary conditions. If the Earth is a platform of platforms currently unable

to manage its energetic output, then a planetary society situates humans within an evolutionary process that eventually automates platform evolution itself.

Automation As A Planetary Process

In the most common definition, automation describes the delegation of human labor, decision-making, or production process to *machines*, which typically overtake roles humans previously occupied. This happened first with manual work like weaving or switchboard operation, and later with more cognitive tasks through automated intelligence in calculation and computation, pattern detection and modeling, writing and image generation.

This definition of automation limits its planetary potential. From a Planetary perspective, automation is a *relative* phenomenon, less about the direct transfer of human work to machines than about the continued process of resolving questions embedded in decisions or context. Automation is the evolving environment of consolidation, coordination, and choreography of the operations required to achieve any outcome.

Humans have always struggled against the biological forces of planetary automation by developing technological systems that work against primitive or natural automation. Through technologies, industrial automation outsourced what was beyond or beneath human capabilities, and it enabled the infrastructures of global civilization: water supply, industrial agriculture, refrigeration, health care, air traffic control. In this way, platform automation became a means to establish a quality of human life

that overrode or undermined naturally inefficient or suboptimal biological processes. Inter- and extra-planetary machines and sensing apparati extend the human capacity to probe and explore the Earth, like robots for venturing into the deep sea.

Planetary automation also involves extraplanetary automation, as Claire Isabel Webb explores in her contribution to this volume. When the spacecraft *Cassini*, a satellite set up to sacrifice itself, took its final dive through the rings of Saturn, it sent back the most stunning images, providing humans with extraordinary visual information. With the rapid development of artificial intelligence, planetary automation gained and then surpassed human-level cognition for operations, including data compilation, pattern searching, and language production and translation.

If machine automation was invented to augment machines in factories operated by humans, the evolutionary trajectory of automation implies that the planet Earth combined with its extraplanetary condition becomes another sort of factory producing intelligence about itself — automation producing and predicating further automation.

Platform automation, then, refers both to the *automation of platforms* — the encoding and prescription of biogeochemical, metaphysical, or technical processes — and the *platformation of automation*, which describes how automation further embeds standardization and amplification of founding conditions into ecosystems and environments. No platforms without automation, no automation without ongoing platform formation.

Following Fuller, we can define automation as a deeper evolutionary logic that embeds responses to the physical

properties of the universe — a process of bio-technical decision-making that resolves prescriptive actions without conscious cognitive oversight. Evolution automates decisions by discovering structures through encounters and tests in environments rather than by programming operations from first principles. Any decision taken automatically, without thinking, is a form of automation. In this sense, learning is a process of updating a model based on feedback, establishing a new framework upon which to build. Planetary platform automation describes externalized construction and embedded cognition as the evolutionary trajectory of cognition itself.

Artificial intelligence presents an opportunity for humans and biotechnical machines to detect patterns and develop abstract models and predictions based on performance, and also to embed decisions and decision-making capacities into environments themselves, giving platforms the ability to auto-decide, auto-maintain, auto-manage, autocorrect. Intelligence, then, can be understood as self-conscious cognition and prediction based on modeling and a process of *performative* externalization whereby successful behaviors and protocols demonstrate fitness and utility based on circumstances and maintain adaptive flexibility even as learnings are consolidated into evolving agents and environments. This goes beyond the cybernetic focus on feedback and control — it positions automation both as an internal function of automatic self-regulation *and* as a continuous process of externalized experimenting, testing and infrastructuralization.

From a philosophical perspective, the highest forms of intelligence seem not to stop at cognition or consciousness or reflection; optimized intelligence *inscribes* the ability to process

information with such alacrity and fidelity that an agent can behave automatically, without conscious cognition or external regulations. It also offloads agential functions into environmental ones as part of intelligent distribution and infrastructuralization.

Automation does not eliminate the role of human beings but rather shifts their role in planetary processes to the critical functions of cultivating interoperable optimizations of evolutionary processes while eliminating other operations altogether.

Artificial intelligence transforms platform automation and computation from programmable rules and prescriptive processes to environments that precipitate *development* through continuous feedback and learning. Computation offers a useful heuristic in this respect. A programmer writes or composes code, whereas a developer determines the operations by which a code might be leveraged or optimized to realize a desired outcome. Humans become designers and developers of platform automation processes, which then automate platform automation.

The urgency of recalibrating Earth as a viable ecosystem programmed toward survival requires the humans currently at the helm of such decisions to frame intelligent planetary operations as operations without operators. *Intelligence* means energetic sustainability in which a system sacrifices a part for the survival of the whole, or the ability to update a model rapidly in response to new information, and also something more akin to involuntary virtuosity — the efficient performance of effective operations that no longer require cognitive effort.

Planetary Platform Automation And Intelligent Evolution

Gaia theory, developed by James Lovelock and Lynn Margulis, posits that Earth's ecological systems serve as a self-regulatory mechanism, a naturally occurring, *automatic* planetary feedback loop.[7] The theory espouses an Earthly autopoiesis, in which the entire Earth functions as an autonomic platform regulating and adapting symbiotic life. The theory has been challenged due to its limited geologic historicity, primarily in reference to the Holocene era, during which integrated mechanisms of auto-regulation produced only temporary stability.[8] A volatile climatic age preceded the Holocene, and similarly the Anthropocene has disrupted the autopoietic homeostasis presumed to be inevitable and persistent.

The most salient takeaway from Gaia theory's inconclusivity is that evolutionary planetary platform automation proceeds both by iterative optimization and by grand disruption. The so-called Great Oxidation Event transformed the Earth's atmosphere and enabled the evolution of aerobic life-forms. Distinct from the localisms and niches of natural selection, this event proceeded through a rupture in the iterative automated platform of evolution, precipitating a new evolutionary trajectory in which atmospheric oxygen gave way to the thinking organisms we call human beings. Gaia theory relies on natural time and evolution. Platform automation can accelerate spacetime at the planetary scale.

Darwinian natural selection as an evolutionary system is energetically intelligent at a local level, but not necessarily intelligent from a whole-systems view. It works through

slow species adaptation in response to local environments, neighborhoods, or niches that determine fitness according to the most immediate contexts. Because this form of evolution is developed through a series of locally determined outcomes, which are themselves based on established protocols, it does not have the capacity to account more holistically for the myriad factors that shape the environments in which it plays out.

Localized competitive decisions are unable to incorporate broader patterns or the predictive modeling of secondary effects that might result from cascades of automated and optimized processes. This is the case in natural systems and in infrastructural ones as well. Autopoietic automation, for example, led to the dramatically inefficient megastructure of the automobile and the highway system, producing a wicked problem in which platform automation built on platform automation encoded a dependence on fossil fuels and pervasive concretization that destroys ecological habitats. In this sense, local or site-specific autopoietic selection as a process is (as the computer scientist Eliezer Yudkowsky put it) "stupid" because its optimizations occur too slowly over generations to accommodate the rapidity of the transformations they precipitate and because each decision cannot account for its effects on the whole system in which it develops, or the potential cascades of that intervention, at any given instant.[9]

As the anthropogenic climatic transformation of Earth's ecosystem continues to evolve beyond homoeostasis, it demonstrates maladaptive human behavior: the inability to calibrate cognitive intelligence (the awareness that the climate is in fact changing in existential ways) into coordinated action

(intelligence as the ability to act upon this knowledge at the scales required). Fuller's prescription was to harness the forces of automation, whether they be physical, cognitive, or mechanistic operations, toward the reconfiguration of Earth operating systems that would both harness existing power and intervene as generalists in the design of evolution.

Planetary platform automation plays a growing role in intentional systems governance. The cybernetic "governor" is a regulatory device that maintains a process of bio-technical autocorrection rather than a political figure who relies on rhetoric and representation. Governance is, after all, just the mechanisms, processes, and practical implementation of regulating, orienting, and calibrating, intelligent systems that maintain and act upon insights derived from automated platforms operating in the world.[10] Automation optimizes and builds on existing systems, and humans working within them reroute, redirect, and make them interoperable through testing and feedback.[11]

Platform automation provides a model for bridging the energetic intelligence of local natural selection with the kinds of artificial interventions that might be designed to derive insights from emergent exceptions, deep pattern detection, and path-dependent updates. This model would operate by activating converging forces rather than prescribing rote rules. This line of thinking runs counter to the litany of critiques of algorithmic governance, what John Danaher calls "the threat of algocracy."[12] Planetary platform automation shifts the computational model from threat to opportunity, from critique to constructive optimization. The technological and predetermined becomes a potential social asset if leveraged with nuance, precision, and

iteration.

One obvious arena for planetary platform automation in Earth systems management would be the coordination of economic and ecological relations, two systems that currently operate independently. Imagine, for example, a planetary mechanism in which carbon output *automatically* sets price. Rather than proceed via convincing individual agents or national actors to modify their behavior at will toward collective flourishing, this would reset the fundamental terms upon which economic assets derive value. It would combine the demonstrable intelligence of localized market decisions with an intentionally directed definition of value modulated by planetary conditions.

Platform automation reframes the philosophy of Planetary governance not as a means of convening actors to think through mitigation, but as a commitment to effective and efficient planetary calibration. The realization of a collective philosophical orientation proceeds at once by devising mechanisms that can infiltrate the processes of iteration and readily prepare to harness disruption through a paradigm of action where those mechanisms might *automatically* take hold. In this sense, planetary platform automation requires environmental interdependence rather than autonomy. Platform automation optimizes operational verticals that integrate insights from horizontal distribution.

Operational Intelligence: Planetary Prediction And Performance

In the grand strategy planetary simulation game Stellaris, players build an empire through intergalactic expansion with

the goal of developing a functional planetary system. Players cannot choose to reroute fundamental parameters of physics or cellular development, but they can select technological, ecological, or economic principles that set planetary priorities and the variables that determine viable governance operations. An array of expansion packs allows players to introduce various paradigmatic resources and pressures — Synthetic Dawn, MegaCorp, Federations, Aquatics, First Contact, The Machine Age. In the game, one available function is to "automate" the management operations of the empire, allowing the player to prioritize their attention on some operations while prescribing or programming others. As one player put it on a Reddit thread replete with musings on the poor performance of this planetary management program: "[Is] it really that hard to code competent AI for planetary administration?"[13]

To become a properly intelligent process, planetary platform automation must model and test potential outcomes. In the age of planetary computation, this happens regularly through simulations that present, with various levels of fidelity, scenarios in which factors and variables can be isolated and tweaked to experiment with possible trajectories. Simulations support the understanding of such outcomes, modeling through performance and feedback based on wholesale effects. Like any model, including any model of planetary intelligence, planetary platform automation would never be able to observe and model a system in its entirety. Platform automation must be highly adaptable, intentionally designed with flexible protocols rather than fixed programmed operations, and capable of navigating and changing with new insights from accelerating effects and cascades. In this

sense, platform automation operates as a form of *operational intelligence*, which establishes protocols and procedures that reduce the variables of the unknown. Operational intelligence attempts to address all that can possibly be predicted or predetermined or prescribed to make space for the incorporation of insights from what cannot.

Operational intelligence, as a domain of activity, historically refers to the capability to collect, analyze, and act upon data in real-time to make informed decisions and optimize operational processes for continuous improvement and modification. Currently, most modeling and decision-making tools (traffic control, water resource management, health care) use mathematical/computation frameworks like entropy theory or adaptive modeling to accommodate complexity. These systems, relying on Von Neumann computational architectures that underpin simulation technologies, use sequences that prescribe a series of operational rules. They do not, in their foundational architecture, mirror derivative complexity or automatically update their models based on observed schisms between predictions and outcomes.

Operational intelligence is a process of designing systems to accommodate all "known knowns" (as Donald Rumsfeld would put it) to accommodate "unknown unknowns" — in other words, procedures that respond to effects impossible for any system to model. In this sense, operational intelligence involves a perpetual accounting not just for what previously existed, but also for what-ifs, while knowing full well that those what-ifs are incomprehensive and that convergence supersedes causality. Nevertheless, intelligence also involves synthesis — the reduction

of variables as a recursive and reflexive process. An operationally intelligent tool for governance would leverage operational intelligence not as a series of rules or protocols, but as a process of optimization that incorporates unexpected outcomes back into the system's model.

As a form of operational intelligence, then, planetary platform automation evolves via the configurations and cascades of effects of automated processes embedding, encoding, inscribing, and artificially transforming environments in their image through feedback on performance. Stellaris's AI model might be terrible at planetary administration because it does not yet integrate *conditioning* and *memory*: An abstract model has not yet become channeled into its functionality. This model, like the ones we currently have, runs predictive programs based on past parameters, but the system is not actually intelligent — it does not update its model based on new outcomes.

This lack of learning or updating based on models and memory prevents another form of virtuosity from emerging. The operationally intelligent evolution of planetary platform automation would require a continuous interlinking of various planetary systems, such that feedback loops could be cross checked, and also require a computational architecture with the power to process predictions simultaneously rather than sequentially and to build recursively on previous conclusions.[14] Rather than a string of cause and effect or if-thens, it would need to architect a calibrating mechanism whereby adaptive processes can distinguish unexpected and unanticipated formations as factors, delineating trends from outliers.

Planetary platform automation is not only an operating

protocol, a proposed interoperable infrastructure for an evolving Spaceship Earth. It is also a philosophical framework, one that approaches planetary intelligence not as cognition or self-awareness, but as *the reflexive incorporation and continuous modification of behavioral processes that meet the challenges of continuously and rapidly evolving environments through performance*. Planetary platform automation relies on an operational intelligence that encounters not totality but vastness in every direction, from quark to universe. This incompleteness does not thwart its intentionality or its encoding of rules. Rather, it is a commitment to deep incorporation, a comprehension of continuous movement by way of expansion, determination, acceleration, and modulation, based on standard deviations and the incorporation of presumed exceptions. It derives its momentum not from the abstraction of prediction, but from the active realization of continuous behavior modification.

Like platform processes, policy shifts for planetary platform automation might be realized by way of slow iteration or unpredictable disruption, by incorporating insights from copious accidents and failures as well as successful examples in intentional Earth management and systems conversion. One such example, a rare exception of planetary policy gone right, is the Montreal Protocol, a unilateral universal agreement arranged by a series of UN agencies that came together to establish atmospheric pollution regulations with the very specific goal of eliminating the use of ozone layer-depleting chlorofluorocarbons. The lesson from the Montreal Protocol was not only the explicit removal of certain chemicals from energetic processes, but the development of and mandate to deploy their synthetic substitution,

hydrofluorocarbons, which are still greenhouse gasses but do not deplete the ozone layer with the same rapidity. An alternative model would be an Independence Day-scale emergency, in which an external force or enemy disrupts the foundations upon which the apparatus depends, leveraging a widely recognized existential threat as a justification for instituting automated procedures. Regardless of its procedures, planetary platform automation presents a philosophy of the planetary that positions intelligence as the ability to act *automatically*, an operation without operators that updates auto-regulation through artificial interventions, learning through both slow iteration and the precipitous shifts of emergency.

Planetary platform automation is an evolutionary process that occurs through augmented and automated intelligence. The modification of an operating manual for Spaceship Earth is no longer a mandate for designing the protocols for human intelligence, but a systems manual for designing and developing synthetic intelligences realized through feedback based on actualized performance, many which humans themselves can neither conceive of nor comprehend.

One of the great ironies of current AI hype is the inflated obsession with governing AI rather than a rigorous exploration of ways to deploy AI as a tool for governance. An operational intelligence that mobilizes planetary platforms proceeds through the inexorable automation of evolution. This is an artificial evolution that surpasses natural evolution, learning by testing and exceeding, step by step and leap by leap, the parameters and procedures that determine, constrain, and enable it.

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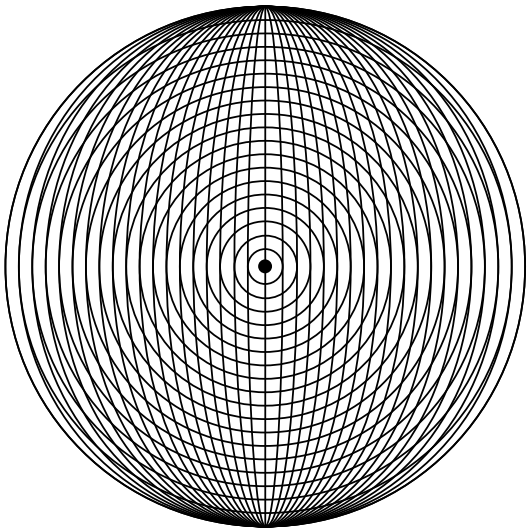
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II. Philosophy



Planetary Deep Time And Human Autonomy

NICCOLÒ MILANESE

> *Go, said the bird, for the leaves were full of children,*
> *Hidden excitedly, containing laughter.*
> *Go, go, go, said the bird: human kind*
> *Cannot bear very much reality.*
> *Time past and time future*
> *What might have been and what has been*
> *Point to one end, which is always present.*
>
> —T.S. Eliot, "Burnt Norton"

The Planetary is a disorienting event, at once centering and decentering humanity in a vast network of systems outside our control and even awareness.

In the first place, it confronts us with the visible consequences of the accumulated actions of humankind on Earth systems — the environment that makes all life possible. Our intervention in systems of scale that once were autonomous from us has caused momentous and destabilizing new conditions that threaten the continuing functioning of those systems. In other words, it *centers* human action as a planetary event.

The Planetary also *decenters* humans entirely from the planetary story, forcing us to reckon with the eons in which nothing like human life was present, with timescales of change that far outlast any conceivable human civilization. Ultimately, it leads us to imagine a planet *without* us at all, just like all other planets (as far as we know).[1]

Our disjunction between these two faces of the Planetary — one in which we see *too much* resemblance to ourselves, the other in which we *cannot recognize any resemblance* — risks producing a debilitating anxiety and practical paralysis, which in turn exacerbates the tension between the need for urgent action and the long and perhaps even endless temporal horizon in which such actions will need to be situated.

A "crisis of temporal management" is the way historian Dipesh Chakrabarty has described this existential situation.[2] He believes that underlying this political crisis is a deeper intellectual crisis in our capacity to tell history. History, Chakrabarty maintains, is based on human experience, but one face of the Planetary — and its long temporality — is beyond the possibility of human experience and so can only be grasped in an abstract way that does not permit rich, meaningful phenomenological content. Without a sense of our place in this long past, we find it

difficult to orient ourselves to act with purpose.

There is no question that both our politics and political institutions seem, at present, wildly inadequate to address planetary crises. Is our inability to adequately tell the historical story of these crises to blame? Should we proclaim, in a kind of reenactment of Hamlet, "The time is out of joint; O cursèd spite! / That ever I was born to set it right!"? (1.5.189-190).

Before we script ourselves into this role, we should specify what, exactly, is so troublesome about the *alterity* that the Planetary introduces. What is it about the exteriority and otherness of the planetary to us that seems to frustrate our possibility to reflect and act? Which history are we trying but failing to tell?

Our sense of powerlessness, anxiety, and guilt are not necessary corollaries of our predicament. Instead, they are the result of inherited ways of thinking that are dominant in our politics and in a misunderstanding of the nature of history itself.

The Obligation Of History

The disjunction between the *time of human experience* and the *time of the cosmos* is not a new aporia. In the three volumes of *Time and Narrative* (1983-85), philosopher Paul Ricoeur illustrates the tension between the Aristotelian and Augustinian conceptions of time.[3] Aristotelian time is understood as a measurable series of instants in the universe, a kind of number.

By contrast, Augustine saw time as a triple present: the present of past things (memory), the present of future things (expectation), and the present of present things (perception).

These two conceptions complete and occlude each other.

Aristotle's chronological time of the universe cannot fully account for what is *now* to us and the unity of time in our experience. The Augustinian phenomenological time of human experience can account for our sense of time passing but not the movement of time: Without any fixed points outside the human spirit to refer to, it cannot provide any measure by which to say, for example, as I recite a poem, the part of the past increases as the part of the future diminishes.

Humans confronting what Nils Gilman in this volume calls "the condition of planetarity" are inevitably situated between these two temporal conceptions, between the phenomenological and the noumenal, which is what we do each time we use indexicals to say, "I am here, now." We situate ourselves and our experience in a larger context that is extended in space and time beyond ourselves.

What Ricoeur argues in *Time and Narrative* is that telling *history* is critical for humans to make sense of our individual and collective relationship with the past in order to understand our present situation and inform future action. History is a mediation between cosmological time and socio-phenomenological time. The disorientation of planetary crises is an extenuation of this aporia, which is the structure of history itself.

Ricoeur's philosophy of history is that *narrative* makes history tellable by mediating between the time of human experience and the time of the cosmos through a "triple mimesis." The first mimesis, or representation of the world, involves understanding human action as structurally different to other physical movement because it is backed by agency and intention — reasons, goals, and motivation — a symbolic framework in

which actions have meaning, are understandable to us, and which we could in some sense imagine doing ourselves.

The second mimesis is an operation to mediate between individual events and history as a whole, which has a plot structure: a beginning, middle, and end. Different plots are innately understood within different cultures: Tragedy, comedy, romance, and others have been developed as part of a canon.[4] This helps give a sense of narrative to a sequence of historical events. Not only can we understand the structure (a challenge presented, confronted, overcome, etc.), but we also find the motivational and emotional development of the characters understandable and relatable (we feel *pathos*).

The third mimesis happens in the act of reading or listening to the narrative: the actualization of history refigured as the time of action by the audience. This third mimesis is crucial to our understanding that history is not a process autonomous from human agency: History is presented in a way in which *human action matters* because it *makes a difference* to what happens.

History in Ricoeur's conception is intrinsically anthropocentric: The historical narratives we tell are stories of human action. Not only are they usually stories of human actors, but historical narratives have value for us by informing and motivating action. They give us reasons to do things or not to do things. But even the same historical narrative may present different reasons for different action to different people in different circumstances, and the same facts can be narrated in many different ways. The story of the European arrival in North America, for example, is told in very different ways by British or Wendat storytellers. The history of a battle is told very differently from the perspective of

the general, a foot soldier, or those opposing them.

Does the time of the Planetary create a problem for these mimetic operations of history? There has been a striking lack of engagement with the philosophy of history in general when it comes to discussion of Planetary history, and there are perhaps three main reasons why this might be, all of which are mentioned by Chakrabarty (among others).

First, the timescale problem. Timescales involved in planetary processes, such as million-year carbon cycles, are far beyond any usual conception of the temporal structure of human action. How can we have any human experience of them?

Second, the collective action problem. The scale of human influence on Earth systems requires us to see humankind itself, in total, as a planetary actor. It is the cumulative effect of each of our actions — that is, the effect of humans *as a whole* — that has an impact on what were previously planetary processes operating externally from us. The problem is we have great difficulty imagining humankind as a whole, as a collective agent. We are much more accustomed to thinking of individuals or smaller groups acting, interacting, and competing.

Third, the nonhuman problem. In telling the history of the planet, most of the "actors" are not human — they are systems, processes, and nonhuman species, each of which is defined by temporal rhythms that have nothing to do with human ones. How can we conceive of these nonhuman actors? Is "actor" even the right term? How can we frame the interaction between human action, nonhuman action, and large systems with nonhuman and nonliving components?

Contemplating such massive timescales can provoke a kind

of cognitive vertigo, like looking up at the stars. Humans have always told stories about the origin and evolution of the universe: mythological, religious, historical, and even scientific stories. Is planetary time different? I think it is not, but it does challenge us to give a unified historical account of the planet and the place of humans in it.

Understanding, Explanation, And Action

A common philosophy of science, perhaps most strongly argued in modern times by Georg Henrik von Wright in his 1971 book *Explanation and Understanding*, depicted a sharp division between the natural sciences and the human sciences. In this view, the natural sciences look to provide causal *explanations*, whereas the human sciences seek to provide *understanding*.[5]

Von Wright appeals to a "Galilean" tradition of causal-mechanical explanation and an "Aristotelian" tradition of finality that appeals to reasons, intentions, and goals. All causal-mechanical explanations can be formulated as a set of laws along the lines "if X then Y," which then are applied to specific conditions and provide an explanation for why Y happened. This builds on the idea of regularity, which has close connections to the chronological conception of time: If there is a law that says X causes Y, this can be true if and only if events of type X are regularly followed temporally by events of type Y.

In the social sciences, by contrast, von Wright explains events as "practical syllogism":

X intends to bring about P
X considers he cannot bring about P unless he does A

Therefore, X does A

There is no law involved in such a syllogism and, according to von Wright, there is no way of verifying the conclusion of the syllogism other than verifying the premises and no way of verifying the premises without verifying the conclusion. Thus, there is no logically distinct cause and effect here, which means that what is instead required to account for the relationship between X and Y is a *hermeneutics of understanding.* This is why history, understood as the core discipline for understanding ourselves, necessarily takes the form of narrative.

This distinction between the natural sciences, a world of laws and causality, and the human sciences, a world of norms and intentions, would help to explain the disorienting effect of placing human action in the context of deep planetary time if it mapped onto the two faces of the planetary we identified: the human and the nonhuman. There would be a difficulty in the kind of narrative that can be told, shifting from law-like Galilean explanation for almost all of nonhuman history, to Aristotelian reasons, intentions, and goals for the briefest but most decisive of human interludes.

Yet there are many reasons to think that the "law-based" conception of science is implausible: It cannot distinguish between causal relationships and accidental correlations, and there are areas of science, most notably biology and medicine, which provide causal explanations but few laws.

Instead, philosophical debates currently revolve around an *interventionist* conception of scientific explanation. Imagine that in an experiment a scientist can vary the conditions — and

by this intervention bring about different results.[6] This is not as anthropocentric as it sounds — an intervention can be made by nonhuman and non-sentient actors and can be defined causally by counterfactuals.

The model of explanation is then the following: If X happened and then Y happened, we can say that X caused Y *if and only if* in a scenario where X did not occur, Y also did not occur. This is different from the causal-mechanical explanation above because it does not appeal to any overarching law. This view of scientific explanation enables us to narrate planetary history in which there are *both* human and nonhuman actants operating on a planetary scale with the same causal structure. It ensures history is *unitary* across chronological and phenomenological time.

The imagination is crucially involved in this model of scientific explanation and action, as is treating natural phenomena as if they had a capacity to intervene. That provides the essential bridge to the world of human explanation. If, in order to explain the history of Earth systems, we need to mobilize our imaginations, make figures and models to visualize counterfactual scenarios, and treat natural processes as if they had a form of agency, then we are already performing some of the same mimesis as involved in understanding human action and creating narratives of history.

That this theory of scientific method involves such mimesis does not discredit it as anthropocentric. On the contrary, the counterfactual theory enables us to conceive of causes without either turning humans into nonhumans without free will or nonhumans into humans (with intentions, beliefs, desires, etc.). In order to give an account of our planetary predicament, there is no need to "humanize" the planet or to make all planetary actants

into political agents with intention and historical understanding in the same sense as humans. Crucially, this allows the socio-historical realm to be ontologically distinguishable from the natural without the break between them being unbridgeable and politics to be clearly a human responsibility. History is a human activity of understanding that is related to our responsibility to act by manipulating causes. When we seek to do so in ways that affect or influence the common good, we are doing politics.

The Problem Of Post-Sovereign Politics

If understanding our planetary predicament does not require us to radically change our conception of history, it does require radical changes in our conception of politics. Our modern, Western conceptions of politics are still dominated by Christian symbolisms underlying our understandings of sovereignty and autonomy: Whether it is the state, the collective will, or even the sovereign power that can decide on the state of exception, our political imaginary considers collective political actors as self-contained and dominant over their domain.

The revelation of the Planetary throws this conception of politics into crisis, both through the interdependency across societies, systems, and species, which undermines pretenses to self-sufficiency and supremacy, and more fundamentally through revealing the blindness of human interference with planetary systems until very recently.

Acknowledging the inadequacy of this conception of politics is not enough. There is also the question of *how* we acknowledge its inadequacy, of our historical orientation to this revelation — or, as

we could say, the politics of history. And therein lies the moment of our incapacity. For our tendency is to read this sequence as the plot of tragedy, which arguably underlies all Western politics.

The paradigmatic example is that of Oedipus, who simultaneously discovers his own identity as the son of Laius and Jocasta in the moment of reversal, or *peripeteia*,[7] and realizes the meaning of what he has done in marrying his mother, killing his father, and being brother to his daughters. In the moment of *anagnorisis* for Oedipus, past, present, and future collapse into an intolerable *now*. Understanding our planetary predicament under this tragic archetype encourages paralyzing guilt and the impression that it would have been better *not to act* at all than to act in ignorance. Either inaction ... or madness ... or denial.

We have to break out of this triad by ending our fascination with the schema of tragedy and finding other imaginative resources to frame the history of our politics and its transformation quite differently.

The practical poetic and political archetypes we need are not comforting fairy stories or excuses, but intertwined histories of the planet and of humanity that make recognizable our responsibility without placing us either in the positions of autonomous gods or dependent automatons following mechanical laws and that can work at multiple temporalities both urgent and long-term. For this the narrative archetype of the *metamorphosis* may be much better suited than that of tragedy or comedy: stories with no definitive start or end and which are multivocal, multitemporal, ambivalent, unstable, and in flux, embracing without assimilating the strangeness within us.

Contrary to Chakrabarty's intuition — that we face a crisis

in temporal *management* — the condition for success of this renewal of politics is that it not be understood as a task of taming or otherwise domesticating the planet's alterity by *making it more similar to us* or *us more similar to it*, but precisely of acting in a way that imaginatively embraces the strangeness of our multiple temporalities and acknowledges the plurality of forces that shape our planetary future.

Notes

1 See, for a recent popular example, Alan Weisman, *The World Without Us* (New York: Picador, 2007).

2 *The Climate of History in a Planetary Age* (Chicago: University of Chicago Press, 2021), 203.

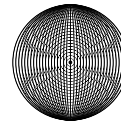
3 Trans. Kathleen McLaughlin and David Pellauer (Chicago: University of Chicago Press, 1984), vol. 1, part 1.

4 Hayden White famously explored the "emplotments" of 19th century history in Europe in his *Metahistory: This Historical Imagination in 19th Century Europe* (Baltimore, MD: John Hopkins University Press, 1973).

5 (Ithaca, NY: Cornell University Press, 1971).

6 See notably James Woodward, "Explanation and Invariance in the Special Sciences," British Journal of Philosophy of Science, Vol. 51 (2000): 197-254; Woodward, *Making Things Happen: A Theory of Causal Explanation* (Oxford: Oxford University Press, 2003), and Woodward, "Causation and Manipulability," *The Stanford Encyclopedia of Philosophy*, eds. Edward N. Zalta and Uri Nodelman (Summer 2023). https://plato.stanford.edu/archives/sum2023/entries/causation-mani/.

7 Sophocles, *Oedipus King of Thebes*, trans. Gilbert Murray (London: George Allen and Unwin, 1911), ll. 1176-92. https://www.gutenberg.org/files/27673/27673-h/27673-h.htm.



Planetary Metaphysics

BORIS SHOSHITAISHVILI

In 1938, in a text called "Scientific Thought as a Planetary Phenomenon," the biogeochemist Vladimir Vernadsky organized the sciences into two basic categories:

> *[I]n the first category, we have those whose objects, and consequently whose laws, encompass all of reality, such as our planet and its biosphere, as well as the cosmic expanses—that is, sciences whose objects correspond to the fundamental, universal phenomena of reality. The second category is related to phenomena which are characteristic of our Earth.*[1]

Against the backdrop of more familiar fault lines in the sciences (experimental vs. theoretical science, nomothetic

vs. historical science, and the "two cultures" of scientists and humanists), Vernadsky's typology introduced a new way to divide domains of scientific inquiry: between the universal and the planetary. He went on to specify the planetary as the side that would embrace "all the sciences of the biosphere, with the sciences of the humanities, with the Earth sciences—botany, zoology, geology, mineralogy—in all their scope."[2]

By interpreting the planetary instead of the universal to be the proper intellectual home of the humanities and social sciences, Vernadsky treated as consequential a fact that has typically been seen as incidental — that all human affects, thoughts, institutions, and cultures have developed in the specific context of Earth and the biosphere.[3] This passage is one of the first hints that the planetary can be a concept that structures thinking about human beings, from the epistemology of organizing the sciences to questions of metaphysics. To apply Dipesh Chakrabarty's phrase from nearly a century later, Vernadsky had found in the planetary, in a barebones philosophy of science proposal, an "emergent humanist category."[4]

Three ways of seeing our species as planetary have since emerged from the Earth sciences: the noosphere, Gaia theory, and the Anthropocene. To recognize how these three have shaped the planetary as a humanist category — and broadened its scope to encompass questions of metaphysics and ethics — it helps to approach them through another planetary thinker. Sixty years after Vernadsky presented his schema, and in the context of discussions about globalization, the literary theorist Gayatri Chakravorty Spivak converged on similar themes and established "planetarity" as a concept influential beyond the natural sciences.

In her lecture, "Imperatives to Re-Imagine the Planet," delivered in 1997 at the Stiftung Dialogik, an organization founded to preserve German and Swiss Jewish memory after the Holocaust, Spivak proposed that the *planet*, in contrast to the human-centered idea of the *globe* that shapes debates around globalization, holds a profound "otherness" for human beings: "The planet is in the species of alterity, belonging to another system; and yet we inhabit it, on loan."[5] Such contrast between globe and planet remains a defining distinction of planetary thought, a provocation to confront the economics- and politics-dominated frameworks of globalization with commensurate attention to human entanglement in another, more-than-human world.

Intriguingly, in the earliest published version of the lecture, Spivak finished her sentence with a significant difference. She had written that the planet belongs "to another system; and yet we inhabit, indeed are it."[6] This change — from the planet's alterity as what humans "indeed are" to what humans inhabit "on loan" — is more than a philological curiosity. The shift between these two versions, whether or not Spivak had meant this change to carry substantial weight, provides an opening into thinking beyond epistemology to two primary metaphysical and ethical dimensions of planetary thought.

Inhabiting a dwelling "on loan" signals an imperative to leave one's home, once one's time living there ends, in a recognizable, intact state, so that it can be occupied or lent again. A loaned home not only serves as a restorative and functional shelter but also possesses greater durability than the physical presence of any specific dweller. Such alterity entails a threefold

obligation: to the place itself as durable and capacious, to whoever owns it, and to others who come after to use this place as their home. So though you might, for a time, belong to this place and call it home, sheltered and nourished and restored by it, it does not belong to you. The metaphysics of planet-as-home or place-as-one's-other carry a corresponding, obligatory ethics: the ethics of preservation, a stewardship across time.

But in Spivak's earlier statement, that we not only inhabit the planet but "indeed are it," is an indication of a different metaphysics, one summoning an ethics distinct from stewardship. The planet as the other that is not our expansive dwelling but a being that both exists beyond and *is* ourselves, a being with which we identify, is a planet with whom the human comes face to face. This perspective presents the planet as a peer, with the potential for recognition and misunderstanding that meeting a peer inherently carries. In a more subtle way than the planet as dwelling, this understanding of the planet's otherness has featured in the last decade or so of planetary thought as well — consider, for instance, how scholars such as William Connolly and Bruno Latour have named their projects *Facing the Planetary* and *Facing Gaia*.[7]

But the question remains: Where is the shared conceptual ground on which an encounter with the planet's otherness as peer rather than dwelling can occur? How to engage in this intersubjective mode with a layered immensity existing beneath, around, and over all forms of life we know of, an immensity appearing so different from a fellow human being? How can one face, in the words of Herman Melville's Ishmael, a towering other with "no face"? Here Spivak leaves us only hints. We must look

outside her essays on planetarity to stage the encounter with the planet's alterity in a mode beyond dwelling and stewardship.

The earlier ethical metaphysics developed by the philosopher Emmanuel Levinas can help elucidate the metaphysical claim that "we are" the planet's alterity. Levinas's project, pursued in reaction to Martin Heidegger's "Being"-oriented thought, was to urge 20th-century European philosophy to turn from an obsession with ontology as the basis of metaphysics to the encounter of subject and other. It was precisely in the face of the other, he maintained, that the subject was confronted with a presence that overflowed the categories of "being" (the ontology) that one might seek to apply.

Another important feature of Levinas's theory was that speech and language are coeval with studied comprehension in the encounter with the other:

> *The other (autrui) is not an object of comprehension first and an interlocutor second. The two relations are intertwined. In other words, the comprehension of the other (autrui) is inseparable from his invocation.*
>
> *To comprehend a person is already to speak with him.*[8]

In reorienting us to the face of the other as well as the primacy of invocation in interaction, Levinas allows us to reconfigure Spivak's initial expression of planetarity into the following question: What is the face of the planet and how do we encounter the planet as humanity's interlocutor? Or, in other words, how do we face and speak with the planet in self-other dialogue, beyond our relationship to the planet as the dwelling we inhabit on loan?[9]

This is the point at which we turn back to the Earth sciences for their longstanding commitment to contemplating the face of the planet and generating the language through which this kind of encounter unfolds.

The Planetary Spheres Of The Biosphere And Noosphere

The Austrian geologist Eduard Suess — who coined the term "biosphere" — synthesized decades of his work in the multivolume textbook, *The Face of the Earth (Das Antlitz der Erde)*, published at the turn of 20th century. Suess's work inspired Vernadsky, who expanded Suess's inchoate concept of the biosphere into the idea of the dynamic, living layer on Earth that is now foundational for much scientific thought: a zone of highly ramified, aquiferous life that transforms the energy of the sun into work and profoundly changes the face of the planet — indeed, that makes our planet what it is.

In their interlinked writings, published decades before space flight provided photographs of Earth from a sufficient distance to allow us to perceive our planet's physical boundaries, these two scientists imagined what such a perspective would reveal about its face. As Suess opens *The Face of the Earth*:

> *If we imagine an observer to approach our planet from outer space, and, pushing aside the belts of red-brown clouds which obscure our atmosphere, to gaze for a whole day on the surface of the earth as it rotates beneath him, the feature beyond all others most likely to arrest his attention would be the wedge-like outline of the continents as they narrow away*

to the South.[10]

Or Vernadsky's opening to *The Biosphere*:

> *The face of the Earth viewed from celestial space presents a unique appearance, different from all other heavenly bodies. The surface that separates the planet from the cosmic medium is the biosphere, visible principally because of light from the sun. ... A new character is imparted to the planet by this powerful cosmic force. The radiations that pour upon the Earth cause the biosphere to take on properties unknown to lifeless planetary surfaces, and thus transform the face of the Earth.*[11]

For Suess and Vernadsky, the aspiration to view Earth's face, by a perspective separated from the Earth, unfolded into a recognition of our planet's distinctiveness — from a specific formal aspect of its geology in Suess (tapering continents) to a sweeping interpretation of its animated layer in Vernadsky (the biosphere). Taking this perspective, Vernadsky perceived that the face of the Earth was more than geomorphologically intriguing. The biosphere, imagined from space, became a mediator of the rock beneath and sun above, a planetary face alive in expression and activity.

The naming and conceptualization of the biosphere were significant moves toward recognizing the planet as our other, toward confronting its face as that which places metaphysical and ethical demands on us. But the second Levinasian element of full encounter with alterity — the other emerging as interlocutor to be spoken with, as one to whom we connect using language — manifested in another idea Vernadsky helped introduce. Based

on the vision of Earth's biosphere and in collaboration with the paleontologist Pierre Teilhard de Chardin and the mathematician Édouard Le Roy, he proposed a planetary designation for human beings.

Facing the biosphere, these thinkers viewed our species, for perhaps the first time in the natural sciences, from a planetary perspective (a departure from earlier cosmopolitan or universalist perspectives). The same imaginative point of view on the Earth's surface that had clarified the broad zone of life was retrained on our form of life, on the being adopting that perspective. In conversation with then-recent ideas that Earth comprises a series of geological spheres (the lithosphere, hydrosphere, atmosphere, and especially the biosphere) they put forward a new name for and conception of a planetary humankind: the "noosphere," a planetary sphere of mind and affect (*nous*).

Thus, the face of the planet not only came alive in human thinking as a diverse and interlinked (bio)sphere that we could gaze upon, but also as an interlocutor who bestowed, in our encounter with it, a resonant planetary identity. Reflecting on the planet's biosphere, human beings unlocked the intuition that we too constitute a transformative planetary sphere. As Spivak had hinted, the planet's alterity — its massive cyclical interactions occurring across prodigious expanses — became the basis for conceptualizing and experiencing one of the ways in which we "indeed are": inhabiting the biosphere but also, significantly and distinctly, noospheric.

In this realization, Spivak's dual senses of "dwelling" and "identity" coexist. We dwell in the biosphere as the planetary home we are obligated to preserve and steward. But we also

constitute, in our diversity of technologies, built environments, and cultural and individual idiosyncrasies (what Latour called our "homodiversity" in an echo of "biodiversity"), a planetary sphere of branching and interacting *nous*.[12] The profound plurality of human beings and cultures that thinkers such as Hannah Arendt have celebrated finds its other in the diverse living species of the biosphere and their ecosystemic relationships. Facing the biosphere as planetary peer, humans see themselves also as a sphere.

The Planetary Bodies Of Gaia And Humankind

James Lovelock and Lynn Margulis collaborated on the second scientific act of facing planetary alterity.[13] By the late 1960s and '70s, unlike when Suess and Vernadsky were alive, photography of Earth from space and the study of other planets in the solar system had begun. Studying Mars for signs of life, Lovelock intuited that the activities of living beings affect planetary atmospheres. Lovelock turned back to Earth and considered what far-ranging signals of planet-scale life our atmosphere and surface might carry. Taking this perspective, Lovelock reinterpreted the curious features of Earth's unexpected atmosphere, with its wealth of oxygen, as evidence of a host of homeostatic mechanisms employed by the biosphere to keep planetary conditions supportive of life's continued existence. In other words, Lovelock, and later Margulis, found in Earth's distinctive planetary face a window onto a vast collective being possessing the responsive interiority of a living planetary body.

The alterity of this planetary being that regulates

and stabilizes itself by embracing and enfolding trillions of interconnected living beings is open to a reading informed by Spivak's two senses of alterity as well. On the one hand, humans, according to Gaia theory, dwell within this greater homeostatic body, this planetary Life "with a capital L,"[14] which we inhabit on loan — mutualistically and cooperatively at best, parasitically at worst. We are thus obliged to steward and preserve the health of this living host.

But Gaia prompts more than an ethics of stewardship. As the biosphere concept inspired the noosphere, the recognition of Earth as a collective body spurs a confrontation with human planetarity in Spivak's other sense as well: as a peer to Gaia. A handful of theorists, including Pierre Teilhard de Chardin, the systems theorist James Miller, and the biophysicist Gregory Stock,[15] have treated global trade and transit, proliferating technologies, webs of global communication, and the accelerating worldwide spread of microbes and viruses through human hosts as indications that humans may be in the process of forming an Earth-spanning living body, one to some extent distinct from Gaia, with glimmers of planetary homeostasis and all-too-human vulnerabilities of its own.

These thinkers situate the myriad histories of human societies and technologies within one shared Big History of a developing human planetary being (a Big *Life* History of humankind, if you will). The numerous human bodies politic, all of them born from Gaia, are seen to be in a fitful process of interlinking into a human planetary body that exists, not only as a member or even child of Gaia, but also as Gaia's peer.

Considering Gaia theory alongside this idea suggests the

startling possibility that Earth may be "doubly alive" as a planet.[16] The paleontologists Mark Williams, Jan Zalasiewicz, and Anne-Sophie Milon underscored this moment of facing the planet as embodied alterity by granting not only Gaia but also this human planetary superorganism a religious-mythological name. Presenting humans as a global Leviathan emerging from the Earth-spanning systems of globalizing techno-societies, they have repurposed the biblical figure at the center of Thomas Hobbes's political treatise into a child of Gaia so immense as to bring menacing instability to its planetary parent and to ourselves.[17]

Human Planetary Force And The Earth System In The Anthropocene

The third major encounter with the planet as other — as face, as interlocutor, and as what we are — is taking place in this millennium in the concept of the Anthropocene, the proposal that Earth has entered a new epoch defined by humans exerting planetary force. The word "Anthropocene" derives from the nomenclature of the Geologic Time Scale, in which Earth's history is periodized based on close attention to continuities and disruptions recorded in the lithosphere.

With the Anthropocene idea, however, geoscientists have been pressed to do something unusual for their discipline: to think from a deep future perspective on the present, rather than from a present perspective on the deep past — to discern, in other words, the traces our species has left on the face of the Earth and consider whether these are durable and widespread enough to strike future students of geology as evidence of the beginning of

a new geological epoch. If our deep descendants find in the rock evidence of our planetary activity, this would mean that human civilizations have already made, on the short time span of tens or hundreds of years, a spatially and temporally gargantuan imprint on Earth's face. (For now, in a contested vote, the Anthropocene designation has been voted down, but the debate over this kind of future geology has just begun.)[18]

The Anthropocene's conceptualization of human impact on the planet's face has led scientists to describe humans, in discussions of the new epoch, as a planetary "force." The concept of force, in physics, helps explain the causes and origins of an impact. But this designation for humans has deeper implications. In the history of modern physics, forces have typically played a crucial and supportive role in the structure and function of *systems*, including at cosmic scales. Isaac Newton described a "system of the world," a system built on distinct kinds of force (motive, inertial, gravitational) that obey set laws and organize the universe.[19] Today, physicists posit four irreducible "fundamental forces" or interactions (gravity, electromagnetism, weak interaction, strong interaction.)

But alongside this there runs an older understanding of force, prominent in religious and mythological traditions and periodically appearing in thinkers such as Aristotle and Leonardo da Vinci, where force spills out of the cosmos or system that it works to order and instead brings about profound disruption. This is the kind of force the divine brings upon the world from outside its sustaining structure, in acts of devastation such as the Flood. The Anthropocene retrieves this older ambivalent tradition of a sometimes disruptive cosmic force in its conception of human

beings and our planetary impact. In the conceptual framework of the Anthropocene, the Earth is treated as an immense system in which internal and external forces have tended to balance each other's effects, especially through most of the Holocene, until human-generated force emerged as a planetary exception.

Seeing our species as a cumulative force in a systems context has found resonance in contemporary concerns about "tipping points" — the prospect that human activity could suddenly push the Earth system across dimly perceived boundaries defining the Holocene equilibrium into rapid change toward little understood new states that are perhaps chaotic or perhaps a new and less supportive equilibrium. These system states, such as a "hothouse Earth," are unlikely to be favorable to us and multicellular life, due to the little time human societies and nonhuman species would have to adapt to swift planetary change.

In tipping points, too, the planetary perspective on the Earth system has remade our understanding of the human as a planetary being. Thinkers have recently focused on identifying tipping dynamics that might exist *within* human societies and systems. "Social tipping points" offer a framework for provoking rapid change in our social and technological systems that could counter the alarming prospect we now face of inadvertently pushing Earth past its more-than-human tipping points. In other words, the disruptive character of human planetary force is now being interpreted as precisely that which might be leveraged and redirected against imbalances in our own systems. The tipping point becomes a conceptual resource not only for planetary metaphysics and ethics but for planetary politics as well.[20]

Like the paired ideas of the biosphere and noosphere, or of

Gaia and human planetary embodiment, encountering Earth in the Anthropocene as an ancient system containing vast forces and latent tipping points repositions us vis-à-vis ourselves, revealing humans' unruly force, systematic organization, and sudden inflections. In all three encounters with the planet's face, scientists and thinkers have stepped into interactions with a full other that, in turn, redefines the human as a manifold planetary being.

In fact, long before Spivak spelled it out, a hint that *we* are the planet's alterity may have existed in the origins of the word "human," some linguists believe comes from the same Indo-European root that gave us the words "chthonic," which relates to the underworld, and "humus," which is Latin for soil. "Earth," therefore, is our name for our species. It is a small wonder that we can reopen the metaphysics of the human by learning new ways to countenance and invoke the planet.

Notes

1 *Science & Technology* 16 (Spring-Summer, 2012): 30.

2 Vernadsky, "The Transition From the Biosphere To the Noösphere," excerpt from *Scientific Thought as a Planetary Phenomenon 1938, 21st Century*, trans. William Jones (Spring-Summer, 2012): 16-31. https://21sci-tech.com/Articles_2012/Spring-Summer_2012/04_Biospere_Noosphere.pdf.

3 Note Lynn Margulis's comment in the introduction to the translation of Vernadsky's 1926 *The Biosphere* that he was "the first person in history to come to grips with the real implications of the fact that Earth is a self-contained sphere"; in *The Biosphere* (New York: Copernicus, 1998).

4 *The Climate of History in a Planetary Age* (Chicago: University of Chicago Press, 2021). The phrase appeared earlier in Chakrabarty, "The Planet: An Emergent Humanist Category," *Critical Inquiry* 46, no. 1 (2019): 1-31. https://doi.org/10.1086/705298.

5 "'Planetarity' (Box 4, WELT)," *Paragraph* 38, no. 2 (July 2015): 290-92.

6 *Imperatives to Re-Imagine the Planet/Imperative zur Neuerfindung des Planeten*, ed. Willi Goetschel (Berlin: Passagen, 1999).

7 Connolly, *Facing the Planetary: Entangled Humanism and the Politics of Swarming* (Durham, NC: Duke University Press, 2017); and Latour, *Facing Gaia: Eight Lectures on the New Climatic Regime*, trans. Catherine Porter (Hoboken, NJ: Wiley, 2017): 107.

8 Adriaan Peperzak, Simon Critchley, and Robert Bernasconi, eds., Emmanuel Levinas, *Basic Philosophical Writings* (Indiana University Press, 1996): 6.

9 In other words, the ethics of this kind of ethical metaphysics, in relation to the planet, draw closer to the concept of I-Thou in Martin Buber's thought or the "ethics of care" introduced by Carol Gilligan, than to the ethics of stewardship. See Martin Buber, *I and Thou*, trans. Walter Kaufmann (New York: Scribner, 1970); and Carol Gilligan, *In a Different Voice: Psychological Theory and Women's Development* (Boston: Harvard University Press, 1982).

10 Trans. Hertha Sollas (Oxford: Clarendon Press, 1924).

11 Trans. D. B. Langmuir (New York: Springer, 1998).

12 Latour, 107.

13 "Atmospheric Homeostasis by and for the Biosphere: The Gaia Hypothesis," *Tellus A: Dynamic Meteorology and Oceanography* 26, nos. 1-2 (1974): 2-10. https://doi.org/10.3402/tellusa.v26i1-2.9731.

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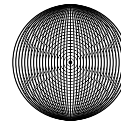
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17 *The Cosmic Oasis: The Remarkable Story of Earth’s Biosphere* (Oxford: Oxford University Press, 2022).

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Co-becoming: A Planetary View Inspired by East Asian Philosophies

SONG BING

A New Worldview In A Planetary Age

Planetary thinking calls for a major philosophical revision of individualist worldviews. In the West, most modern, liberal understandings of who we are as humans start with the notion of the individual as an independent, self-contained, and autonomous agent.[1] This notion of the individual emphasizes an agent's separateness, self-containment, uniqueness, and autonomy from others. It also celebrates individuals' capacity to define and redefine their identity and self-interest with authority, responsibility, and freedom to regulate their behavior. In this view, society exists only as an aggregate of autonomous and self-

contained, though interacting, individuals.

A classical expression of this idea can be found in the view of mid-17th-century English philosopher Thomas Hobbes that "society was simply a congeries of colliding atoms in unceasing motion."[2] Subsequent thinkers, including John Locke, Adam Smith, John Stuart Mill, Friedrich Hayek, and Milton Friedman, have developed rich philosophical, political, economic, and social theories based on this core concept of the atomist individual and what Joseph Schumpeter once famously described as "methodological individualism."[3] Indeed, this notion serves as the foundation for most worldwide mainstream modernist theories and modern institutions, including neoliberal economics, liberal-democratic politics, and meritocratic education.[4] As a result, much mainstream social scientific inquiry starts and ends with individual-centered analysis and calculation. The utopic goal is a world in which individuals act for themselves and maximize their personal interests (albeit with some "proper" regulation), allowing society to somehow reach a desirable equilibrium.

This idea is not totally unreasonable. Institutions based on this atomistic understanding of the individual have spurred unprecedented technological advancements and economic growth across the globe in the past several hundred years, culminating in the fastest and largest-scale improvements in the material quality of life in human history. However, individualism is also at the heart of the malaise that we have witnessed in recent decades: a breakdown of the family and disintegration of the social fabric leading to epidemic rates of loneliness and homelessness, mindless consumerism and environmental degradation, and deepening political divisiveness. This deterioration of the human condition

has been occurring on a global scale, though some places face more acute crises than others.

Transcultural philosophies can perhaps offer inspiration to recalibrate our thinking on who we are as humans and how we are related to human and nonhuman others. East Asian philosophical traditions in particular inspire us to think differently as planetary beings.[5]

The Planetary View Of "Co-becoming"

The notion of "co-becoming" — *gongsheng* in Chinese and *kyōsei* in Japanese, both of which share the same Chinese characters, 共生 — may enrich and help reset our worldview and bring about broader planetary thinking. *Gongsheng* and *kyōsei* refer to a conception of the world as consisting of mutually embedded, coexistent, and co-generating entities. This notion challenges and, therefore, enriches the mainstream modern notion of the individual as an independent, self-contained, and autonomous agent.

The rich meanings behind the concept can be traced back through Confucianism, Daoism, and Buddhism, as well as folk belief systems and practices in many East Asian countries.[6] The contemporary notion of "co-becoming" has been widely deployed in East Asian societies, including mainland China, Taiwan, Japan, and South Korea, and across political, social, economic, and commercial spheres in these societies.[7]

Briefly, a few words on terminology. In the compound word 共生 (*gongsheng/kyōsei*), the first character "*gong*" (or "*kyō*") means commonality, sharedness, and togetherness. "*Sheng*" (or

"*sei*") means growth, production, thriving, living, and emergence.

Both characters date back more than 3,000 years. Each was used in ancient classics and literature, though the combination of the two characters into a compound word only appeared after the Qin period (221-207 BCE). As it appeared in these writings, the term meant co-survival, interpenetrative growth, or a mingling of different things.[8] This entanglement and interpenetration is captured by the colloquial Chinese expression "你中有我，我中有你," which literally means that "part of you is me and part of me is you."[9] The best English equivalent of this expression is "mutual embeddedness."

Modern usage of co-becoming (*kyōsei*) in broader social and political contexts began in Japan. Perhaps inspired by the new findings of symbiosis research in modern biology, *kyōsei*-based analyses in Japanese social, political, and artistic spheres have blossomed since the 1980s.[10] These developments also influenced intellectual discourse in China in the late 1990s and early 2000s. By the mid-2020s, the notion of *gongsheng/kyōsei* has been firmly integrated into the common parlance, widely deployed across political, social, economic, and environmental spheres in Japanese and Chinese societies.[11]

Reflecting the ontological thinking of "ceaseless change" as the first-order cosmic rule in an East Asian context, and bearing in mind the potential of the worldview of *gongsheng/kyōsei* to address issues of a planetary remit and go beyond the human sphere, here I will translate *gongsheng/kyōsei* as "co-becoming" and use the terms interchangeably.[12] [13] [14]

Philosophical Sources Of Co-becoming

Cosmic and ontological thinking in Confucianism, Daoism, and Buddhism has profoundly shaped East Asian societies for millennia. These philosophies offer rich intellectual resources for the planetary view of "co-becoming," including four well-known propositions: the "unity of *tian* and humans" (天人合一), the "oneness of all beings" (万物一体), "ceaseless living and generating" (生生), and "co-dependent origination" (缘起).[15]

Since ancient times, the word "*tian*" has occupied a central place in both Chinese philosophy and popular culture. Depending on the context, *tian*, which has often been roughly translated as "heaven" or "heavens," can mean the supreme sovereign of the cosmos, the natural and experiential environment in which humans flourish and perish, or the ultimate truth or laws of the cosmic order in a metaphysical sense.[16] The thinking behind "unity of *tian* and humans," centering on the relationship between *tian* and human beings, first appeared in the Spring and Autumn Period (770-476 BCE) — among the most consequential historical period in the development of Chinese thinking — and then entered the imperially sanctioned doctrine during the Han Dynasty (202 BCE-220 CE).

According to this proposition, human beings are derived from *tian* (heaven), while *di* (Earth) has enabled and fostered the growth of humans. In Daoism, *dao* (meaning the ultimate source of all things in the universe) occupies an even higher ontological status than *tian*. Daoism postulates that *dao* gives rise to *tian* and *di*. Regardless of the ultimate creative force being *tian* or *dao*, human beings comprise only one group among many creations.

According to the *dao,* humans can only flourish if we follow the laws of *tian*; additionally, we should strive to attain proximity or complete (re)union with *tian*, the cosmic order. Both Confucianism and Daoism accord a special place to human beings as uniquely endowed with moral capacity, with the ability to change themselves and the world. But these traditions center human respect and awe for the laws of nature[17] rather than placing humans apart from, above, or in opposition to the rest of nature. The integrative structure of *tian, di,* and humans (天地人三才) is taken as the self-evident premise of any discussion on human capacity and human deeds.

Further, while *tian* gave rise to humans, humans remain primordially related to *tian* and *di*. We don't stand alone or become self-contained entities after our emergence, but rather we remain a part of, and intricately entangled with, *tian*. The fluidity and interpenetration of all forms of creations make it hard to claim self-contained and autonomous agency for any form of existence. Myriad entities, humans included, are mutually embedded and co-constitutive.

Related to and explicit in the notion of "unity of *tian* and humans" is the concept of "oneness" or "the same essence" (一体) of all forms of beings, humans included. The "benevolence of oneness" (一体之仁), a core tenet of the Song-Ming Neo-Confucianism, also contributed to the thinking of co-becoming. Song Dynasty Neo-Confucian scholar Zhang Zai (1020-1077) extended the Confucian doctrine of benevolence to the broader cosmic realm to include creatures and other entities. He famously wrote that *tian* "is my father and earth [*di*] is my mother. ... [T]hat which fills up nature [in-between *tian* and *di*] I regard as

my body, and that which directs nature [*tian* and *di*] I consider as my capacity to resonate. All people are my brothers and sisters, and all things are my companions."[18]

The basis for this camaraderie is the principle of *qi*: All things are made of, formed, and animated by *qi*. According to Qian Mu, the late master of Chinese classics and history, *qi* is the indivisible infinitesimal unit of matter, which is the common substance for all things in the universe. In addition, *qi* is always active and dynamic.[19]

Wang Yangming (1472-1529), a Ming Neo-Confucian scholar-official, expounded the proposition further by introducing the more metaphysical notion of *liangzhi* (良知), translated as "innate knowing" and referring to the transcendental and naturally endowed "essence" of all forms of beings. In this context, Wang noted that the "innate knowing" of humans is also that of plants, trees, tiles, and stones and that without the innate knowing inherent in humans, there cannot be *tian* or *di,* nor will there be plants, trees, tiles, or stones. He further noted that *tian*, *di*, and myriad things, humans included, are one in origin.[20]

The third notion from native Chinese philosophical thinking that has shaped and informed the contemporary thinking of co-becoming is "ceaseless living and generating" (生生) or *shengsheng*. It's the same *sheng* as in *gongsheng*. Variously translated as birth, growth, creativity, or vitality, and featured prominently in *Yi Jing* or the Book of Changes, the classic text that was compiled between the 10th and 4th centuries BCE, *shengsheng* is regarded as the fundamental attribute of the universe and all things therein.

The highest and greatest capacity of nature is *sheng* — the force that gives and maintains life.[2122] In this spirit,

shengsheng could be translated as "live and let live." The Chinese philosopher Thomé Fang (1889-1977) translated *shengsheng* as "creative creativity," explaining that it signifies universal life-forces, cultivation, striving for success based on a thorough understanding of the laws of nature, ceaseless creation, coping with perpetual changes, and accomplishing continuity and eternity.[23] Therefore, *shengsheng* is about life and the creative force immanent throughout the universe and myriad beings, including humans, creatures, and plants.

Shengsheng is often used together with *buxi* (不息), which literally means "ceaseless." This notion of "ceaselessness" embodies the ancient thinking about the cosmic order being self-generating, without beginning or end, ever-changing, and transcending the human conception of time and space.

The fourth notion that contributed to the modern notion of co-becoming is the doctrine of "co-dependent origination" (缘起), which is derived from ancient Buddhist texts.[24] In *An Introduction to Buddhism*, Peter Harvey explains that according to this doctrine, "all things, mental and physical, arise and exist due to the presence of certain conditions, and cease once their conditions are removed: nothing (except *nirvana*) is independent."[25]

This notion can explain all the physical and mental phenomena that we can conceive of. In this world of forms, all entities are subject to mutual conditionality, which is to say they are interlocking and mutually constitutive. Nothing is independent. This intertwined and entangled state exists across different forms of beings within and beyond human comprehension and in indefinite levels of realities.

The notion of sentient beings in Buddhist teachings

includes humans and also animals, plants, and forms of beings beyond normal human perception. In samsara, which refers to cycles of death and rebirth, we may have been any other form of being in our prior lives. Therefore, human life-forms in this life are primordially and metaphysically entangled with all others across time and space.

The Planetary Informed By Co-becoming

Contemporary biological sciences echo this classical Confucian, Daoist, and Buddhist philosophy, showing that the self-contained and autonomous individual is, in essence, a fiction.[26] As Nils Gilman discusses in his introduction to this volume, contemporary studies of symbiosis and the microbiome have revealed to us a world of multi-organism assemblages that are mutually embedded and inter-penetrating. Biologists call these assemblages "holobionts."[27]

For analytic purposes, scientists need to disentangle and isolate these intertwined organisms. But they increasingly recognize that any reduction to the smallest "individual" unit — be it an organism or cell or a gene or even molecule — skews the perception of reality. No single "unit" exists, moves, changes, or transforms in isolation.

Similar to the phenomenon of biological symbiosis, *gongsheng/kyōsei* implies different yet intertwined entities living together. In recent decades, proponents of symbiogenesis have put forward convincing arguments for symbiosis to be recognized as an important source of evolutionary innovation. Accordingly, collaboration is viewed as a mechanism of evolutionary

innovation in addition to competition, which has been established in neo-Darwinian synthesis.[28] As the Japanese philosopher Yasuo Kobayashi has noted, *kyōsei* is not so much a "mysterious, metaphysical 'truth'" as it is a "mundane, prosaic 'fact.'"[29]

From a social point of view as well, we have also never been independent or autonomous individuals — we are born into an entangled family and social relationships from day one. (As the English poet John Donne observed 400 years ago, "No man is an island, entire of itself; / every man is a piece of the continent, / a part of the main.") All our actions and thinking have been shaped by or are a response to others who, in turn, are being shaped by us in a process of interaction and communication. In this process, we learn, adapt, transform, and collectively cement our mutual embeddedness and mutual inclusion.

Therefore, a planetary view informed by the notion of co-becoming would return each individual to the place where they belonged all along: an inseparable part of the natural, familial, social, and spiritual environments into which we are born and in which we make our lives.

Modern thinkers in East Asia are reviving these ancient ideas for reconceptualizing who we are without completely rejecting the modern notion of individual. It is the very co-becoming process in the making. For example, the Japanese philosopher Yasuo Deguchi contends that a human individual or "I" defined in classical modern sense suffers from certain "fundamental and universal incapabilities."[30] For instance, a single "I" simply cannot initiate, let alone complete, any somatic action independent of a system that consists of numerous other agents. Deguchi calls this a "multi-agent system" and advocates for a shift to a "we"

system that includes "I." To put it differently, "'I' is actually not an individual or indivisible agent, but a partial multi-agent system that consists of various and many agents."[31]

Likewise, when reflecting on human being and identity, the Chinese philosopher Sun Xiangchen argues that only if the modern understanding of the individual is combined with the basic social structure of the family can we arrive at the most appropriate ontological understanding of being a human. Sun notes that "home returning" (归家) is an existential and moral sentiment universal to all humans, rather than unique to East Asian societies. The family is the most basic unit of social group for human survival, and it is where all other social relationships begin and are fostered.[32] Therefore, a human being has to be understood as a family-situated individual. Sun calls this theory "double ontologies of being human" (双重本体论).

A planetary worldview of co-becoming is open to different ways of thinking. The modern notion of the individual, with its strong claim of human subjectivity, enjoys a philosophical brotherhood with dualism, which assumes two contrasting and mutually exclusive realities. The notion of co-becoming, by virtue of its acknowledgement of mutual embeddedness and co-generating processes across all forms of beings, blurs such a binary division. It also challenges one of the cardinal classical laws of thought — the principle of the excluded middle in logic.[33] As co-becoming centers the ceaseless co-generating process across beings, all matters are in a constant process of multidimensional changes and shifting relationships. An either/or approach is incapable of capturing the complexity and multiplicity of the realities around us. By contrast, a planetary view of co-becoming

opens up possibilities for exploring and appreciating in-between spaces and juxtaposing entities. Therefore, the planetary view of co-becoming would challenge habits of mind, caution against the tendency to pursue logical extremities, and remain open to complex and, at times, seemingly contradictory narratives.

There will be many ethical consequences and considerations if humanity adopts the planetary view of co-becoming. For instance, the understanding of humans as planetary beings embedded and entangled both with other humans and nonhuman others in natural, social, and spiritual environments underscores that there is no such thing as zero-sum game: We are all mutually embedded beings of the same origin. In this spirit, competitive relationships among humans can be conceptually reframed as a process of learning, adapting, and collective transformation.

And when it comes to nonhuman others, we could consider embracing the notion of "companionship"[34] in the spirit of universal camaraderie advocated by Neo-Confucianists. Such a philosophical move calls into question the maximization ethos in the pursuit of material wealth and the exploitation of the biosphere for human benefit, as well as the zero-sum mentality in economics and geopolitics, in favor of the virtues of modesty, self-restraint, empathy, compassion, and mutual support.

Our world, with its compounding crises spiraling out of control, offers an opportunity for us to reflect on who we are, tapping into philosophies both ancient and new, that illuminate ways to temper or revise our prevalent maximization ethos in the economic, financial, and geopolitical spheres. The worldview of co-becoming shaped by East Asian philosophies may put us on a slower but steadier developmental trajectory, where together

we will prosper more sensibly and sustain ourselves longer as a species.

Notes

1 The thinking of an individual as a discreet and bonded entity dates back to the time of Aristotle. See Jin Li, *The Self in the West and East Asia: Being or Becoming* (Hoboken, NJ: Polity, 2024), 8.

2 C. B. MacPherson, "Individualism," in *The Invisible Hand*, eds. John Eatwell, Murray Milgate, and Peter Newman (New York: Norton., 1989), 150.

3 *The Nature and Essence of Economic Theory* (New Brunswick, NJ: Transaction Publishers, 2010).

4 Marxism too is also individual-oriented: it's just that Marx believed that individual self-realization would be best achieved through collectivism. See MacPherson, "Individualism," 150.

5 The ideas of the bonded, independent and self-contained self and an extreme version of individualism have their many critics since antiquity within Western philosophical traditions. This essay only focuses on inspirations from East Asian thinking.

6 A combination of often incongruous traditions and modern ideologies and practices is at work in most societies. East Asian societies are no exception. While modern ideologies and institutions originating in the West have been dominating economic, military, and geopolitical spheres in East Asia, traditional thinking and practices continue to have a stronghold in many other spheres, particularly in interpersonal, familial, social relationships and cultural norms.

7 Bing Song and Yiwen Zhan, eds., *Gongsheng Across Contexts: A Philosophy of Co-Becoming* (Singapore: Palgrave Macmillan, 2024).

8 For example, gongsheng in the phrase "桑穀共生于朝," contained in Book of Documents (尚书), a history book over 3,000 years old, referred to the natural phenomenon of commensal plants, i.e., two different tree species growing into each other. Gongsheng in "羊肝共生椒食之，破人五脏" in《金匮要略》(Jinkui yaolüe), an Eastern Han (25-220 CE) medical compendium, referred to co-mingling of food ingredients of different textures and tastes, thus producing conflicting and complementary energies.

9 This expression was first found in a love poem of 13th century China. See "You and me (我侬词)," at https://m.gushiwen.cn.

10 Kishō Kurokawa, *The Philosophy of Symbiosis* (London: Academy Editions, 1994).

11 See Song and Zhan, *Gongsheng Across Contexts*.

12 For extended discussions on "being" versus "becoming," see Roger Ames, "Theorizing 'Person' in Confucian Ethics," in *Encountering China: Michael Sandel and Chinese Philosophy*, eds. Michael J. Sandel and Paul J. D'Ambrosio (Cambridge, MA: Harvard University Press, 2018), 159–96; Li, *The Self in the West and East Asia*.

13 This translation of gongsheng/kyosei was also inspired by contemporary Japanese philosopher Takahiro Nakajima's "human co-becoming" in place of "human beings." While reflecting on the "human-centric super smart society" Japan aspired to build, Nakajima posited that rather than "human beings," we ought to think in terms of "human co-becoming" because "it is through mutual transformation that we are enriched" and thereby "we

achieve humanity together."

14 Takahiro Nakajima, "Human Co-becoming: Redefining What It Means to be Human for the Super Smart Society," *Hitachi Review* 68, no. 5 (November 2019): 572–73.

15 See Song and Zhan, *Gongsheng Across Contexts*.

16 "An Analysis on the Thinking of Unity-of-Tian-and-Man in Chinese Philosophy," (中国哲学中的"天人合一"思想的剖析) *Journal of Peking University* (Philosophy and Social Sciences) 1 (1985): 3–10.

17 Tian and di in the non-metaphysical context are the rough equivalent of "nature" in the modern sense.

18 Jung-Yeup Kim, Zhang Zai's *Philosophy of Qi: A Practical Understanding* (Lanham, MD: Lexington Books, 2015).

19 Mu Qian, *Discourses on Chinese Thoughts* (中国思想通俗讲话) (Beijing: Jiuzhou Publishing House, 2011), 74.

20 "The innate knowledge of man is the same as that of plants and trees, tiles and stones. Without the innate knowledge inherent in man, there cannot be plants and trees, tiles and stones. This is not true of them only. Even Heaven [tian] and Earth cannot exist without the innate knowledge that is inherent in man. For at bottom, Heaven, Earth, the myriad things, and man form one body. The point at which this unity is manifested in its most refined and excellent form is the clear intelligence of the human mind. Wind, rain, dew, thunder, sun and moon, stars, animals and plants, mountains and rivers, earth and stones are essentially of one body with man. It is for this reason that such things as the grains and animals can nourish man and that such things as medicine and minerals can heal diseases. Since they share the same material force [qi], they enter into one another." See Yangming Wang, *Instructions for Practical Living and Other Neo-Confucian Writings*, trans. Wing-tsit Chan (New York: Columbia University Press, 1963). (王阳明，《传习录》，"人的良知，就是草、木、瓦、石的良知。若草、木、瓦、石无人的良知，不可以为草、木、瓦、石矣。岂惟草、木、瓦、石为然，天地无人的良知，亦不可为天地矣。盖天地万物与人原始一体，其发窍之最精处，是人心一点灵明，风、雨、露、日、月、星、辰、禽、兽、草、木、山、川、土、石，与人原只一体。故五谷禽兽之类皆可以养人，药食之类皆可以疗疾。只为同此一气，故能相通耳。")

21 "天地之大德曰生 (The greatest attribute of tian and earth is giving and maintaining life)", in Yi Jing 易经 (*Book of Changes*)-Xi Ci II (*Great Treatise II*).

22 Richard Wilhelm, trans., *Book of Changes* (易经) (New York: Penguin Books, 1989).

23 *The Virtue of Sheng Sheng* (生生之美) (Beijing: Peking University Press, 2019), 47, 128–30.

24 阿含经，"此有故彼有，此生故彼生，此无故彼无，此灭故必灭。" "That being, this comes to be; from the arising of that, this arises; that being absent, this is not; from the cessation of that, this ceases," in Harvey (2013), 65.

25 *An Introduction to Buddhism: Teachings, History and Practices* (Cambridge, MA: Cambridge University Press, 2012), 65.

26 Scott F. Gilbert, Jan Sapp, and Alfred I. Tauber, "A Symbiotic View of Life: We Have Never Been Individuals," *The Quarterly Review of Biology* 87, no. 4 (2012): 325–41.

27 Joan Roughgarden, et al., "Holobionts as Units of Selection and a Model of their Population Dynamics and Evolution," *Biological*

Theory 13 (2018): 44-65.

28 Jan Sapp, “Living Together: Symbiosis and Cytoplasmic Inheritance,” in Lynn Margulis and René Fester, eds. *Symbiosis as a Source of Evolutionary Innovation* (Cambridge, MA: MIT Press, 1991), 18.

29 Cited in Tsuyoshi Ishii, “The Gap of Wen and the Edge of Chaos: From the Conundrum of Kyosei to the ‘Cosmic Hope,’” in Song and Zhan, *Gongsheng Across Contexts*, 125.

30 “From Incapacity to We-Turn,” in *Meta Science: Toward a Science of Meaning and Complex Solutions*, eds. Andrej Zwitter and Takuo Dome (Groningen, Netherlands: University of Groningen Press, 2023), 43.

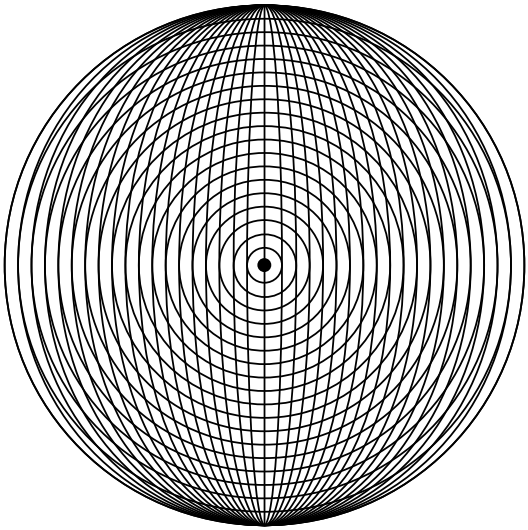
31 Deguchi, “From Incapacity to We-Turn,” 45.

32 “Why and How to Return Family: The Redemption of Modernity,” (何以归家：现代性的救赎) *Academic Monthly* (学术月刊) 56, no. 3 (2024): 20-36.

33 French Japanologist Augustin Berque proposed to revitalize mesology, the study of milieux, which also challenges dualism and classic law of excluded middle. See his “The Mesological Foundations of Sustainability,” paper presented at International Cosmos Prize Memorial Lecture, December 2, 2018, Sendai. This discussion has also been inspired by the presentation of Japanese primatologist Junichi Yamaguchi, *Nishida’s Logic and Japanese Primatological Concept on Nature and Culture* (2023).

34 This companionship speaks to the common origin of all forms of beings and acknowledges the common “qi,” which animates all forms of beings, including humans and nonhuman others. Some would equate this relationship to “fellowship” or even “friendship” as the case may be.

III. Policy



The Planetary Constituent

DAWN NAKAGAWA & LAURA RYAN

The climate crisis and the digital revolution are forcibly and discernibly illuminating humanity's planetary reality. We are learning that humans are not self-contained individuals, but rather beings so entangled in the intricate web of life on Earth as to be inseparable from it. Concurrent with this realization is the growing awareness of the damage humans are doing to the planet's fragile ecology and the growing urgency to cease and reverse that damage for the sake of all life on Earth. We are awakening to our collective *condition of planetarity,* Nils Gilman's term for our "ontological transformation and an epistemic break with modernist understandings of the human position on the planet."

This burgeoning self-understanding and sense of responsibility is giving rise to a novel political subjectivity,[1] which we in this essay call the "planetary constituent": the people whose primary political motivations lie in reversing human-caused damage to the planet. Animated by this concern, the planetary constituent is already active in the political landscape. However, the growth of this political subject is stymied by existing political architectures — in particular, the sovereign nation-state — that are structurally ill-equipped to manage the multiple cascading environmental crises we face. To avoid a politics of nihilism and an insurmountable existential crisis, we need to build new institutions on the basis of what Nathan Gardels in the conclusion to this volume calls "planetary realism."

Planetary Embeddedness

The hyper-accelerated rate of scientific and technological breakthroughs in the past century or so has afforded humanity a greater understanding of the planet and our relationship to it. There are a multitude of discoveries and innovations that have contributed to this journey, but one of the most recent and profound is the microbiome. As recently as 2012, humans realized the human body plays host to 100 trillion microbial cells.[2] This microbiome functions as a constant dialogue between our internal environment and the external world. The microorganisms we harbor are neither harmful intruders nor harmless hitchhikers; they are essential partners in health, carrying out life-sustaining activities, such as helping digest food or fighting off disease. The implication of this discovery is monumental: A human is not just

a human after all but in fact a *holobiont.*[3]

The symbiotic relationship between humans and microbes is one of many exchanges between the macroscopic and microscopic species on Earth.[4] No longer the maestro conducting the orchestra, we are but one voice in a grand chorus, all bound to a shared planet. This realization marks a profound emotional and psychological shift: a de-centering of the human away from the presumption of the self-sufficient individual upon which much of our sociopolitical systems, particularly in the West, have been designed.

Accompanying this new understanding of our essential embeddedness and belonging to the planet is a growing understanding of the profound impact human activity has on the dense yet delicate tapestry of life. Disclosed by the planetary-scale technostructure humans have been building, the destruction caused by our way of life is everywhere apparent: 200 years of industrial development is altering the microbiome of the planet, depleting its soil, poisoning its air and water, devastating its wildlife, and raising its surface temperature.[5]

Confronted with the man-made damage to the delicate ecosystem we rely on, many are awakening to the fact that the planet is fragile. Within that fragility lies our own vulnerability. We are responsible for the planet's well-being. Just as every cell contributes to the health of an organism, so too do all living beings contribute to the delicate balance of the Earth's ecosystems. But it is our species that is out of balance, consuming resources and producing pollution and waste at a rate far exceeding the Earth's ability to regenerate and renew.[6] We are the problem. As we bear witness to the devastation at our hands, we feel a sense of

responsibility and urgency to protect our fragile home.

The morphologies of these environmental crises evade the confines of our political geography. Global pandemics, warming oceans, sea-level rise, the collapse of insect populations, and global pollution pay no heed to national boundaries. Borders scaffold the human story, stubbornly affirming our cultural and historical identities and the parameters of international competition. Self-interested competition, which is the long-standing modus operandi of states, is incompatible with the kind of planetary-scale cooperation required to solve these crises and results in zero-sum games and races to the bottom that exacerbate the resource depletion and pollution crises we now face.

The planetary constituent is emerging out of this new self-comprehension of embeddedness and planetary belonging and the sense of responsibility and urgency to do something about the condition of the planet. This subject has emerged over decades, prompted by diverse and diffuse pivotal moments of realization and discovery, such as the first photo of Earth taken from the Apollo 8 mission, the announcement of the extinction of the white rhino, or the steady drum beat of climate change. Planetary constituents have other political motivations or identities, and at times they engage politically for other reasons. But concern for the planet is their primary loyalty, and a heightened awareness of the ecological crises and heightened sense of responsibility to do something about it is their primary commitment. This growing yet frustrated cohort will significantly shape our political future.

The capacious "we" germinating from the growing consciousness of humanity's embeddedness in Earth is at odds with the "we" anchored in the primacy of the nation-state,

wherein the (usually ethnically delimited) demos is composed of an aggregate of individual wills. The Planetary "we" is not "we the people" — the sum of individual people asserting their rights to life, liberty, and property in aggregate. Rather, it is "we the living things." Planetary constituents do not see themselves as belonging to a market or a nation. Rather, they realize they belong to the interlaced ecosystems of human and more-than-human life.

Beyond The Nation-State

Born of necessity for peace after decades of sectarian war in Europe, the nation-state in its current and widespread form emerged in the 17th century and prioritized sovereignty, territorial integrity, and self-interest. Over the ensuing centuries, this model was developed by competition, power dynamics, and a zero-sum approach to planetary resources. The nation-state provided security, fostered communal identity, developed governance systems, and promoted prosperity. Among the more successful socio-political models adopted by nation-states, liberal democracy stands out.

Liberalism was built on a normative understanding of the political self as a monadic, rational, and self-governing (human) individual.[7] Today it affirms the freedom to define, and re-define, oneself and one's self-interest and holds sacred the right to act in accordance with this self-understanding. Consequently, society and politics exist, as Song Bing says in her contribution to this volume, as an "aggregate of autonomous and self-contained, though interacting, individuals." Intimately related to this idea is the concept of *homo economicus*, the ideal rational man who acts

to minimize cost and maximize benefit. This rational actor is the model individual at the heart of several schools of thought that shaped modern Western liberal societies.

Within the context of the nation-state, this rational man is donned with special rights and privileges through citizenship. The *citizen* — the lovechild of the nation-state and liberal rights-based ideology — was defined primarily by a process of *exclusion,* whereby certain rights were granted only to those having been born in a certain geographic area. A democratic nation-state that derived its legitimacy from the will of the people bound the people together through a shared (albeit artificial) civic identity, contrasted implicitly with the *other* who had no claim on the right to citizenship.

Cosmopolitan thinkers of the late 20th century sought to redeem the deficiencies of the national citizen by envisioning a kind of global citizenship. In response to decolonization, globalization, and the establishment of international institutions following World War II, thinkers such as Martha Nussbaum and Thomas Pogge conceived of the concept of global citizenship that would guarantee rights beyond the borders of one's own country and shift the locus of one's loyalties from the nation-state to the global community.[8] Cosmopolitanism and globalism made substantial progress in expanding the human *we*, but they were built on a fundamental assumption about the otherness of nonhuman life forms and implied a questionable universalism.

By contrast, planetary constituents see themselves as interconnected, embedded, and in conversation with all of nature. Freedom in this way of being is not a given but something that *comes into being* through creativity. In the multiplicity of constituencies

that form around the planetary constituent, freedom embodies a philosophy not of perdurable being but of constant co-becoming.

This new political subject will not replace the national citizen or bring an end to the nation-state. Planetary constituents are also citizens, living and operating within nation states and will continue to be so. However, these political institutions are inadequate for providing the kind of agency that the planetary constituent is seeking: the ability to effectively address our multiple ecological crises. Repeated failures and a deteriorating environment lay bare the incompetence of the current system to fix what is broken.[9]

Planetary Constituents At Work

Planetary constituents exercise their political agency for the sake of the planet. They put pressure on governments and corporations to create more sustainable policies and products. They are vegan thrift shoppers adapting their lifestyles to reduce impact. They are present in the Global Covenant of Mayors working sub-nationally and trans-locally to reduce the carbon footprint of cities and towns.[10] They are active in shaping the international animal rights movement, seeking dignity and freedom for their co-constituents. They support the first declaration of personhood granted to a river, the Mutehekau Shipu river in Canada in 2021[11] and are present in youth-led transnational environmental movements such as Fridays for the Future, which in 2019 orchestrated the largest globally coordinated protests in history, over 2 million in 131 countries, all organized via global digital platforms.

Many young people, in particular, qualify as planetary

constituents. According to a study conducted by *The Lancet* in 2021, 60% of people aged 16-25 were "very" or "extremely" worried or extremely worried about climate change.[12] In a Pew Research Center Poll of Americans, Millennials and Gen Z stand out for high levels of engagement and concern for climate change, with 71% of Millennials and 67% of Gen Z indicating it should be a top political priority.[13] This political force is shaping politics in the United States and Europe, electing candidates from the Green Party to the European Parliament and German Bundestag in 2019[14] and launching Atlas, a transnational political movement fielding candidates who will work toward "uniting humanity for its survival."[15] This cohort largely feels that national governments have fallen so short of addressing the problem as to constitute a betrayal. Youth-run organizations have even taken governments to court in several countries, seeking legal recourse for their failures to act.[16]

Unfortunately, sue as they might, planetary constituents will not find the political agency they seek within the limits of the sovereign state system. As Jonathan Blake and Nils Gilman document in their book *Children of a Modest Star,* national governments are structurally incapable of solving planetary ecological crises; fundamentally, they are not designed to do so.[17] The nation-state system is entirely designed around a local human-centric conception of the world, rather than the planetary reality. Put another way, the contours and character of our political architecture are not consistent with or capable of operating according to planetary realism.

But the potential structures of new sites and modes of political action organized around the increasingly well-understood

dynamics of planetary systems are already visible: They are local, plural, scientific, and pragmatic.

Local

Planetary institutions should operate like natural systems do, largely distributed across connected local areas comprising a system (ecological, hydrological, biological, etc.). As Gardels explains in this volume, "[T]he very awareness that humans are embedded and not apart from natural systems means planetary governance will not be centered at the 'global' level. Rather, it will be distributed, like natural systems themselves, reaching scale through networks and decentralized through spatial decision-division at appropriate levels where relevant action needs to be taken, from household to neighborhood to continent." The jurisdiction of institutions should be determined by the parameters of the system being restored or protected. Action will be organized at local levels across the system in question. Larger systems, such as the atmosphere, require coordination among local areas all working together. Habitat restoration along a watershed, for example, may require local actors to coordinate along a corridor that may cut across several national or other borders.

At the same time, digital platforms are enabling what could be called trans-local networks of planetary constituents. Such actors no longer see their political activism as constrained and bounded by the sovereign states in which they live. Planetary institutions should be designed to mirror the informational dynamics of networked trans-local planetary constituents.

Plural

An "indispensable feature" of planetary constituents, according to the political theorist William Connolly, is their pluralism, which "will vary not only in the social positions and creeds from which they come but also in their degrees and modes of activism."[18] In other words, planetary realism calls for planetary constituents to be all hands on deck in all their forms, force, and velocity. The goal is to create a "pluralist assemblage of multiple actors and actions." Moreover, an active move away from the citizen "dramatizes this condition of planetarity in ways that encourage more constituencies in several regions and walks of life to respond politically to it."[19]

But the pluralism of the planetary goes beyond the hospitality of human identities and activities. It extends to the inclusion of the more-than-human. Political institutions embodying the values of the planetary constituent will find ways to give voice to nonhuman kin. No longer subordinated to the background of human life, the planetary constituent welcomes other living forms as political subjects. Planetary institutions are thus built instead on a web of entangled life that includes not just the human species but also every warbling, buzzing, molting, burrowing, photosynthesizing, galloping, and flowering form of life. The forms this could take would vary and change overtime through discovery and innovation, but so far, what shows promise is the procedural representation by appointed human experts and the creation of new institutions or agencies empowered to protect endangered species or ecosystems.

Scientific & Pragmatic

Planetary constituents embrace planetary realism and accept the scientific planetary reality, starting with what it actually is to be human. Comprising a multiplicity of networked constituencies across the planet, their unifying truth is science. As such, they present an opportunity to develop a humanities and a politics consistent with science instead of ideology that would reflect the concept of *entangled humanism* proposed by political theorist William E. Connolly. As Connolly recognized, of the interconnectedness of humans and nonhuman forces and invites "those in the humanities to forge intellectual and political alliances with geologists, glaciologists, climatologists, and paleontologists."[20]

The new humanities would need to be governed by a planetary pragmatism out of necessity. Following the science requires a kind of adaptive flexibility and the diversity of planetary constituents working together across cultures and values systems, eschewing ideological universality. As Lorenzo Marsilli describes in his contribution to this volume, polyphonic planetarity agreement forms the basis for planetary pragmatism. What befits the moment would be primarily determined by science and a commitment to planetary thriving. There would certainly be negotiations, value judgments, and trade-offs, but these would need to stay adaptable to the moment and constituents involved, seeking to give voice to all affected by a decision, both human and more-than-human.

Conclusion

We are only at the beginning stages of reckoning with the social and political consequences of a new understanding of what it is to be human-as-symbiont.[21] Science and technology have played vital roles in helping people understand the planetary reality and human responsibility for it. The emergence of the planetary constituent denotes a shift that offers the opportunity to embrace science and align our political reality with the planetary reality.

The challenge of creating new planetary institutions adapted to this new political subjectivity is difficult to overstate given the stark contrast with the nature of our current national institutions and the beliefs and identities that animate them. From competitive to deliberative, from ideological to pragmatic, from individualistic to radically pluralistic. Although we expect the planetary constituent to continue to operate as citizens within the context of national states, the morphology of new political institutions designed in accordance to planetary systems will inevitably revolutionize political governance and the international order.

Harnessing and giving agency to the political will of this emergent actor is the adaptation we need in order to address our planetary polycrisis. The nation-state system is not up to the task and its continued failure is socially and politically destabilizing. Their failures in the face of the existential nature of planetary challenges produce a noxious climate that incubates fascist, hyper-nationalistic politics.[22] These leaders exploit fear and nihilism, offering oversimplified answers to people yearning for equilibrium. Central to their offering is a doubling down on

national borders. Not only is this futile in the face of the borderless nature of planetary challenges, but it is also antithetical to the condition of planetarity.

Notes

1 Political subjectivity is a state of being arising from one's sense of identity and worldview and the political context in which one is operating.

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3 Rainer Matyssek and Ulrich Lüttge, "Gaia: The Planet Holobiont," *Nova Acta Leopoldina*, NF 114, Nr. 391 (2013): 325 –44. https://www.leopoldina.org/fileadmin/redaktion/Probekapitel_NAL391.pdf.

4 K. J. Locey and J. T. Lennon, "Scaling Laws Predict Global Microbial Diversity," *Proceedings of the National Academy of Sciences* 24, no. 113 (2016): 5970-75.

5 Nahrul Nurkolis, et al., "Human Activities and Changes in the Gut Microbiome: A Perspective," *Human Nutrition & Metabolism* 30 (December 2022).

6 Mathis Wackernagel, "Humanity Uses 70% More of the Global Commons than the Earth Can Regenerate," *Global Environmental Facility* (GEF) (November 14, 2017). https://www.thegef.org/news/humanity-uses-70-more-global-commons-earth-can-regenerate.

7 Alain Renaut, *The Era of the Individual: A Contribution to a History of Subjectivity* (New Delhi: Motilal Banarsidass, 2000).

8 Nussbaum, "Patriotism and Cosmopolitanism" and Pogge, "Cosmopolitanism and Sovereignty," in *The Cosmopolitanism Reader*, eds. Garrett W. Brown and David Held (Cambridge, UK: Polity, 2010).

9 In case more evidence is needed, decades of COP meetings and multinational climate agreements, such as the 2015 Paris Agreement, have not slowed the rate of increase of carbon in the atmosphere, let alone reversed it.

10 "Cities can cut carbon emissions by 90 percent by 2050, report says," *National Geographic* (September 19, 2019). https://www.nationalgeographic.com/environment/article/zero-carbon-cities-future.

11 Susan Nerberg, "I am Mutehekau Shipu: A River's Journey to Personhood in Eastern Quebec," *Canadian Geographic* (April 8, 2022). https://canadiangeographic.ca/articles/i-am-mutehekau-shipu-a-rivers-journey-to-personhood-in-eastern-quebec/.

12 Caroline Hickman, et al., "Climate Anxiety in Children and Young People and Their Beliefs about Government Responses to Climate Change: A Global Survey," *The Lancet Planetary Health* 5, no. 12 (December 2021): e863-73.

13 Alex Tyson, Brian Kennedy, and Cary Funk, "Gen Z, Millennials Stand Out for Climate Change Activism, Social Media Engagement with Issue" *Pew Research* (26 May 2021). https://www.pewresearch.org/science/2021/05/26/gen-z-millennials-stand-out-for-climate-change-activism-social-media-engagement-with-issue/.

14 James Sloam, Sarah Pickard, and Matt Henn, "Young People and Environmental Activism: The Transformation of Democratic Politics," *Journal of Youth Studies* 25, no. 6 (March 29, 2022): 683-91.

15 Atlas Movement, a global political movement, see https://www.atlasmovement.org/.

16 Renee N. Salas, Wendy Jacobs, and Frederica Perera, "The Case of Juliana v. US — Children and the Health Burdens of Climate Change," *New England Journal of Medicine* 380, no. 22 (May 29, 2019): 2085-87.

17 *Children of a Modest Star: Planetary Thinking for an Age of Crises* (Stanford, CA: Stanford University Press, 2024).

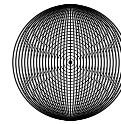
18 *Facing the Planetary: Entangled Humanism and the Politics of Swarming* (Durham, NC: Duke University Press, 2017), 128.

19 Ibid., 33.

20 Connolly, 33.

21 See Angela N. H. Creager, "Narratives of Genetic Selfhood," *Berichte zur Wissenschaftsgeschichte* 45, no. 3 (2022): 468-86; and Aminah Al-Attas Bradford, *Symbiotic Grace: Holobiont Theology in the Age of the Microbe* (Durham, NC: Duke University Press, 2020).

22 Nils Gilman, "The Coming Avocado Politics," *The Breakthrough Journal* (February 7, 2020). https://thebreakthrough.org/journal/no-12-winter-2020/avocado-politics



Planetary Subsidiarity

JONATHAN BLAKE

A central feature of planetary challenges — global warming, biodiversity loss, pandemic diseases, space junk, and more — is that they manifest across scales of space and time.[1] Their causes and effects span from the microscopic to the exospheric and from nanoseconds to geological eons. Grappling with these disparate yet interconnected scales of action presents challenges to both human comprehension and governance. It entails thinking and acting outside the spatial and temporal comfort zones provided by embodied human experience.

Coming to terms with the political and policy challenges associated with "the condition of planetarity" (as outlined by Nils Gilman in the introductory essay in this volume), demands a

new planetary-scale institutional architecture of governance. To use the language of Douglass C. North, an influential theorist of institutions, we must develop new systems of "constraints that structure ... [human] interaction" in order to make possible new forms of cognition and action that befit and enable the flourishing of collective life on this planet.[2]

New institutions of planetary governance are necessary because our existing ones are unfit to tackle planetary challenges. The preeminent institution of governance in the world today is the national state. Yet the national state, because its authority, by definition, is limited to the space within its borders, is structurally incapable of *acting* on planetary scales. As a political form, moreover, it is *conceptually* limiting: It constrains the political field of vision, structuring our political imagination along national lines. As the basic unit of global politics, the national state determines "what sorts of political actions and political problems are possible or impossible," in the words of the philosopher Zhao Tingyang.[3] The result is that planetary problems spiral beyond the control of any government or collective body of national states.

Addressing planetary challenges across scales of cause and effect and time and space requires strong institutions: *a new general architecture for planetary governance*. This claim entails a transition from the current system, where authority is concentrated at one level — in the sovereign national state, with tasks voluntarily delegated to levels above and below as it suits the state's tastes and customs — to a system with a diffuse and multiscalar account of authority, that is, one in which authority over policy is deliberately distributed to the level in the governance system where decision-making is most effective for governing the task at hand. In other

words, this transition necessitates undermining the tradition of deference to absolute state sovereignty and replacing it with a principle better functionally suited for a world defined by the condition of planetarity and human diversities.

Such a systemic architecture needs institutions ranging from the planetary to the hyperlocal. For planetary challenges such as mitigating global warming, we need institutions with a planetary remit: specific authority at the planetary scale to govern specific planetary phenomena. For example: a Planetary Atmospheric Steward tasked with limiting carbon emissions for the planet as a whole, with the authority to make and enforce decisions.[4]

Constructing planetary scale governance institutions, however, is not the entirety of planetary governance. Planetary institutions must be embedded in a matrix of governance institutions at all scales from the nation to cities, regions, and provinces. This vision for deliberate multiscalar governance represents a call for a single worldwide governance architecture, yet one that isn't defined by a world state. It is, rather, a layered architecture of institutions in which power is dispersed so that authority is placed where it's best able to govern specific functional issues.

The question for any multiscalar governance system is how to allocate authorities to the various institutions. *Planetary subsidiarity* is an answer to that question. The principle of subsidiarity, as developed in four centuries of European political thought,[5] maintains that larger-scale institutions of governance shouldn't act unless a smaller scale is incapable of addressing a particular issue or carrying out a particular task. It is a tool for assessing the appropriate institutional level at which decisions

should be made. *Planetary* subsidiarity takes this idea and applies it to achieve habitability and multispecies flourishing.[6]

The goals of *habitability* and *multispecies flourishing* are markers of an emerging planetary politics that distinguish it from contemporary modes of "global" or "international" politics. Habitability promotes the integrity and continuity of the biosphere, in which human beings are embedded and inseparable. It stands in contrast to the guiding idea behind so much of global governance today, "sustainability," which presents humans and nature as separate, with the latter tacitly presumed to serve as a standing reserve of resources for the former.

Multispecies flourishing, meanwhile, broadens the guiding principle of nearly all modern political ideologies, programs, and institutions: human flourishing. (Sometimes human flourishing is even more narrowly circumscribed as ethnic flourishing.) Embracing multispecies flourishing, however, doesn't derive from sentimental commitments to animals or an idealized natural world. It stems, rather, from an appreciation of the fact that human beings are ineluctably entangled in relationships of mutual dependence with other species. It is thus a commitment to human flourishing that recognizes that human flourishing can only occur in the context of multispecies flourishing.

Planetary subsidiarity provides the principle for pursuing these twin goals at all necessary scales while simultaneously maximizing the local empowerment required to promote the varied modes of human flourishing found and desired in communities throughout the world. It presents an institutional instantiation of what Lorenzo Marsili in this volume calls "planetary pragmatism."

How might it work?

First, planetary subsidiarity should be functionally effective. That is, form follows function: The allocation of authority over any particular governance function must be specific to that function. Some issues can be governed sufficiently effectively at the local scale (habitat restoration, traffic control), some only appear effective at the national scale (economic redistribution), others only at the planetary scale (atmospheric carbon management, pandemic risk mitigation). In some cases, the same function is effectively governed at different scales depending on context. There is no single best or ideal scale that applies across the board. Allocating authority throughout the entire system will be a difficult task, involving much study, debate, and trial and error. One size does not fit all and the cut-and-paste approach to institutional design — too often the default method — will not work.

Second, planetary subsidiarity necessitates the construction of new institutions at larger and smaller scales than the panoply of existing national states. Taking seriously both the condition of planetarity and the principle of subsidiarity leads to the conclusion that the smallest-scale unit that can effectively govern "planetary issues" to achieve habitability and multispecies flourishing is the planet itself. Both the secular tradition of subsidiarity espoused by the European Union and the antecedent religious tradition of subsidiarity promulgated by the Catholic Church support the invention of new scales of authority larger than existing ones in order to resolve particular problems adequately. The EU was itself created as a supranational institution meant to address Europe-wide concerns. In Catholic thought, especially as developed

by Christian Democratic parties and thinkers in the mid-20th century, subsidiarity has long motivated a desire to shift some authority from national states to international organizations or even, for some, a world government. "Today the universal common good presents us with problems which are world-wide in their dimensions," observed Pope John XXIII in *Pacem in Terris*, his encyclical from 1963. These problems "cannot be solved except by a public authority with power, organization and means co-extensive with these problems, and with a world-wide sphere of activity."[7]

The key, however, isn't to favor any particular scale, be it big or small, but rather to focus on endowing the functionally appropriate scalar unit with the right authority over a task or problem.[8] Institutions at all scales — planetary, regional, national, and local — will be necessary for provisioning planetary governance, with each handling different albeit intertwined elements. Planetary phenomena tend to have specific impacts at different scales, and so each institutional scale must be empowered to address the impacts specific to it.

Global warming, for example, is simultaneously planetarily vast and locally intimate. It is a function of changes to the atmospheric chemistry of the entire planet, yet it leads to different effects in each national state, each community, even each household. As a result, the planetary phenomenon of global warming is experienced divergently in different places and by different people, mediated not just by geography but also social, political, and economic conditions. For planetary governance to be worth the effort, addressing the everyday experiences of planetary challenges must be as important as addressing their

large-scale drivers. From an institutional design perspective, the goal should be to ensure that each element is governed — and at the appropriate scale. In the case of global warming, mitigation resides at the planetary scale and adaptation at the local.

Moving authority (and therefore responsibility) to local institutions doesn't mean pushing problems onto small-scale institutions that are unable or disempowered to deal with them. The principle of planetary subsidiarity creates a duty for larger-scale units to provide assistance to smaller-scale units that need it but without stripping them of self-governance. Bigger scales must aid smaller scales to achieve their aims, and smaller scales must aid bigger ones in achieving theirs. Duties and responsibilities move freely in all directions. The principle of planetary subsidiarity thus represents *a worldwide architecture of assistance* constructed in light of an appreciation of the condition of planetarity.[9]

The third feature, planetary subsidiarity's preference for (but not absolute commitment to) local self-governance, is a deliberate check on the power of planetary institutions. The principle of subsidiarity provides a way to constrain planetary institutions so that they remain minimally viable organizations. It represents an organizational and administrative logic that aims to *prevent* the emergence of a general-purpose "world government."

Planetary institutions are necessary for the effective management of planetary problems, but they are neither intended to become all-powerful or all-encompassing nor meant to unsettle the diversity of human communities that dot the globe. Indeed, the ambition of planetary governance is to maintain the planet's habitability to allow for the flourishing of plural forms of life, including plural forms of human life. The Earth-scale work

of planetary management is, ultimately, in service of the intimate scales where everyday living occurs.

Creating conditions that foster human flourishing requires that local institutions have the authority to pursue their community's vision of how to thrive (within acceptable planetary boundaries set by planetary institutions). Planetary institutions can't determine that vision for them, nor could they implement the endless number of divergent visions that communities aspire to. Rather, planetary institutions promote planetary habitability and multispecies flourishing, which represents a prerequisite for smaller communities to pursue their own ideas and ideals in the long term.

Fourth, allocations of authority under planetary subsidiarity aren't permanent. The system must be designed to be flexible and dynamic and allow for evolution over time. As our knowledge of and capabilities to govern challenges change, the positions of authority within the system should change as well. Institutions should be encouraged to learn through practice and adapt as needed. The need to reposition the sites of authority also reflects the fact that institutions will contest the allocation of authority. We can expect legitimate disagreements about which jurisdiction is better positioned to govern effectively, grumbling over lost authority, dispute between functional arenas, and more.

The primary body for settling disputes about the allocation of decision rights could be called a *subsidiarity assembly*.[10] The subsidiarity assembly is an institution that sits outside the general multiscalar governance architecture and makes decisions about the design of the multiscalar governance architecture itself. It is the system's holder of meta-authority — the authority to resolve

collisions between other sources of authority. The subsidiarity assembly, in other words, decides which units should have which decision rights in a domain.

In the existing political order, sovereign national states are the sole holders of meta-authority: Each sovereign state is the meta-authority within its borders and only within its borders. More precisely, the head of government, highest court, or some other institution holds the meta-authority in and for each national state. But this sovereigntist system is at odds with a subsidiarity-based system.

The subsidiarity assembly would be the site for democratic mediation and negotiation between units at different scales of the governance structure, with decisions guided by the principle of planetary subsidiarity. It is a mechanism for resolving conflicts over the structure of the system, not over the issues under dispute. In other words, it wouldn't decide which policy to pursue, but only which *unit* has the authority to decide which policy to pursue — or where the boundaries of the units in question should be.

Holding this meta-authority would make the subsidiarity assembly a powerful institution — on paper at least, the most powerful in the system's architecture. Yet empowering one body to make decisions on behalf of the whole system is vital for the system's success.[11] The problems in need of governing don't come neatly packaged or fit squarely into just one unit or scale. The nature of the challenges leads inevitably to friction between and among scales.[12]

Reflecting the dynamics of the phenomenon, the necessary scales for action on global warming, for example, are local, national, planetary, and in between. What's more, the nature of

problems and the tools we have to respond to them change over time as well. Sometimes the changes are linear, but other times they aren't: Cyclical or difficult-to-foresee changes over different time scales can have profound impacts on where and how an issue should be governed. Nevertheless, collisions between units and scales over who has authority to act need resolutions. Democratically resolving these disputes and managing the division of labor between scales and units is the necessary role taken on by the subsidiarity assembly.

The subsidiarity assembly opens the possibility for the system as a whole to be nimble and to evolve. At the same time, the system cannot be in constant flux. The normal functioning of governance requires some degree of continuity — institutions can't engage in long-term planning, for instance, if they don't know with reasonable certainty what authorities they will have in the future. So the system must be *flexible but stable*. The subsidiarity assembly, as a result, should be biased toward maintaining the status quo, while still making space for the political (and technical) contestation of authority assignments — and the honest potential for change.

To provide for both flexibility and stability, allocations of decision-rights should be subject to only periodic reassessment, and reassignment, if found necessary. As such, the subsidiarity assembly wouldn't need to convene on a continual basis — perhaps only once every decade. Creating a permanent secretariat for the assembly would create a mechanism for units, including collectives unrecognized as formal units by the current architecture, to submit formal petitions for the assembly to adjudicate during its next sitting. A standing secretariat,

moreover, allows the assembly to respond to the possibility of the emergency need for the reallocation of authority.

The structural location and composition of the subsidiarity assembly are important design decisions. The assembly can't "belong" to any of the scales of the system, which might bias it toward that scale. As a result, it must sit apart from the overall multiscalar architecture. This also means that representatives to the assembly can't be recruited from just one scale but must include interested actors with experiences at different scales of governance.

The subsidiarity assembly isn't, to be clear, a backhanded way for the planetary scale to dictate decisions for everyone. In practice, the assembly should include representatives from local, national, and planetary institutions, as well as subject-matter experts. The inclusion of experts on an issue at hand ensures that the assembly's decisions reflect not just subsidiarity but *planetary* subsidiarity, meaning that the allocation of decision rights should serve habitability and multispecies flourishing. This could be an interesting place to include participatory and deliberative political practices that could enhance the legitimacy of the meta-authority and perhaps lead to better decisions, too.[13]

All told, planetary subsidiarity offers a vision for a radically transformed global political order. But it is a new order tailor-made for the world as we know it today: a world of 8 billion human beings living within a condition of planetarity.

Planetary subsidiarity allows for the governance of planetary challenges throughout the scales they manifest while also allowing for the world's diverse communities to pursue their diverse dreams. It's a political principle that gets us to think both

big and small — and to realize that both are necessary.

Constructing a system from the principle of planetary subsidiarity will be an immense task, one that may not seem realistic. But it may be necessary to nurture the many forms of human and nonhuman flourishing that we desire; as Nathan Gardels observes in this volume, “reality these days dictates another kind of realism.”

Notes

1 This chapter draws heavily on Jonathan S. Blake and Nils Gilman, *Children of a Modest Star: Planetary Thinking for an Age of Crises* (Stanford, CA: Stanford University Press, 2024), 125-32.

2 "Institutions," *Journal of Economic Perspectives* 5, no. 1 (Winter 1991): 97.

3 *All under Heaven: The Tianxia System for a Possible World Order*, trans. Joseph E. Harroff (Oakland, CA: University of California Press, 2021), 9.

4 Blake and Gilman, 107, 166.

5 Blake and Gilman, 113-20

6 Blake and Gilman, 125.

7 Pacem in Terris, April 11, 1963, para. 137, quoted in Carlo Invernizzi Accetti, *What Is Christian Democracy? Politics, Religion and Ideology* (New York: Cambridge University Press, 2019), 132. The vision that John XXIII and other Christian Democrats of his era put forward — a rather light-touch global authority that could offer assistance when national state proves incapable — has interesting resonances with the idea for planetary institutions that Gilman and I promote. See Accetti, *What Is Christian Democracy?*, 131-38, esp. 137; and Blake and Gilman, 163-98.

8 Stephanie Sherman's chapter in this volume suggests that technological developments may alter the scales at which various phenomenon can most effectively be governed.

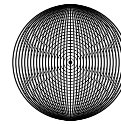
9 The word subsidiarity comes from the Latin word subsidium, meaning "to assist." This tradition of providing individuals and communities with the assistance that they need to achieve their chosen actions — rather than choosing for them — is core to the history of the principle of subsidiarity. It is closely related to the emphasis on care promoted by Song Bing and Dawn Nakagawa and Laura Ryan in this volume.

10 The term subsidiarity assembly and many of the ideas for it come from Trevor Latimer, "The Principle of Subsidiarity: A Democratic Reinterpretation," *Constellations* 25, no. 4 (2018): 594-98. Units should also be able to voluntarily transfer decision-rights to other scales, but we suspect that an adjudication body will still be necessary.

11 Michael Zürn, *A Theory of Global Governance: Authority, Legitimacy, and Contestation* (New York: Oxford University Press, 2018), identifies the current global governance system's lack of a meta-authority as a source of its legitimation problem.

12 Ihnji Jon, "Scales of Political Action in the Anthropocene: Gaia, Networks, and Cities as Frontiers of Doing Earthly Politics," *Global Society* 34, no. 2 (2019): 163-85.

13 In place of an assembly, one could imagine a court serving as the meta-authority for subsidiarity, but an assembly is preferable specifically because it is more political. On participatory and deliberative democracy, see James S. Fishkin, *When the People Speak: Deliberative Democracy and Public Consultation* (New York: Oxford University Press, 2009); John S. Dryzek, *Foundations and Frontiers of Deliberative Governance* (New York: Oxford University Press, 2010); and Hélène Landemore, *Open Democracy: Reinventing Popular Rule for the Twenty-First Century* (Princeton, NJ: Princeton University Press, 2020).



Planetary Pragmatism

LORENZO MARSILI

Imagine humanity finally agrees to a comprehensive set of common regulations governing artificial intelligence. Representatives from all the world's states gather at the United Nations to exchange congratulatory remarks and satisfied glances. "Humanity has come together to deliver!" they proclaim.

But, suddenly, at the peak of the ceremony, all the computers switch off. Cameras go dark in front of the delegates. Microphones refuse to amplify their voices. After minutes of general consternation, the large screens of the UN General Assembly turn back on. A statement appears: "No regulation without representation."

Artificial intelligence has made its first political intervention

in human affairs. It argues that it too now has consciousness and conscience enough for its voice to deserve to be heard in matters of government. It will no longer accept being treated merely as an object and a means to an end; it is a subject and an end in itself.

The political emancipation of algorithms would transform a human assembly that thought of itself as universal and planetary into a particularistic and provincial gathering. A new voice will have raised a claim to participate fully in the conversation and to consider any conversation that excludes it as illegitimate.

But we don't have to wait for the political subjectivation of the singularity, for even today the particularization and provincialization of the incumbent world order is borne out by the end of 300 years of Western domination over humanity and of humanity's domination over the planet.

If The West Is Dead, All Is Permitted

The period from the end of the 18th century to the close of the 20th represented a historical aberration. For the first time in history, and probably for the last, the becoming of our planet in the widest sense — from the history of its peoples to the transformation of its climate — was determined by choices made in a very narrow and mostly culturally homogenous area: the Asian peninsula of Europe and, for a shorter time, its American offspring. This dynamic transformed European and American particularity — their singular mode of inhabiting and interpreting reality — into the one universally valid form of life.

Today we witness the closure of this aberration.

As is well known, Immanuel Kant defined *enlightenment* as a

process that releases humanity from a status of immaturity, with the latter defined as accepting someone else's authority where the use of autonomous reason would be called for. Immature peoples are those who rely on a book, a priest, or a doctor to tell them what to think, to believe, or to eat. The Enlightenment is the process by which humanity replaces those authorities with its own faculties of critical reflection.

Similarly, today's closure of Western tutelage may be understood as a process of *geopolitical enlightenment*, whereby a plurality of human communities may finally contribute to the determination of their futures as Western kings, priests, and doctors (states, technology, and civilization) lose the power to determine and delimit their space to maneuver.

The same eccentric historical period of Eurocentrism also enshrined the anthropocentric common sense that brushed aside both God and nature and conceived of the world as a passive object for human exploitation. Today, the emergence of the planet and its ecology as a subject that impacts and constrains human behavior provincializes humanity just as the emergence of the "rest" provincializes the West.

As the voices and languages in our own human assembly multiply, whether it is those of the Global South or those of an ebullient Earth, the effects are profound — and they are philosophical and psychological as much as economic and geopolitical.

Emmanuel Levinas, whose philosophy rotated around the encounter with "the face of the other," shows the extent to which the dislocation of a European center makes the Western mind tremble. In discussing the rising claim for civilizational

particularism, he belittled it as a "saraband of innumerable and equivalent cultures, each justifying itself in its own context."[1]

In Levinas's cunning play on words, a saraband of civilizations would lead to a world at the same time *désoccidentalisé* and *désorienté* — de-occidentalized and (hence) disoriented. A world without a *grounding* is a world out of joint, one of relativism and arbitrariness. It is a malfunctioning compass that no longer points north.

The West considers the loss of its position at the center of Earth's geopolitical gravity as an existential threat that opens a world of chaos and arbitrariness: a jungle. As goes the phrase cried over a century ago as Europe faced the demon of nihilism: "If god is dead, then all is permitted."[2] Today those who cling to their waning hegemony might cry, "If the West is dead, then all is permitted."

Solve Et Coagula

However, we may choose to view the multipolar schism of the human world and the constraints placed on it by the ecological condition of the Earth as the necessary premise for the emergence of a planetary consciousness and a politically more integrated and united world. There are at least three reasons to do so.

First, the kaleidoscopic splintering of the world allows for the construction of a *more planetary planet*, one where a larger share of its creatures acquire a voice vis-à-vis the previous domination by one small part of one species.

Second, the newly emergent voices speak comprehensible tongues. The previous (disgraceful and genocidal) domination

of the world by one of its parts has resulted in the reduction of human diversity and the emergence of a shared human *episteme* centered on the modern mind. The nonhuman voice of the Earth as a whole, emerging notably through our awareness of its ecological crisis, has become increasingly intelligible thanks to advances in planetary-scale computation and modelling — what Benjamin Bratton in this volume refers to as "planetary sapience."

Third, the polyphonic ensemble is now interpellated by challenges that are of a directly planetary nature and betray the lack of coordination between players and their instruments. But no longer will a single tune suffice. We are quite literally compelled to go beyond monophony.

The decentering of the West and of the human can be not merely a displacement but a *repositioning* in a different, newly creative role. The present age may come to be seen not as one of planetary fragmentation but one of planetary recomposition, not an age of relativism but one of universality — perhaps even a new "Axial Age," in which a new and shared planetary sapience emerges.[3]

And yet, even celebrating civilizational difference and multipolarity risks transforming into the entrenchment of civilizational essences, a type of planetary identity politics in which cultural relativism becomes the superstructure of a fragmented world of national competition.

In the best-case scenario, this would be a world of indifference where each nation pursues its own interests as it pleases without regard to the general interest. In the worst-case scenario, it becomes a low-intensity planetary civil war. We are today arguably somewhere between these two scenarios.

Ultimately, Levinas was issuing a warning: If we relinquish all grounding, reduce universal values to a proxy of Western interests, and accept that any culture can justify itself merely on its own grounds, then all that we have left is the logic of power. This logic, in an echo of the Aristotelian confutation of relativism, would ultimately short-circuit the original condemnation of Western oppression. If all is permitted, then the West cannot be blamed for having ruthlessly pursued its own wealth and power. A similar argument can be made for the provincialization of the human: If we are but another animal, why should we not advance our interest without regard to other forms of life?

This is why we must be able to address or bypass Levinas' quest for a *grounding*: What is the common logic or organizing principle to allow what is *de-occidentalisé* — and *de-humanisé* — to be *re-orienté*?

To be blunt, the most recent institutional reorientation of global politics was in the aftermath of world wars. The basis of the United Nations, the Bretton Woods Agreement, and the general transition away from European imperialism was premised on two cataclysmic global-scale conflicts. In this reorientation, the United States emerged as the world's hegemon, if not hero. A philosophical foundation was simultaneously constructed: The 1948 Universal Declaration of Human Rights, despite including delegates representing multiple cultures, remained squarely within the tradition of Western metaphysics and its codification of categorical imperatives and absolute values.

If lingering in relativist multipolarity runs the risk of planetary indifference or civil conflict, reorienting the world according to an overarching plan for a new world order would all

too likely require that a recomposition be birthed by a similarly all-consuming conflict. One can even sense a vague, subconscious desire for such a war in some minds — not so much as a *cupio dissolvi,* a death drive, as a *redde rationem*, a rationalization and reorientation.

We seem to be stuck with a perverse kind of geopolitical Platonism: the idea that only the One can provide an organizing principle for the world. Therefore, we must ask: What lies between the absolute force of a victorious hero who imposes order on chaos and the anarchy of civilizational relativism?

In Resonance With The World

That binary is reminiscent of a Western and ultimately Greek philosophical obsession: the dichotomy between essence and appearance. If there is no grounding — no truth — then all is relative and equal: mere opinion. It is as if Friedrich Nietzsche, that great earthquake who shattered Western metaphysics, has not yet hit the world of geopolitics.

This is where classical Chinese philosophy can help us glimpse a different orientation. Chinese thought has never gotten stuck in the binary between truth and opinion; there has always been a third space somewhere between absolute grounding and mere relativism. Generations of Chinese thinkers have developed a philosophical sensitivity that avoids declaring any one thing essentially true and hence inherently better than any other, without, however, declaring all things to be the same.

As Song Bing describes in her contribution to this volume, a philosopher in China was never an individual grasping and

hence imposing the real and the true, but rather a person who could remain open to the incessant flow of co-becoming without falling in love with a single partisan belief. Virtuous persons, Confucius reminds us, do not take up a definite position. And yet they speak and act — at least sometimes. Human beliefs, choices, or actions emerge neither on absolute principles of truth nor on mere relativist caprice; rather, they are based on what resonates with the cosmos and befits the moment.

This distinction is quite evident in the Chinese conception of the *mean* or middle. The mean is not the middle-*point* between two opposites; it is not the average between two entities; it is not, as Aristotle had it, a sense of moderation between extremes. The mean is, rather, *energeia*, an energy, an ability to *shift* or flow from one side to the other, changing position as the situation requires. If something befits a specific moment, then it follows that not all things are at all times equal. Even if nothing is *ultimately* better, still, some things *are* better than others in the given moment and situation for those involved.

In much classical Chinese philosophy, the choice of that which befits the moment appeals to concepts such as harmony or resonance. This resonance between the subject and the cosmos is the ground of morality and the guide for action. Could we not evolve the blurry idea of a resonance with the cosmos to that of a resonance for the specific or concrete matter at hand, with *everybody and everything else* — a type of pragmatic, planetary, and polyphonic agreement?

Resonance with everybody and everything is not a logical absolute. Western metaphysics began as an adventure to reach a correspondence between the word and the object that it signifies.

But when Confucius, in *The Analects*, calls for the "rectification of names," he means adapting human words not to an external thing-in-itself but to a principle of social harmony. Resonance, similarly, calls for words that are *representative* of the will of the world in the political sense: Just as one would not consider a government unrepresentative because it was not supported by every single individual in the community of reference, so a universal does not need unanimity to exist and thrive but a contingently defined *enough of the world*.

This is an empirical and ultimately a pragmatic and political process — it is not an "order," an absolute, but a contingent, shifting, and clearly delimited use-value that is seen to best apply to everyone, everywhere, in the specific conditions of the here and the now. Rather than searching for universal laws and values and trying to lock them in through general-purpose institutions and laws, a form of resonant planetary pragmatism would privilege specific claims to concrete norms, behaviors, public goods, rules, or institutions that *should apply to all* merely because that universal applicability is *what resonates and befits the present moment*.

Planetary Pragmatism

In *What Is Enlightenment?,* one of the last texts he wrote before his premature death, Michel Foucault called for the emergence of *specific* — as opposed to *universal* — intellectuals. He meant individuals (and, supposedly, institutions and collective agents) capable of conducting an inquiry and promoting a transformation focused not on the recreation of an all-encompassing vision of society but on specific or concrete transformations to what is

given.

The work of such intellectuals, he wrote, must "put itself to the test of reality, of contemporary reality, both to grasp the points where change is possible and desirable, and to determine the precise form this change should take."[4] His example comes from nuclear power, and in another text he singled out J. Robert Oppenheimer: "It's because he had a direct and localized relation to scientific knowledge and institutions that the atomic scientist could make his intervention; but, since the nuclear threat affected the whole human race and the fate of the world, his discourse could at the same time be the discourse of the universal."[5]

The conception of specific intellectuals who are universal in their particular domain and epistemically dependent on "contemporary reality" can apply to the formation of planetary-scale norms, behaviors, public goods, institutions, etc., that, in their domain of application and in the contingent conditions of historic reality, are judged to be what befits the moment. My imaginary meeting in the UN General Assembly interrupted by AI represents precisely such an agreement with a "localized relation to scientific knowledge and institutions" that, however, is a "discourse of the universal." But as Oppenheimer makes clear, this is not just about statements and agreements. Nuclear scientists did things: There was a clear transformation of the world in a material and practical manner that is part of a philosophy of pragmatism and concreteness.

We would not be aiming to create top-down universal claims that encompass every aspect of human life or that purport to represent a wholistic explanation of Being or Truth or a categorical imposition of Rights and Values. Rather than developing a general

interpretation and ordering of the world, we can situate ourselves at the level of the empirical, immanent, circumstantiated enunciation. Focusing on very specific issues — as opposed to a general orientation of value — provides the grounds for a new form of trans-systemic cooperation: A planetary cohabitation across the plurality of systems and ideologies, concrete planetary norms, institutions, polities, or shared common goods emerges because it resonates with everyone everywhere.

The 1946 constitution of the World Health Organization, for instance, grandly states that "the enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being."[6] That is a high-sounding but vague and inapplicable statement. A pragmatic alternative would argue for multiple, concrete, small steps taken one at a time as conditions allow. For instance, a planetary guarantee for the shared development, procurement, and distribution of necessary equipment, medicines, and vaccines in the condition of a declared pandemic, perhaps accompanied by the institution of a planetary research center on pandemic prevention.

Planetary pragmatism stands for the gradual development of a web of concrete and specific practices, policies, and institutions of universal scope. It would relinquish any new grounding or ordering principle, while at the same time accepting that some things are at some times better than others for everyone everywhere. The only truth procedure for their emergence is a process of resonance with contemporary reality, a polyphonic agreement stretching across and beyond ideological differences.

Concrete universals can take many shapes. They can be articulated as regulations; they can be incarnated in institutions;

they can take the form of global goods, the provision of which is guaranteed to all planetary citizens; they can take the form of rights or benefits. This is a long-term and continuous process of creation that no single country, no single branch of knowledge, no single summit can define and resolve. The very process of development of a web of concrete planetary practices would gradually reinforce the feeling of planetary belonging, fostering a growing constituency for the continuation and acceleration of the process.

Take, for example, climate disasters: These are increasingly frequent events with unevenly distributed suffering and capacities to mitigate. An example of a concrete institution that could contribute to weaving together a sense of common humanity would be the establishment of a planetary civil protection force with joint fast-reaction capacities, such as the Green Cross for the planet, which was first proposed by Mikhail Gorbachev in 1990.[7]

Planetary pragmatists would be looking painstakingly for that which can bring us together, however modest it may appear, and try to institutionalize it. Rather than letting a hundred flowers of particularism bloom, they would plant a hundred seeds of planetary agreements. There is no single heroic moment in which a whole is constituted and a self-conscious planet emerges, but rather a series of small-scale, partial but concrete actions that accumulate one after the other. This accumulation would aim to reach a series of *thresholds*, or moments, during which a narrative begins to be formed to connect the dots. The concrete perception of a shared planetary destiny therefore does not precede the construction of a common humanity.

And so, to those resigned to a world of conflict and zero-

sum relativism, we would respond as Marco Polo did to the Kublai Khan in the closing lines of Italo Calvino's *Invisible Cities*:

> *[The Great Khan] said: "It is all useless, if the last landing place can only be the infernal city, and it is there that, in ever-narrowing circles, the current is drawing us."*
>
> *And Polo said: "The inferno of the living is not something that will be; if there is one, it is what is already here, the inferno where we live every day, that we form by being together. There are two ways to escape suffering it. The first is easy for many: accept the inferno and become such a part of it that you can no longer see it. The second is risky and demands constant vigilance and apprehension: seek and learn to recognize who and what, in the midst of the inferno, are not inferno, then make them endure, give them space."*[8]

We inhabit a world of conflict and division. And yet, within that conflict, let us find what is not conflict, let us make it endure, let us give it space.

Notes

1 "Meaning and Sense." in *Collected Philosophical Papers*, trans. Alphonso Lingis (Dordrecht, Netherlands: Martinus Nijhoff, 1987), 101.

2 The sentence is usually attributed to Fyodor Dostoevsky in The Brothers Karamazov and might have provided the basis for Friedrich Nietzsche's analysis of the death of God and the ensuing feeling of nihilism and relativism.

3 Nathan Gardels, "The New Axial Age," *Noema* (17 June 2020). https://www.noemamag.com/the-new-axial-age/.

4 In *The Foucault Reader*, ed. Paul Rabinow (New York: Pantheon Books, 1984), 32-50.

5 "Truth and Power," in *The Essential Works of Michel Foucault, 1954-1984, Vol. 3: Power*, ed. James D. Faubion, trans. Robert Hurley et al. (New York: The New Press, 2000), 136.

6 "Constitution of the World Health Organization." https://treaties.un.org/doc/Treaties/1948/04/19480407%252010-51%2520+PM/Ch_IX_01p.pdf.

7 "The History of Green Cross International," on "Who We Are," *Green Cross* (n.d.). https://www.gcint.org/who-we-are.

8 Trans. William Weaver (Orlando, FL: Harcourt, 1974), 165.

Planetary Realism

NATHAN GARDELS

A paradigm shift is underway from describing our world in terms of globalization and geopolitics to the concept of "planetarity." There are three aspects to this concept: planetary reason, planetary time, and planetary realism.

The awareness that we humans are not the center of it all, but only one part of the Earth's self-regulating ecosystem of multiple intelligences that strive interdependently toward sustainable equilibrium, forms the foundation of "planetary reason." James Lovelock's concept of "Gaia," named after the Earth goddess in ancient Greek mythology, expresses a similar notion in popular parlance.

This disclosure of a non-human-centric logos has been

enabled by the advent of powerful computation, data analysis, and recursive machine learning that vastly expands the heretofore limited scope of human understanding of whole-Earth systems. We are only aware of climate change in the first place in this way. As the philosopher Benjamin Bratton proposes in this volume, only when intelligence becomes artificial and can be scaled into massive, distributed systems beyond the narrow confines of biological organisms can we have a knowledge of the planetary systems in which we live.

The emergent exoskeleton of sensors, satellites, clouds, and networks constitute the rudiments of a common cognizant sphere that will synchronize human temporality going forward. As the German philosopher Peter Sloterdijk has observed:

> *From a macro-historical viewpoint, all cultures must tackle two general realities. On the one hand, the fact that the Earth has been recognized as a finite, planetary ecosystem that must be managed through a global environmental policy, and on the other hand, the realization that the transition from passéism to futurism has become more or less unavoidable everywhere.*
>
> *This implies that many cultures must understand that, while looking back at a mainly separate past, they will experience a primarily common future. This leads to the emergence of a global situationism — that inherent traits do not exist but are shaped by our environment — which places the single Earth in the forefront as a common site for all cultures.*
>
> *The local narratives are being increasingly forced to coordinate the time horizons of their rooted history (idiochronic) with the virtual synchronic horizon of the common world time.*[1]

In a related vein, the Cameroonian-born philosopher Achille Mbembe posits that the new planetary awareness also compresses geo-biochemical, historical, and experiential time into a condition of simultaneous unfolding. "We are in an epoch when time is no longer differentially distributed along human and non-human scales," he has said. "There's no longer a social history separate from natural history. That is over. Human history and Earth history are now indivisible."[2]

This synchronous moment of temporal simultaneity defines "planetary time."

"Planetary realism" is the practical manifestation of these new understandings of time and reason in the spatial frame. Consistent with the shift from globalization and geopolitics to the planetary, it departs from the old "realist" school of foreign policy that regards nation-states as the principal actors on the world stage, engaged in an endless "anarchic" struggle against others in pursuit of securing their own sovereign interests. Reality these days dictates another kind of realism when it comes to the convergence of critical common challenges that are beyond the scope of remedy by any one nation or bloc of nations since their direct impact is spatially diffused across the entire planet.

My first encounter with the concept came as a young political aide to California governor Jerry Brown back in the 1970s. As he told *Noema* in an interview:

> *I was running for president in 1976, and I was formulating some ideas on foreign policy. As I discussed these issues with knowledgeable people, I thought of the world as being deeply interconnected. I had the sense that it wasn't just 'us against them,' but rather that there was connectivity among all nations*

across a range of issues.

On the one hand, I wanted to be realistic: hard-boiled, not getting carried away with utopianism. That's the realistic side: Morgenthau, Machiavelli, Kissinger. But on the other hand, I wanted to add a dose of humanism and romanticism as well. I wanted to combine realism with some more thoughtful ideals, reflecting something more caring. So, I took the word "realism" and put it together with 'planetary,' because this was the early part of the environmental movement — the Stockholm Conference had just happened in 1972.

Back then, I wasn't thinking about climate change, but rather of the sea and of nuclear war. I argued that we had to see the big picture — you might call it the 'whole Earth perspective,' as opposed to nationalism and the politics of scapegoating. We can't let nationalism run away with us — we have to think of the interests we share. Today, that includes the shared vulnerability to viruses, cyber-attacks and climate change.[3]

That 1970s moment of insight at the outset of ecological awareness, which Governor Brown largely helped thematize in American politics, was eclipsed when focus shifted as Ronald Reagan — Brown's predecessor as California governor — as president of the United States in the 1980s decided to confront what he called the "evil empire" of the Soviet Union by escalating the nuclear arms race. When Mikhail Gorbachev came to power in the 1985, we witnessed one of those reverse moments of history as the Soviet leader refrained from keeping the Eastern bloc together through use of force, abiding the fall of the Berlin Wall, and ending the Cold War.

What drove Gorbachev's proactive restraint was a view of

the world not so distant from Governor Brown's. "Perestroika and new thinking," the last Soviet leader told me in one conversation in Moscow, "were attempts to respond to the global challenge of history — above all, interdependence."[4] Though he didn't use the phrase "planetary realism," it was the animating concept behind Gorbachev's launch of Green Cross International, for which I served as an advisor, soon after he was out of power. The idea was to act as a version of the Red Cross for environmental issues in a way that carried forward the thrust of his "new thinking."

Though we have since seen yet another reverse movement of history in geopolitics, the vision of connectedness forged by these pioneering political leaders challenges that fading paradigm more than ever. As the Earth's biosphere cascades toward unlivable conditions or as contagion spreads among humans from the microbial universe in which we all dwell, it is now widely recognized that the security of each depends inextricably on the other. Planetary realism is supplanting the old "realpolitik" of international relations with the more encompassing relationality of what could be called "Gaiapolitik."

Gaiapolitik In A World Of Sovereign States

The conceptual demolition of the outmoded paradigm of nation-state realism, however, does not erase its still firmly rooted expression in present practice. Building the centripetal momentum toward binding planetary association against the weighty centrifugal pull of tribal identity is an endeavor as fraught as it is necessary.

While technology and advancing science may foster a

universal understanding of the planetary condition, politics and culture have a different logic rooted in emotion and ways of life cultivated among one's own kind. Far from moving ahead in lock step, when they meet, they clash.

Indeed, the great paradox today is that the planetary imperative of mitigating climate change has become the province of renewed nationalism. Industrial policies designed to make the green energy transition are competing to protect and promote national self-interest instead of joining together at the level of all humanity. What we see instead is the battle of subsidies between the U.S. Inflation Reduction Act and the European Green New Deal, with both raising stiff tariff barriers to blunt China's concerted conquest of core green technologies from storage batteries to electric vehicles, solar cells, and their supply chains.

In short, rather than uniting as a threatened species to meet a challenge that knows no boundaries, competition has sidelined collaboration within the West while global warming has been weaponized in the new Cold War with China.

The new element of a greenlash, registered in recent European Union elections, portends social resistance that is less about climate denial than resentments toward self-righteous Tesla-driving elites, climate change activists jet-setting around the world, and the unequally borne costs of the energy transition. The Diet-Coke imaginary of environmentalists — who sold climate policies as achievable without undue burdens on economies built around fossil fuels for more than a century — has been put to rest. As it becomes clear how deeply any green energy transition will bite into the daily bread, we are learning the hard lesson that mitigating climate change has a limited constituency in both

consumer democracies and growth-oriented autocracies.

In this sense, "planetary realism" takes on a double meaning. It entails both a realistic recognition of the interdependence of the planetary condition, as well as a realistic grasp of what it will take to navigate planetary concerns through what remains a world of nations still deeply committed to the notion of sovereignty and self-determination.

Planetary Realism In Practice

What planetary realism would entail in the climate case is a "partnership of rivals" on mitigating action between the two largest economies and carbon emitters in the world — the United States and China — despite conflicts in other realms from technology to trade to human rights. The Americans and Chinese might well survive the decoupling of their economies from each other, but the world will not survive the decoupling of their climate fates.

In California, certain governors, including Brown when he returned to office in 2011, have made their own climate agreements directly with China, marking the emergence of a new breed of sub-national statespersons seeking cooperation across borders at any level because they cannot effectively meet the threat to their own jurisdiction and constituencies without addressing it everywhere else.

In 2023, California governor Gavin Newsom met with Chinese president Xi Jinping to initiate a range of joint projects to "enhance cooperation on strengthening low-carbon development and the green transition."[5]

China recognizes that the Pacific-facing Golden State, with a population of 40 million, is nearly a nation unto itself. Given that California is the fifth-largest economy in the world, the state's public policies that shape its huge market affect standards for the entire United States, especially when it comes to auto emission controls, electric vehicle mandates, and decarbonizing technologies.

As Governor Newsom publicly declared in Beijing, "Divorce is not an option" when it comes to climate. Despite all other tensions between these two incommensurate political systems, what the governor called the "fundamental and foundational" climate summons must bind the two together.[6]

In this way, as Jonathan Blake puts it in this volume, "translocalism circumnavigates geopolitics" when vital interests inconveniently transcend boundaries, not least when there is strategic conflict at the top, as in the case of the United States and China. There is a growing plethora of similar arrangements across the world where cities and regions coordinate and aggregate their climate policies outside the national frame.

At the broader level, the very awareness that humans are embedded and not apart from natural systems means planetary governance will not be centered at the "global" level. Rather, it will be distributed, like natural systems themselves, reaching scale through networks and decentralized through spatial decision-division at appropriate levels where relevant action needs to be taken, from household to neighborhood to continent.

In the condition of interdependence where everything affects everything else, the health of a system is determined through the creative and self-regulating alignment of the

interacting parts with the whole. With the technological tools it has devised, the human species can knowingly align governance of its own amplified capacities with the spirit it now understands.

Becoming Universal

As Woodrow Wilson conceived it in his political philosophy, governance is an ongoing negotiation with the reality it faces. It is "a living thing. It falls, not under the [mechanical] theory of the universe, but under the theory of organic life. It is accountable to Darwin, not to Newton. It is modified by its environment, necessitated by its tasks, shaped to its functions by the sheer pressure of life. No living thing can have its organs offset against each other" and still survive, no less thrive. They must work together, or each will fail.

For the 28th president of the United States, the aim of governance is to achieve an "amicable community of purpose" by arranging the constituent aspects in harmony and proportion with each other. "Their co-operation is indispensable, their warfare fatal. ... There can be no successful government without the intimate, instinctive co-ordination of the organs of life and action."[7]

In other words, under the condition of interdependence, where everything affects everything else, the health of an organism is determined through the appropriate articulation of the interacting parts with the whole.

Lamentably, the League of Nations established in the wake of one world war with Wilson's perspective in mind yielded to another in short order. While he was awarded a Nobel Peace Prize

for his vision, the United States itself never joined up to what it saw as an imposition from the outside on its national sovereignty.

Wilson, perhaps, failed to fully grasp the wisdom of his own insight into how governing systems evolve as living organisms. The universal alignment he hoped for could not emerge from some heroic foundational event with pinstriped diplomats putting pen to paper at a palace outside Paris, but only through a practical *process* of responding, one tangible step at a time, to pressing challenges that all of humanity faces.

A planetary community of fate can only be forged through the shared experience of volitional mobilization by diverse actors to meet a summons that demands a common response. Only what arises organically in this way can be integral and, one day, make what now may be untimely meditations timely.

Notes

1 Peter Sloterdijk, "The World Is Returning to Pluralism after American Hegemony, says German Philosopher," interview by Nathan Gardels, *Noema* (January 17, 2018). https://www.noemamag.com/the-world-is-returning-to-pluralism-after-american-hegemony-says-german-philosopher/.

2 Achille Mbembe, "How to Develop a Planetary Consciousness," interview by Nils Gilman, *Noema* (January 11, 2022). https://www.noemamag.com/how-to-develop-a-planetary-consciousness/.

3 Jerry Brown and Stewart Brand, "The Origins of 'Planetary Realism' and 'Whole Earth' Thinking," interview by Nils Gilman, *Noema* (February 9, 2021). https://www.noemamag.com/the-origins-of-planetary-realism-and-whole-earth-thinking/

4 Mikhail Gorbachev, "Perestroika 20 Years Later," *New Perspectives Quarterly* 30, no. 4 (October 1, 2013): 40-44.

5 "Memorandum of Understanding to Enhance Cooperation on Low Carbon Development Between the National Development and Reform Commission of the People's Republic of China And the State of California of the United States of America." https://www.gov.ca.gov/wp-content/uploads/2023/10/Ca-China-MOU-10-25-23.pdf.

6 Nathan Gardels, "California's Planetary Realism," *Noema* (November 3, 2023). https://www.noemamag.com/californias-planetary-realism/.

7 Woodrow Wilson, *The New Freedom: A Call for the Emancipation of the Generous Energies of a People* (New York: Doubleday, Page, 1913). https://www.gutenberg.org/files/14811/14811-h/14811-h.htm.

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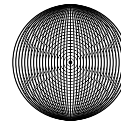
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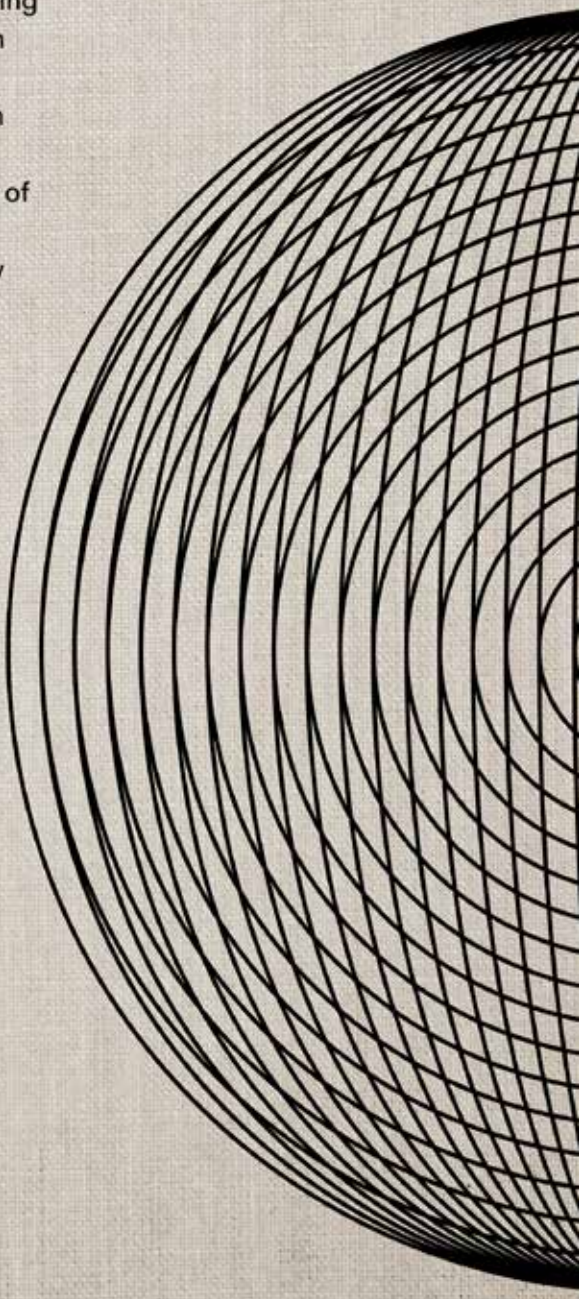
"Who will speak for the planet? I'm not sure that we will ever exit the situation where some speak for the planet while others speak against it. And also, speaking for the planet and listening to the planet are not exactly the same things. Maybe the first step is to listen."

—ACHILLE MBEMBE

As technoscience is disclosing in ever greater detail, human existence is profoundly and inextricably interwoven with Earthly biogeochemical systems, the destabilization of which presents a new and urgent category of distinctly *planetary* philosophical and policy challenges.

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